Exemple d'utilisation du package DegradSim

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Préparation

Installation du package depuis GitHub :

```
remotes::install_github("rytracetito/levy_packages", subdir = "DegradSim")
```

Les parties I et II utilisent les sept **processus de Lévy** du package (`A = NULL` par défaut). La partie III montre les **processus de Lévy à échelle de temps** (*time-scaled Lévy processes*) $X_t = Y_{A(t)}$, obtenus avec une fonction d'échelle de temps A (*time-scaling function*) : gamma, Wiener et inverse gaussien.

Paramètres utilisés :

Processus	Fonction	Paramètres
Gamma	`sim_gamma`	`a = 2`, `b = 3` (taux)
Wiener	`sim_wiener`	`mu = 0.5`, `sigma = 1.5`
Inverse gaussien	`sim_ig`	`mu = 0.5`, `eta = 2`
Variance-gamma	`sim_vg`	`theta = 0.5`, `sigma = 1.5`, `nu = 2`
Meixner	`sim_meixner`	`a = 1`, `b = 0.3`, `d = 1.5`, `m = 0.2`
Normal inverse gaussien	`sim_nig`	`alpha = 2`, `beta = 0.5`, `delta = 1`, `mu = 0.2`

Processus	Fonction	Paramètres
Poisson composé	`sim_cpoisson`	`lambda = 0.5`, sauts exponentiels de moyenne 1 (`rjump` par défaut)

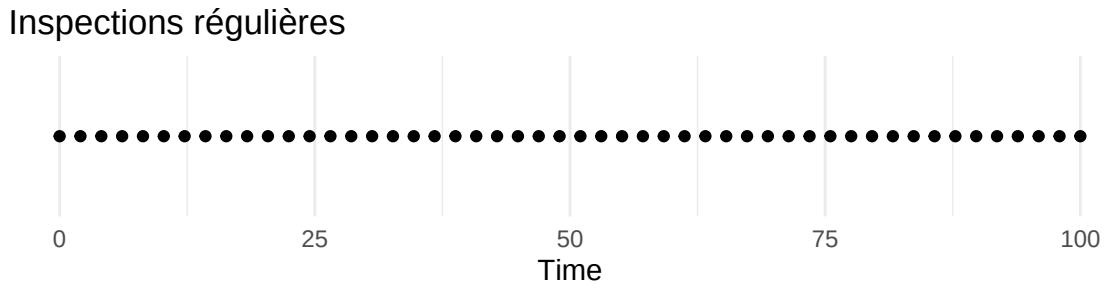
Dans chaque cas, les trajectoires simulées sont d'abord **stockées** dans un objet `x`, puis utilisées : premières valeurs, accroissements de la première trajectoire (arrondis à 3 décimales), puis graphique. Les accroissements montrent qu'un palier presque plat sur le graphique correspond à de petits accroissements, pas à des accroissements nuls (sauf pour le Poisson composé, constant entre deux chocs). Quand toutes les trajectoires ont les mêmes instants d'inspection, `x` est une matrice : une colonne `t`, puis une colonne par trajectoire (`X1`, `X2`, ...).

I - Une trajectoire

Une trajectoire inspectée 50 fois entre 0 et 100. `x` est une matrice à deux colonnes : `t` et `X1`.

Cas 1 : inspections régulières

```
t1 <- inspection_times("regular", n = 49, tmax = 100)$time
plot_plan(t1, "Inspections régulières")
```



Modèle 1 : gamma

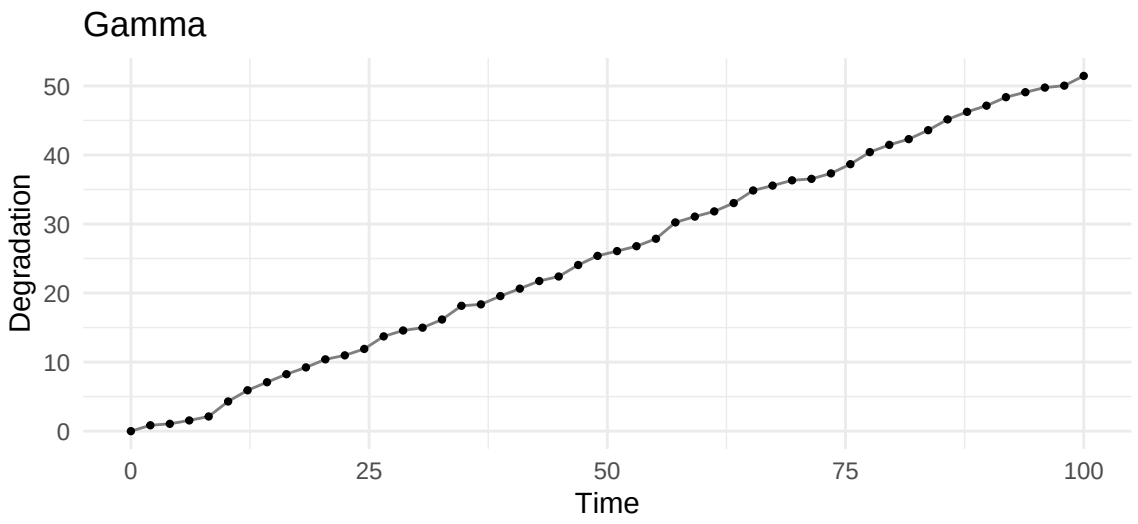
```
set.seed(11)
x <- sim_gamma(t1, a = 2, b = 3)
head(x)
```

```
##              t        X1
## [1,]  0.000000 0.0000000
## [2,]  2.040816 0.8501413
## [3,]  4.081633 1.0605764
## [4,]  6.122449 1.5495729
## [5,]  8.163265 2.1242214
## [6,] 10.204082 4.2990767
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 0.850 0.210 0.489 0.575 2.175 1.621 1.165 1.163 0.984 1.167 0.579 0.929
## [13] 1.820 0.850 0.408 1.184 1.987 0.208 1.198 1.078 1.106 0.654 1.663 1.324
## [25] 0.691 0.708 1.078 2.368 0.839 0.757 1.207 1.823 0.707 0.766 0.213 0.789
## [37] 1.337 1.744 1.053 0.829 1.296 1.561 1.097 0.890 1.232 0.718 0.669 0.281
## [49] 1.414
```

```
plot_traj(x, "Gamma")
```



Modèle 2 : Wiener

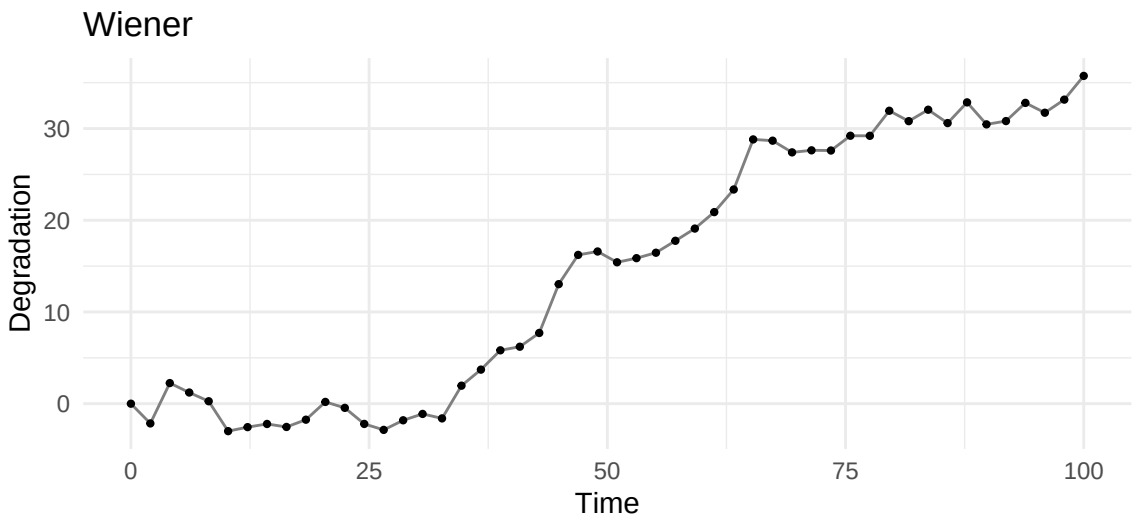
```
set.seed(12)
x <- sim_wiener(t1, mu = 0.5, sigma = 1.5)
head(x)
```

```
##              t         X1
## [1,]  0.000000  0.0000000
## [2,]  2.040816 -2.1522367
## [3,]  4.081633  2.2478203
## [4,]  6.122449  1.2180618
## [5,]  8.163265  0.2670301
## [6,] 10.204082 -2.9932234
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] -2.152  4.400 -1.030 -0.951 -3.260  0.437  0.345 -0.326  0.792  1.938
## [11] -0.646 -1.752 -0.650  1.046  0.694 -0.487  3.568  1.750  2.107  0.392
## [21]  1.500  5.322  3.189  0.372 -1.177  0.447  0.594  1.301  1.333  1.796
## [31]  2.465  5.460 -0.139 -1.274  0.222 -0.019  1.609 -0.007  2.731 -1.132
## [41]  1.245 -1.457  2.259 -2.399  0.359  1.984 -1.073  1.428  2.588
```

```
plot_traj(x, "Wiener")
```



Modèle 3 : inverse gaussien

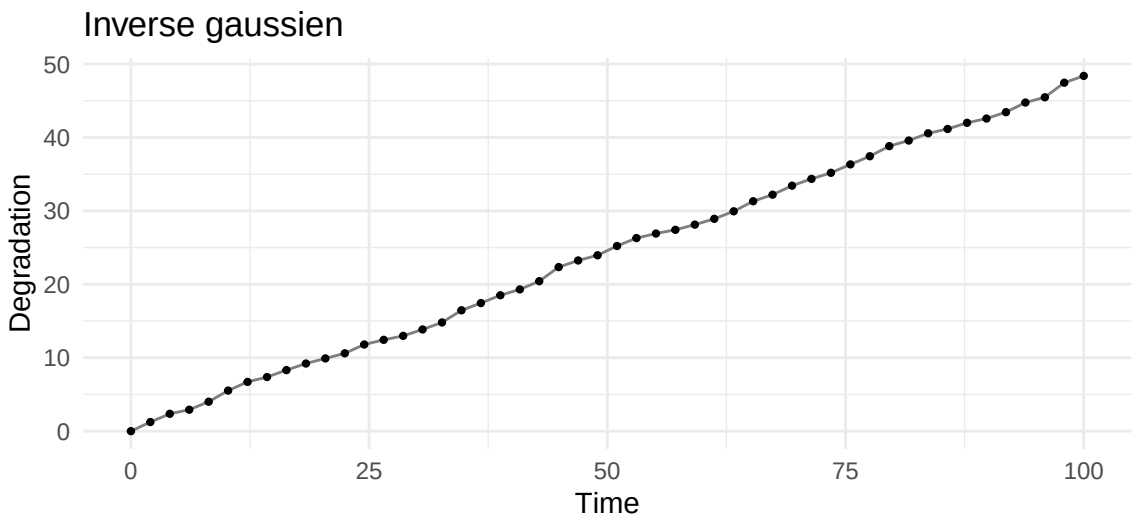
```
set.seed(13)
x <- sim_ig(t1, mu = 0.5, eta = 2)
head(x)
```

```
##              t       X1
## [1,]  0.000000 0.000000
## [2,]  2.040816 1.238516
## [3,]  4.081633 2.364051
## [4,]  6.122449 2.917536
## [5,]  8.163265 4.007073
## [6,] 10.204082 5.525188
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 1.239 1.126 0.553 1.090 1.518 1.180 0.666 0.939 0.898 0.695 0.697 1.199
## [13] 0.637 0.539 0.875 0.953 1.656 0.985 1.062 0.799 1.119 1.920 0.900 0.709
## [25] 1.267 1.075 0.616 0.509 0.706 0.791 1.023 1.371 0.892 1.227 0.927 0.826
## [37] 1.146 1.110 1.379 0.759 0.985 0.589 0.837 0.584 0.866 1.315 0.715 1.986
## [49] 0.926
```

```
plot_traj(x, "Inverse gaussien")
```



Modèle 4 : variance-gamma

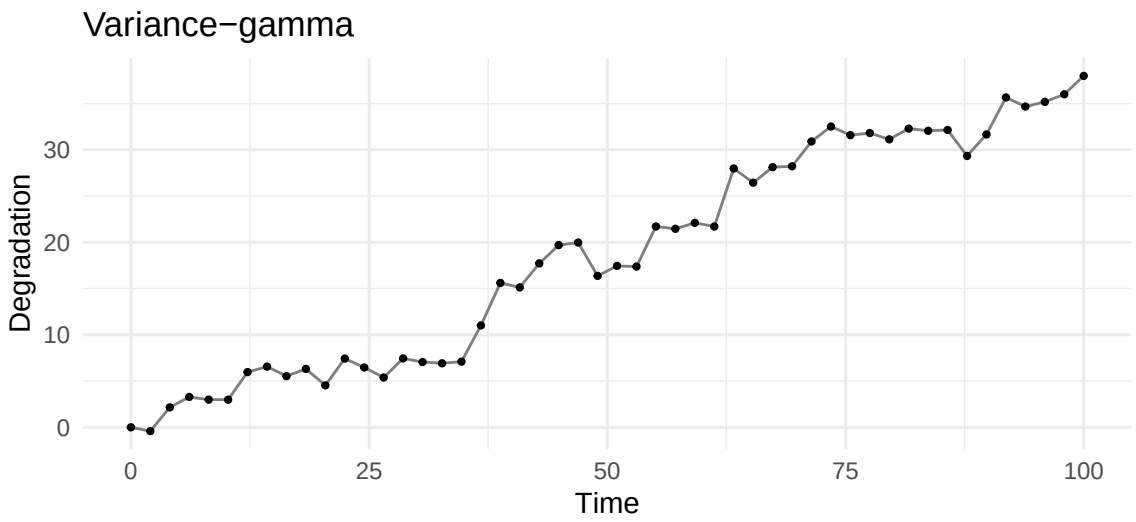
```
set.seed(14)
x <- sim_vg(t1, theta = 0.5, sigma = 1.5, nu = 2)
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  2.040816 -0.414327
## [3,]  4.081633  2.158973
## [4,]  6.122449  3.277011
## [5,]  8.163265  2.991563
## [6,] 10.204082  2.988429
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] -0.414  2.573  1.118 -0.285 -0.003  2.979  0.594 -1.029  0.777 -1.774
## [11]  2.894 -0.959 -1.087  2.062 -0.388 -0.133  0.176  3.916  4.592 -0.478
## [21]  2.590  1.973  0.280 -3.604  1.078 -0.075  4.339 -0.256  0.645 -0.399
## [31]  6.276 -1.532  1.670  0.098  2.692  1.608 -0.931  0.225 -0.678  1.162
## [41] -0.236  0.087 -2.809  2.324  4.000 -0.985  0.504  0.818  1.991
```

```
plot_traj(x, "Variance-gamma")
```



Modèle 5 : Meixner

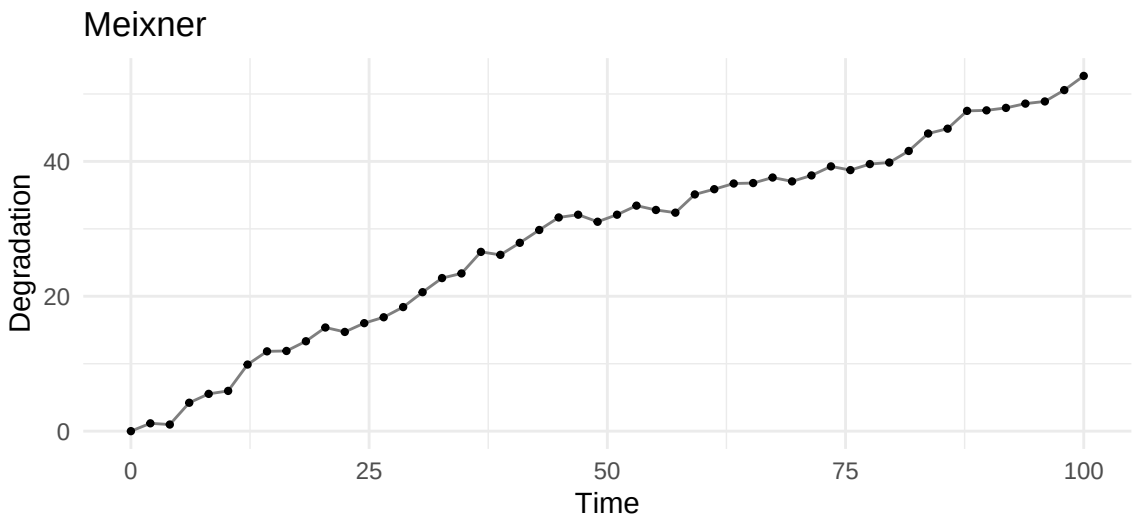
```
set.seed(15)
x <- sim_meixner(t1, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x)
```

```
##              t        X1
## [1,]  0.000000 0.0000000
## [2,]  2.040816 1.1594818
## [3,]  4.081633 0.9808625
## [4,]  6.122449 4.2100728
## [5,]  8.163265 5.5272998
## [6,] 10.204082 5.9679779
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  1.159 -0.179  3.229  1.317  0.441  3.909  1.957  0.057  1.441  2.038
## [11] -0.657  1.301  0.874  1.510  2.209  2.092  0.688  3.195 -0.438  1.783
## [21]  1.903  1.856  0.410 -1.047  1.046  1.344 -0.641 -0.400  2.704  0.775
## [31]  0.842  0.067  0.821 -0.574  0.885  1.340 -0.549  0.895  0.227  1.715
## [41]  2.588  0.718  2.640  0.076  0.366  0.633  0.322  1.685  2.104
```

```
plot_traj(x, "Meixner")
```



Modèle 6 : normal inverse gaussien (NIG)

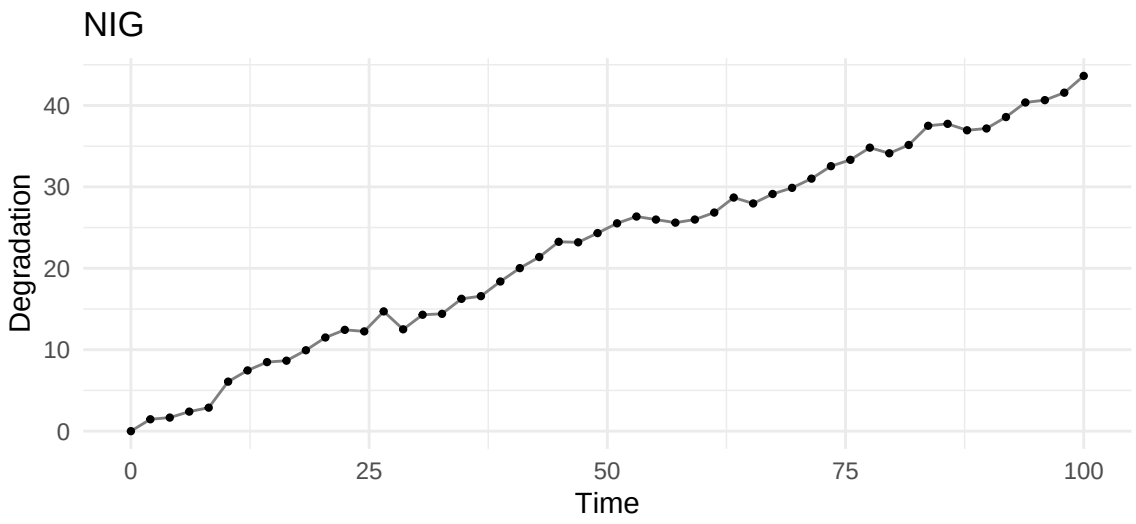
```
set.seed(16)
x <- sim_nig(t1, alpha = 2, beta = 0.5, delta = 1, mu = 0.2)
head(x)
```

```
##              t       X1
## [1,]  0.000000 0.000000
## [2,]  2.040816 1.457556
## [3,]  4.081633 1.666554
## [4,]  6.122449 2.398280
## [5,]  8.163265 2.882498
## [6,] 10.204082 6.078801
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  1.458  0.209  0.732  0.484  3.196  1.378  1.021  0.175  1.282  1.565
## [11]  0.941 -0.190  2.454 -2.200  1.789  0.113  1.844  0.328  1.799  1.642
## [21]  1.366  1.868 -0.064  1.134  1.198  0.835 -0.374 -0.378  0.381  0.858
## [31]  1.844 -0.726  1.163  0.754  1.131  1.533  0.777  1.494 -0.689  1.020
## [41]  2.358  0.235 -0.788  0.219  1.406  1.789  0.290  0.913  2.062
```

```
plot_traj(x, "NIG")
```



Modèle 7 : Poisson composé

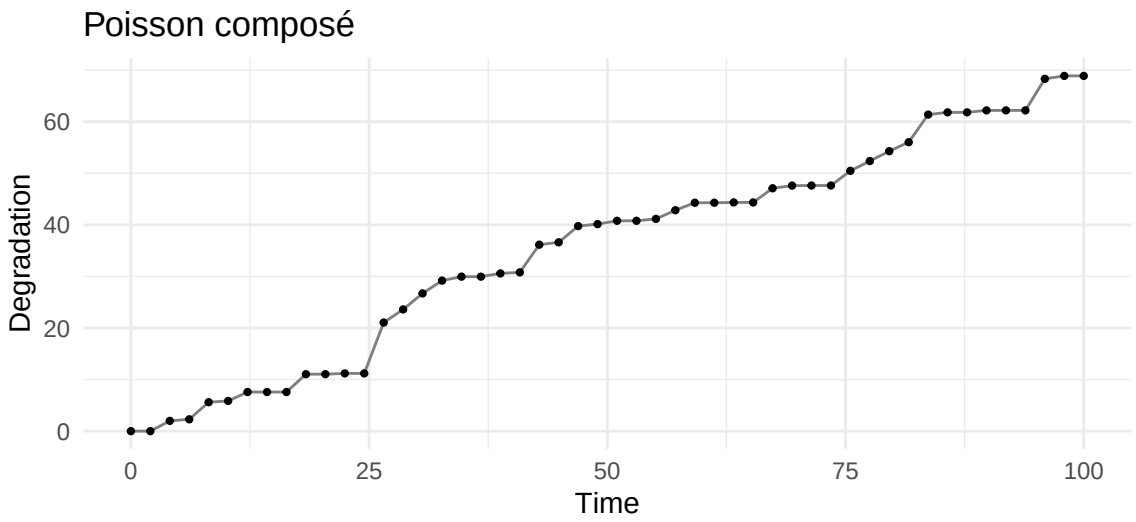
```
set.seed(17)
x <- sim_cpoisson(t1, lambda = 0.5)
head(x)
```

```
##              t       X1
## [1,]  0.000000 0.000000
## [2,]  2.040816 0.000000
## [3,]  4.081633 1.995153
## [4,]  6.122449 2.292960
## [5,]  8.163265 5.614836
## [6,] 10.204082 5.851075
```

```
round(diff(x[, "X1"]), 3)
```

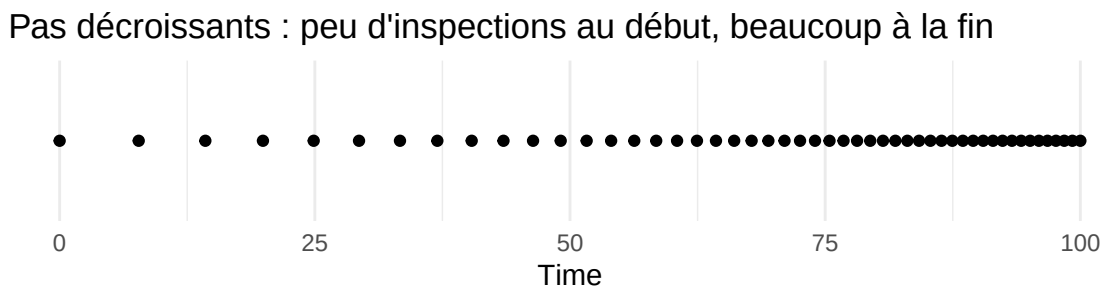
```
##  [1] 0.000 1.995 0.298 3.322 0.236 1.734 0.000 0.000 3.458 0.000 0.151 0.000
## [13] 9.866 2.540 3.108 2.481 0.768 0.000 0.629 0.201 5.374 0.448 3.153 0.388
## [25] 0.640 0.000 0.368 1.677 1.457 0.000 0.059 0.000 2.747 0.513 0.022 0.000
## [37] 2.848 1.900 1.910 1.746 5.355 0.425 0.000 0.383 0.000 0.000 6.123 0.565
## [49] 0.000
```

```
plot_traj(x, "Poisson composé")
```


Poisson composé
Degradation
0
20
40
60
0
25
50
75
100
Time

Cas 2 : pas décroissants

```
t2 <- inspection_times("decreasing", n = 49, tmax = 100)$time
plot_plan(t2, "Pas décroissants : peu d'inspections au début, beaucoup à la fin")
```



Modèle 1 : gamma

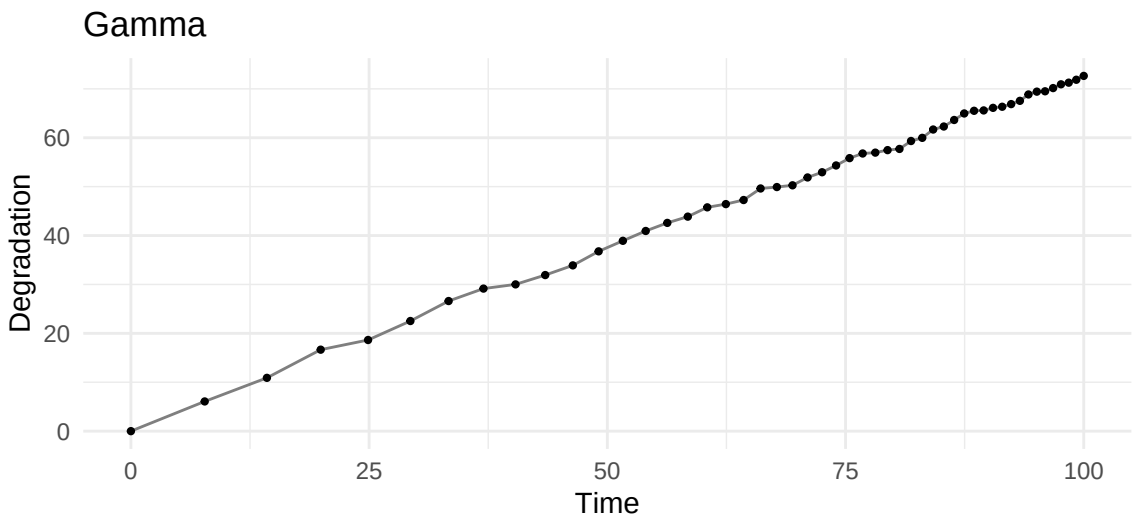
```
set.seed(21)
x <- sim_gamma(t2, a = 2, b = 3)
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  7.745007  6.072512
## [3,] 14.274444 10.898459
## [4,] 19.918616 16.660539
## [5,] 24.889163 18.640144
## [6,] 29.329869 22.520703
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 6.073 4.826 5.762 1.980 3.881 4.086 2.544 0.861 1.907 1.981 2.869 2.156
## [13] 2.010 1.653 1.277 1.902 0.663 0.835 2.359 0.293 0.358 1.605 1.065 1.385
## [25] 1.499 0.950 0.177 0.510 0.244 1.620 0.647 1.706 0.612 1.337 1.335 0.545
## [37] 0.074 0.526 0.224 0.542 0.668 1.304 0.576 0.072 0.658 0.771 0.335 0.596
## [49] 0.788
```

```
plot_traj(x, "Gamma")
```



Modèle 2 : Wiener

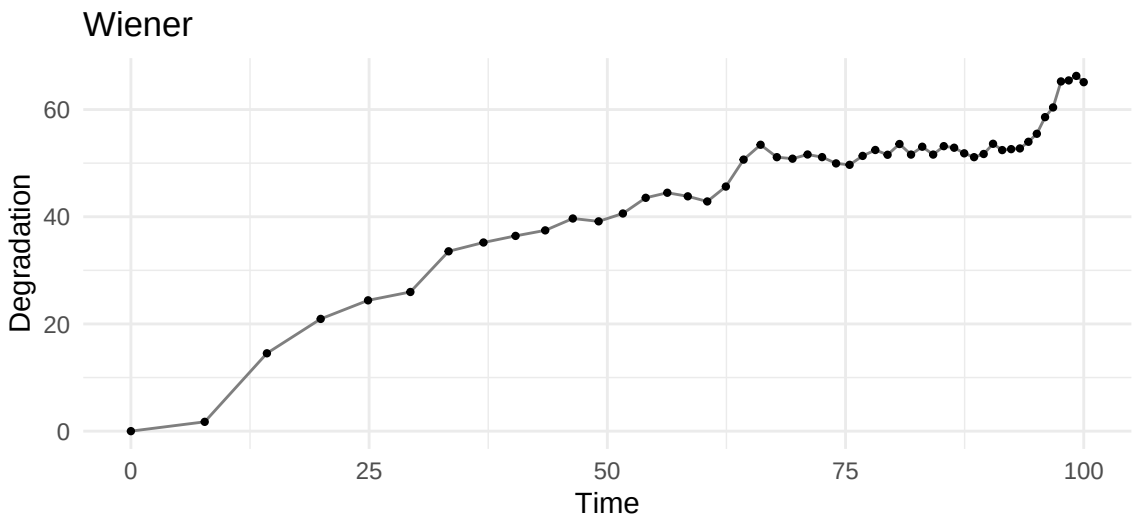
```
set.seed(22)
x <- sim_wiener(t2, mu = 0.5, sigma = 1.5)
head(x)
```

```
##              t       X1
## [1,]  0.000000  0.00000
## [2,]  7.745007  1.73459
## [3,] 14.274444 14.52480
## [4,] 19.918616 20.93840
## [5,] 24.889163 24.40291
## [6,] 29.329869 25.96275
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  1.735 12.790  6.414  3.465  1.560  7.590  1.641  1.235  1.028  2.216
## [11] -0.531  1.468  2.924  0.944 -0.669 -0.956  2.779  5.031  2.767 -2.314
## [21] -0.284  0.782 -0.490 -1.174 -0.261  1.663  1.101 -0.891  2.006 -1.971
## [31]  1.465 -1.464  1.572 -0.294 -1.044 -0.715  0.589  1.905 -1.176  0.177
## [41]  0.131  1.244  1.493  3.102  1.810  4.853  0.187  0.845 -1.172
```

```
plot_traj(x, "Wiener")
```



Modèle 3 : inverse gaussien

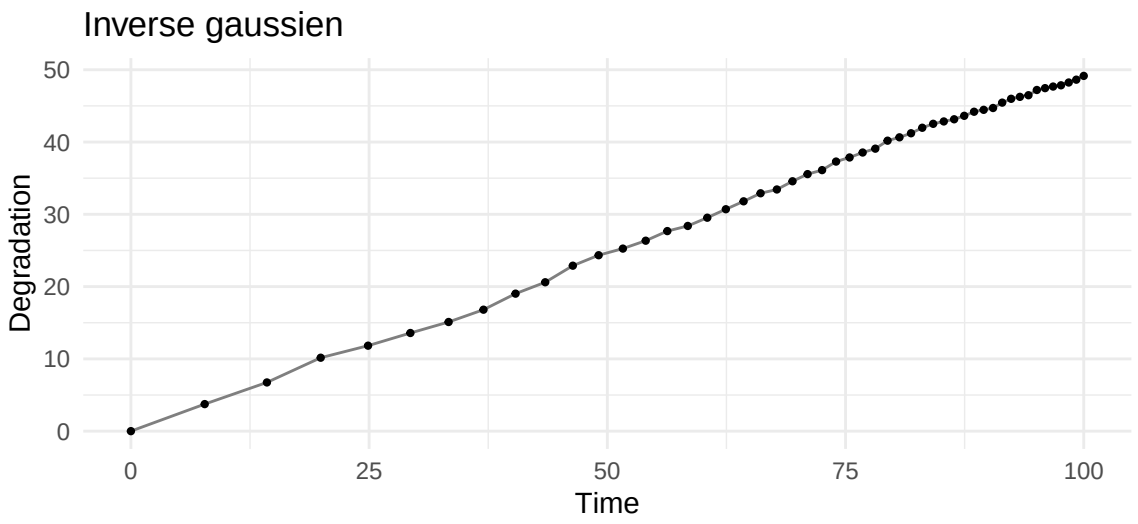
```
set.seed(23)
x <- sim_ig(t2, mu = 0.5, eta = 2)
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  7.745007  3.740389
## [3,] 14.274444  6.738983
## [4,] 19.918616 10.158120
## [5,] 24.889163 11.824819
## [6,] 29.329869 13.578556
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 3.740 2.999 3.419 1.667 1.754 1.523 1.702 2.219 1.577 2.292 1.447 0.917
## [13] 1.092 1.330 0.712 1.139 1.174 1.095 1.115 0.524 1.136 0.989 0.554 1.183
## [25] 0.568 0.686 0.532 1.106 0.465 0.562 0.760 0.547 0.326 0.301 0.480 0.563
## [37] 0.265 0.253 0.743 0.530 0.268 0.226 0.725 0.255 0.220 0.174 0.385 0.393
## [49] 0.522
```

```
plot_traj(x, "Inverse gaussien")
```



Modèle 4 : variance-gamma

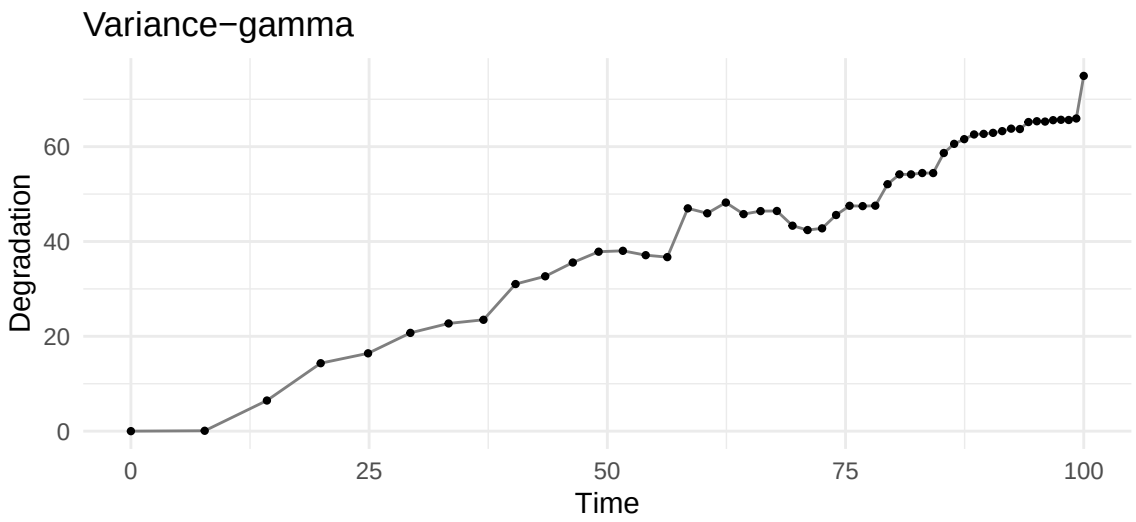
```
set.seed(24)
x <- sim_vg(t2, theta = 0.5, sigma = 1.5, nu = 2)
head(x)
```

```
##              t          X1
## [1,]  0.000000  0.00000000
## [2,]  7.745007  0.07862292
## [3,] 14.274444  6.45985339
## [4,] 19.918616 14.32829297
## [5,] 24.889163 16.41061317
## [6,] 29.329869 20.72743266
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  0.079  6.381  7.868  2.082  4.317  1.987  0.772  7.536  1.638  2.903
## [11]  2.283  0.186 -0.918 -0.412 10.290 -1.046  2.267 -2.435  0.632  0.025
## [21] -3.105 -0.916  0.358  2.813  1.947 -0.070  0.090  4.533  2.075 -0.001
## [31]  0.280 -0.007  4.232  1.928  0.995  0.994  0.118  0.215  0.365  0.520
## [41] -0.080  1.475  0.170 -0.070  0.298  0.078 -0.039  0.315  8.983
```

```
plot_traj(x, "Variance-gamma")
```



Modèle 5 : Meixner

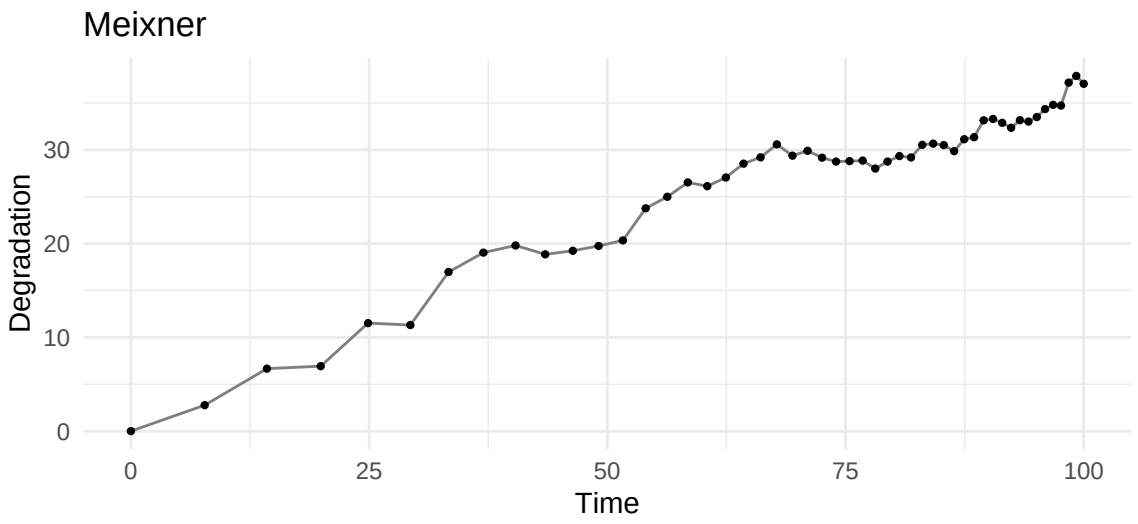
```
set.seed(25)
x <- sim_meixner(t2, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  7.745007  2.770050
## [3,] 14.274444  6.664740
## [4,] 19.918616  6.928232
## [5,] 24.889163 11.519308
## [6,] 29.329869 11.315784
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  2.770  3.895  0.263  4.591 -0.204  5.659  2.067  0.758 -0.944  0.382
## [11]  0.512  0.591  3.424  1.230  1.534 -0.404  0.924  1.478  0.672  1.386
## [21] -1.208  0.522 -0.734 -0.415  0.046  0.060 -0.852  0.746  0.585 -0.144
## [31]  1.341  0.130 -0.160 -0.646  1.275  0.218  1.802  0.141 -0.411 -0.529
## [41]  0.809 -0.143  0.480  0.848  0.451 -0.075  2.449  0.715 -0.842
```

```
plot_traj(x, "Meixner")
```



Modèle 6 : normal inverse gaussien (NIG)

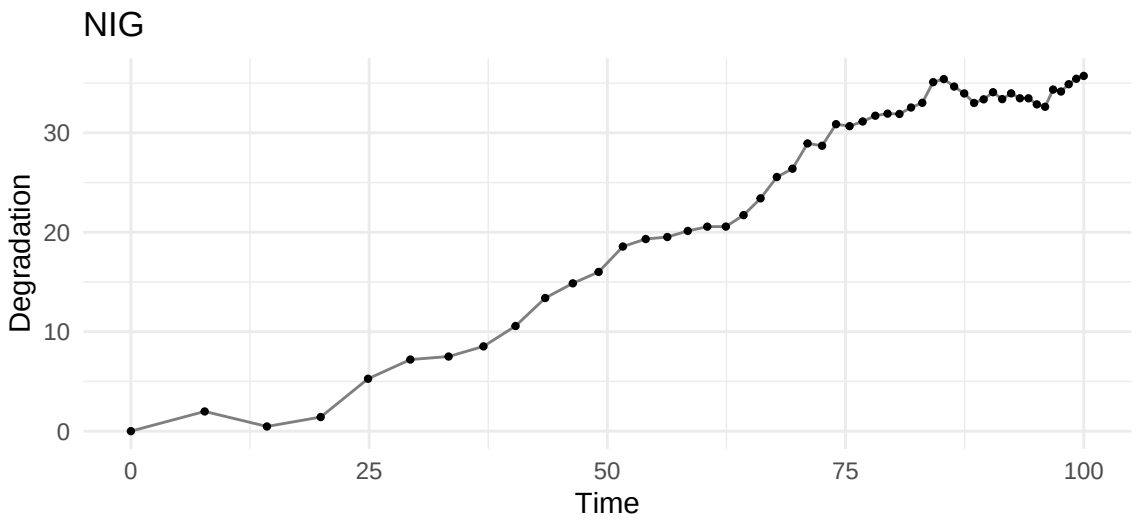
```
set.seed(26)
x <- sim_nig(t2, alpha = 2, beta = 0.5, delta = 1, mu = 0.2)
head(x)
```

```
##              t        X1
## [1,]  0.000000 0.0000000
## [2,]  7.745007 1.9802078
## [3,] 14.274444 0.4694236
## [4,] 19.918616 1.4150693
## [5,] 24.889163 5.2673012
## [6,] 29.329869 7.1993188
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  1.980 -1.511  0.946  3.852  1.932  0.304  1.021  2.047  2.817  1.475
## [11]  1.147  2.555  0.753  0.203  0.614  0.427  0.008  1.152  1.689  2.143
## [21]  0.839  2.541 -0.236  2.171 -0.198  0.472  0.577  0.209 -0.032  0.648
## [31]  0.471  2.079  0.308 -0.750 -0.692 -0.958  0.365  0.716 -0.694  0.580
## [41] -0.494 -0.003 -0.611 -0.243  1.731 -0.195  0.732  0.551  0.287
```

```
plot_traj(x, "NIG")
```



Modèle 7 : Poisson composé

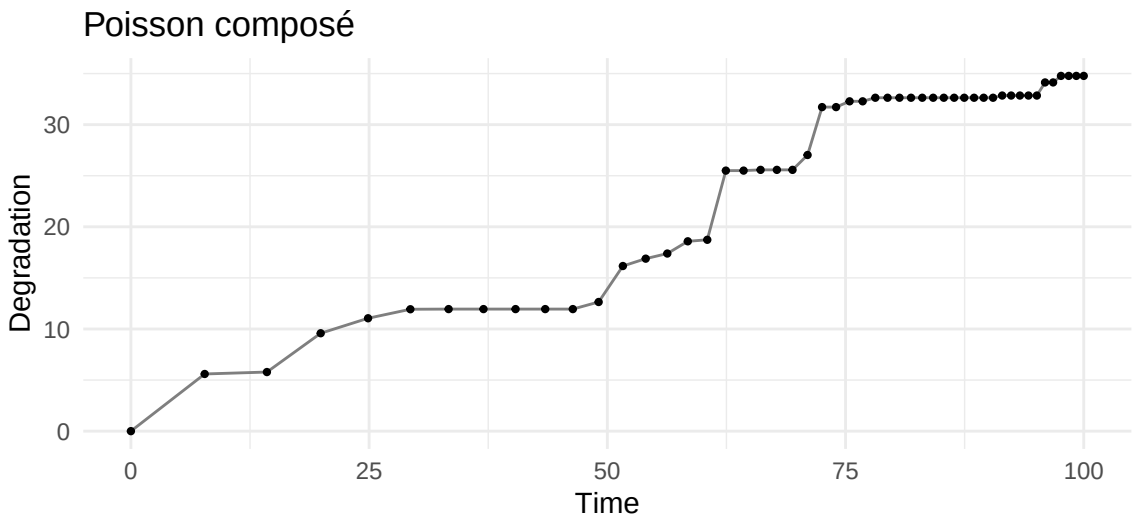
```
set.seed(27)
x <- sim_cpoisson(t2, lambda = 0.5)
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  7.745007  5.594785
## [3,] 14.274444  5.783156
## [4,] 19.918616  9.582708
## [5,] 24.889163 11.048961
## [6,] 29.329869 11.930607
```

```
round(diff(x[, "X1"]), 3)
```

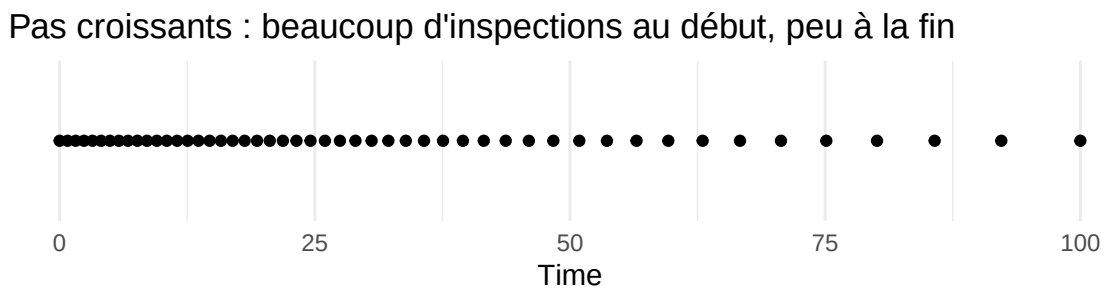
```
##  [1] 5.595 0.188 3.800 1.466 0.882 0.015 0.000 0.000 0.000 0.000 0.694 3.522
## [13] 0.720 0.502 1.196 0.148 6.772 0.000 0.074 0.000 0.000 1.455 4.688 0.000
## [25] 0.567 0.000 0.349 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000
## [37] 0.000 0.000 0.215 0.000 0.000 0.000 0.000 1.283 0.000 0.640 0.000 0.000
## [49] 0.000
```

```
plot_traj(x, "Poisson composé")
```


Poisson composé
Degradation
0
10
20
30
0
25
50
75
100
Time

Cas 3 : pas croissants

```
t3 <- inspection_times("increasing", n = 49, tmax = 100)$time
plot_plan(t3, "Pas croissants : beaucoup d'inspections au début, peu à la fin")
```



Modèle 1 : gamma

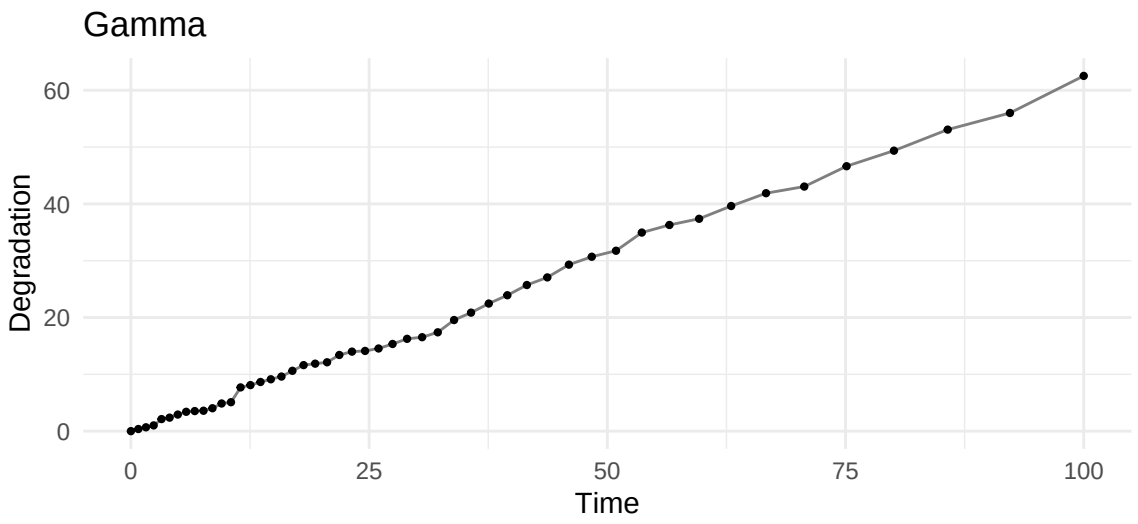
```
set.seed(31)
x <- sim_gamma(t3, a = 2, b = 3)
head(x)
```

```
##              t        X1
## [1,] 0.0000000 0.0000000
## [2,] 0.7809824 0.3733343
## [3,] 1.5768699 0.6759188
## [4,] 2.3882425 1.0089026
## [5,] 3.2157149 2.1225410
## [6,] 4.0599388 2.3735653
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 0.373 0.303 0.333 1.114 0.251 0.545 0.486 0.126 0.059 0.441 0.835 0.225
## [13] 2.605 0.405 0.546 0.480 0.480 1.013 0.998 0.258 0.244 1.286 0.590 0.121
## [25] 0.431 0.795 0.906 0.283 0.863 2.160 1.303 1.599 1.461 1.797 1.346 2.242
## [37] 1.393 1.054 3.203 1.335 1.089 2.246 2.255 1.170 3.581 2.738 3.706 2.927
## [49] 6.527
```

```
plot_traj(x, "Gamma")
```



Modèle 2 : Wiener

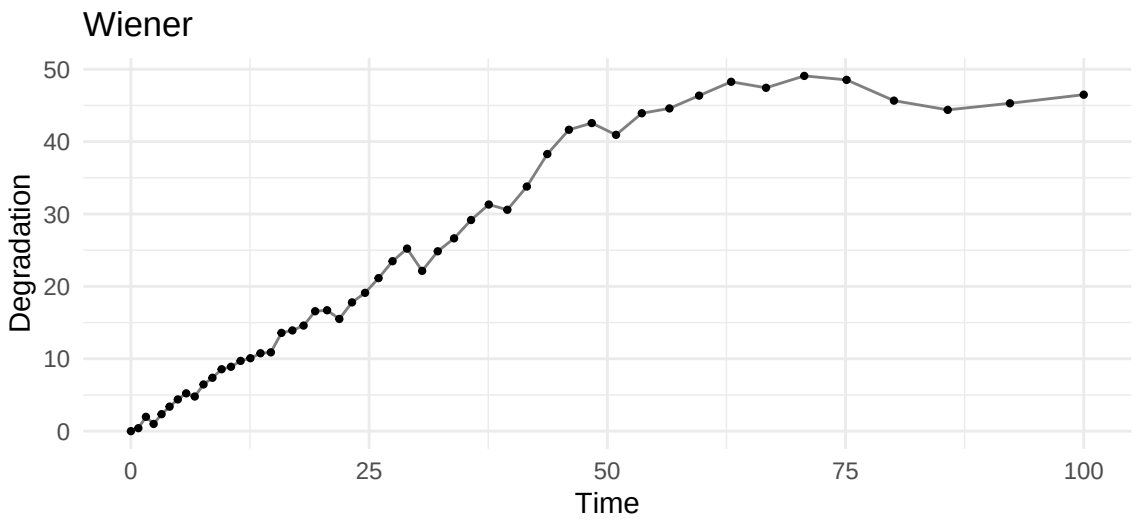
```
set.seed(32)
x <- sim_wiener(t3, mu = 0.5, sigma = 1.5)
head(x)
```

```
##              t        X1
## [1,] 0.0000000 0.0000000
## [2,] 0.7809824 0.4098987
## [3,] 1.5768699 1.9764668
## [4,] 2.3882425 0.9932504
## [5,] 3.2157149 2.3425640
## [6,] 4.0599388 3.3841013
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  0.410  1.567 -0.983  1.349  1.042  0.998  0.841 -0.438  1.666  0.922
## [11]  1.179  0.343  0.810  0.373  0.684  0.124  2.690  0.338  0.670  1.983
## [21]  0.137 -1.202  2.288  1.320  2.032  2.337  1.739 -3.076  2.709  1.783
## [31]  2.534  2.139 -0.726  3.210  4.490  3.356  0.923 -1.629  2.987  0.667
## [41]  1.769  1.910 -0.831  1.645 -0.547 -2.871 -1.275  0.908  1.190
```

```
plot_traj(x, "Wiener")
```



Modèle 3 : inverse gaussien

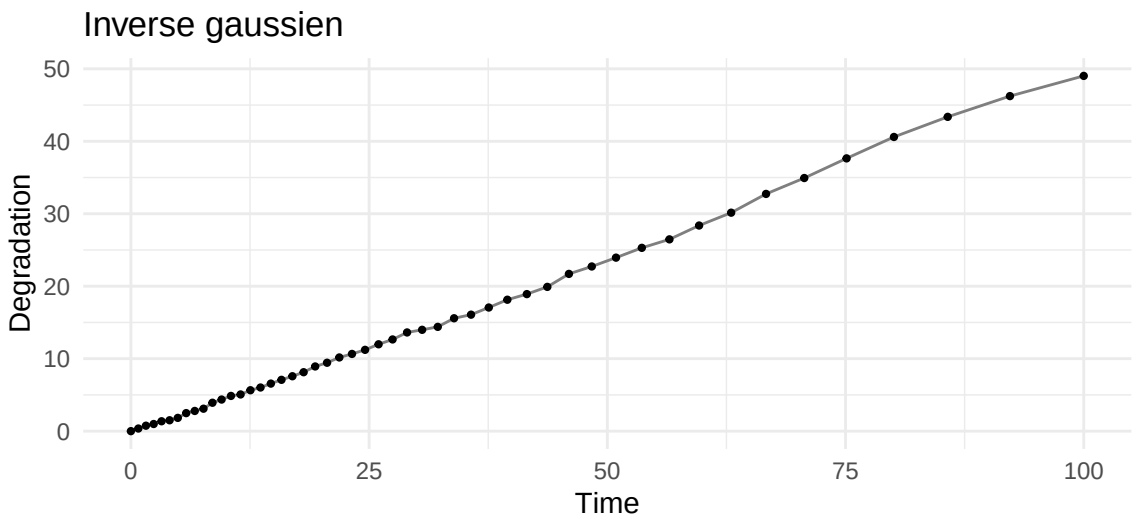
```
set.seed(33)
x <- sim_ig(t3, mu = 0.5, eta = 2)
head(x)
```

```
##              t        X1
## [1,] 0.0000000 0.0000000
## [2,] 0.7809824 0.3615935
## [3,] 1.5768699 0.7505417
## [4,] 2.3882425 0.9837075
## [5,] 3.2157149 1.3629841
## [6,] 4.0599388 1.5014091
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 0.362 0.389 0.233 0.379 0.138 0.330 0.656 0.299 0.311 0.822 0.441 0.499
## [13] 0.204 0.582 0.377 0.539 0.521 0.493 0.557 0.793 0.512 0.731 0.486 0.564
## [25] 0.767 0.667 0.967 0.363 0.407 1.187 0.497 0.986 1.075 0.777 0.996 1.792
## [37] 1.024 1.218 1.353 1.171 1.909 1.772 2.590 2.193 2.711 2.956 2.775 2.862
## [49] 2.789
```

```
plot_traj(x, "Inverse gaussien")
```



Modèle 4 : variance-gamma

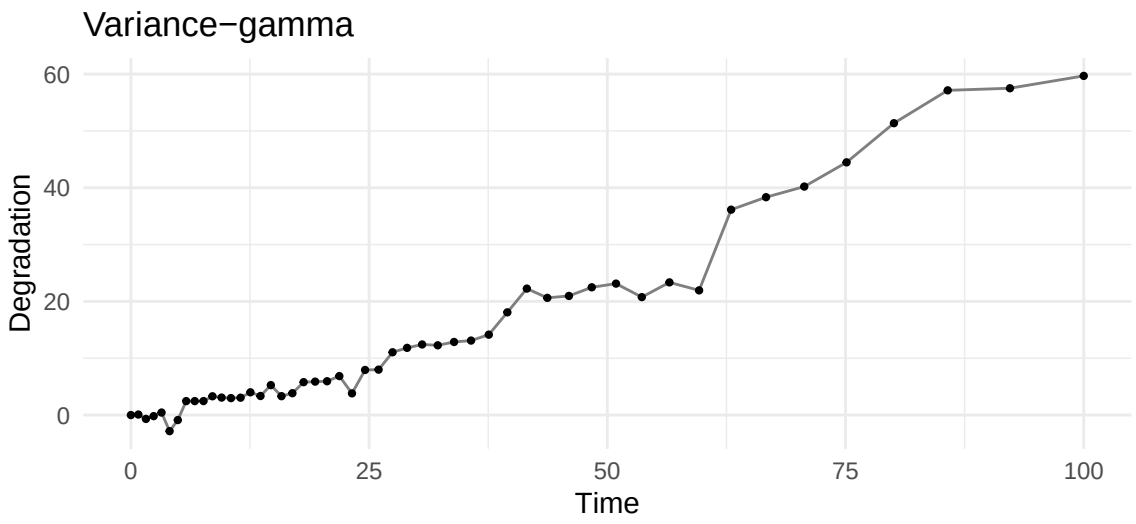
```
set.seed(34)
x <- sim_vg(t3, theta = 0.5, sigma = 1.5, nu = 2)
head(x)
```

```
##              t          X1
## [1,] 0.0000000  0.00000000
## [2,] 0.7809824  0.09322591
## [3,] 1.5768699 -0.68209708
## [4,] 2.3882425 -0.18811757
## [5,] 3.2157149  0.44419564
## [6,] 4.0599388 -2.82976129
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  0.093 -0.775  0.494  0.632 -3.274  1.952  3.334  0.000  0.005  0.846
## [11] -0.229 -0.094  0.073  0.949 -0.650  1.915 -1.948  0.526  1.943  0.090
## [21]  0.068  0.908 -3.039  4.118  0.056  3.058  0.774  0.597 -0.147  0.587
## [31]  0.247  1.039  3.927  4.181 -1.631  0.341  1.518  0.649 -2.375  2.592
## [41] -1.407 14.203  2.194  1.870  4.262  6.893  5.780  0.368  2.173
```

```
plot_traj(x, "Variance-gamma")
```



Modèle 5 : Meixner

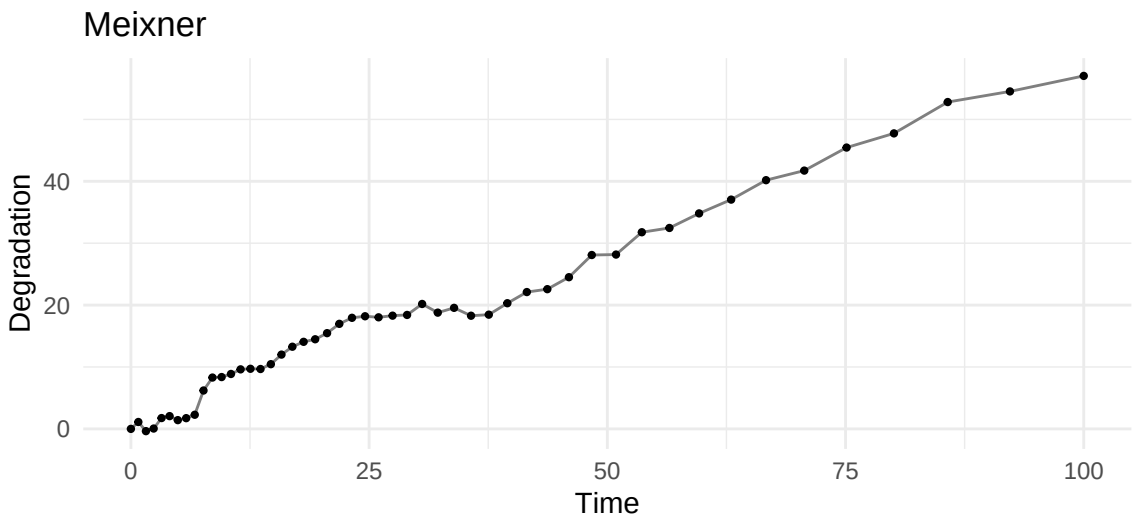
```
set.seed(35)
x <- sim_meixner(t3, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x)
```

```
##              t          X1
## [1,] 0.0000000  0.00000000
## [2,] 0.7809824  1.11330276
## [3,] 1.5768699 -0.36680705
## [4,] 2.3882425  0.05187965
## [5,] 3.2157149  1.74654076
## [6,] 4.0599388  2.05897099
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  1.113 -1.480  0.419  1.695  0.312 -0.647  0.319  0.550  3.918  2.094
## [11]  0.079  0.500  0.745  0.092 -0.037  0.780  1.549  1.273  0.793  0.413
## [21]  0.987  1.506  0.972  0.251 -0.170  0.267  0.116  1.773 -1.401  0.772
## [31] -1.273  0.172  1.850  1.802  0.468  1.935  3.584  0.074  3.607  0.691
## [41]  2.350  2.230  3.139  1.549  3.722  2.290  5.062  1.718  2.509
```

```
plot_traj(x, "Meixner")
```



Modèle 6 : normal inverse gaussien (NIG)

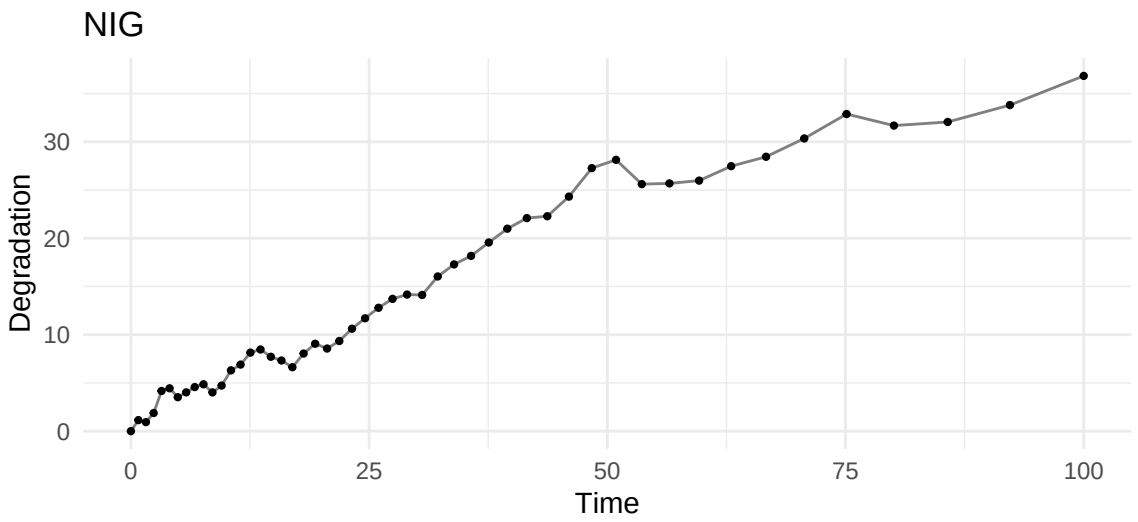
```
set.seed(36)
x <- sim_nig(t3, alpha = 2, beta = 0.5, delta = 1, mu = 0.2)
head(x)
```

```
##              t        X1
## [1,] 0.0000000 0.0000000
## [2,] 0.7809824 1.1428735
## [3,] 1.5768699 0.9335355
## [4,] 2.3882425 1.8781197
## [5,] 3.2157149 4.1653651
## [6,] 4.0599388 4.4414531
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  1.143 -0.209  0.945  2.287  0.276 -0.920  0.500  0.548  0.289 -0.841
## [11]  0.709  1.577  0.596  1.243  0.325 -0.755 -0.387 -0.699  1.421  1.017
## [21] -0.500  0.783  1.275  1.081  1.090  0.923  0.447 -0.041  1.917  1.249
## [31]  0.882  1.393  1.428  1.101  0.192  2.033  2.953  0.862 -2.520  0.076
## [41]  0.289  1.497  0.975  1.899  2.533 -1.199  0.375  1.751  3.026
```

```
plot_traj(x, "NIG")
```



Modèle 7 : Poisson composé

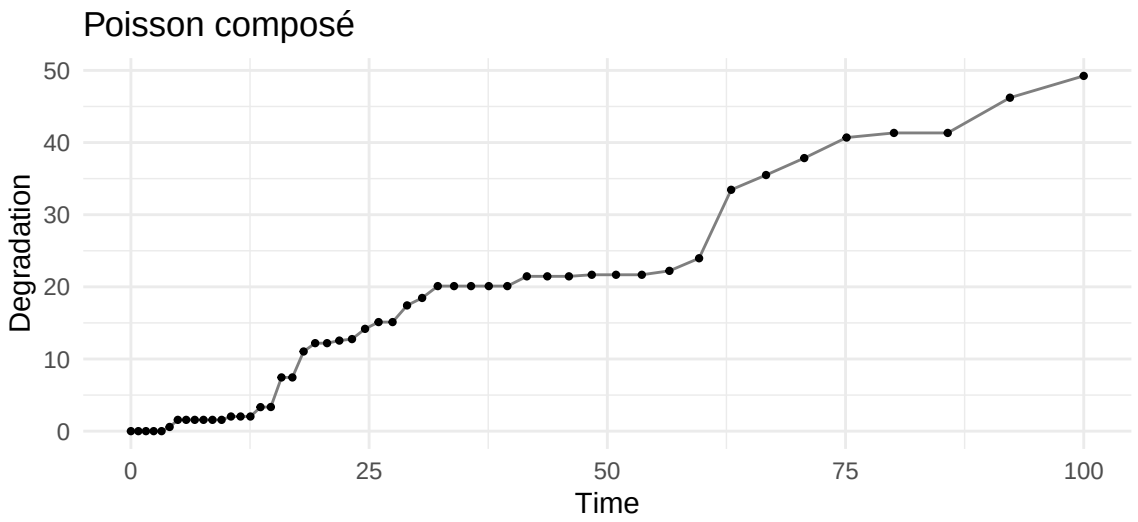
```
set.seed(37)
x <- sim_cpoisson(t3, lambda = 0.5)
head(x)
```

```
##              t        X1
## [1,] 0.0000000 0.0000000
## [2,] 0.7809824 0.0000000
## [3,] 1.5768699 0.0000000
## [4,] 2.3882425 0.0000000
## [5,] 3.2157149 0.0000000
## [6,] 4.0599388 0.5790226
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 0.000 0.000 0.000 0.000 0.579 0.986 0.000 0.000 0.000 0.000 0.000 0.463
## [13] 0.000 0.000 1.300 0.019 4.098 0.000 3.598 1.152 0.000 0.357 0.209 1.419
## [25] 0.937 0.000 2.313 1.033 1.643 0.000 0.000 0.000 0.000 1.352 0.000 0.000
## [37] 0.217 0.000 0.000 0.545 1.743 9.491 2.047 2.347 2.846 0.640 0.000 4.882
## [49] 3.023
```

```
plot_traj(x, "Poisson composé")
```


Poisson composé
Degradation
Time
0
10
20
30
40
50
0
25
50
75
100

Cas 4 : mixtes, beaucoup au début et à la fin

```
t4 <- inspection_times("inc-dec", n = 49, tmax = 100)$time
plot_plan(t4, "Beaucoup au début et à la fin, peu au milieu")
```



Modèle 1 : gamma

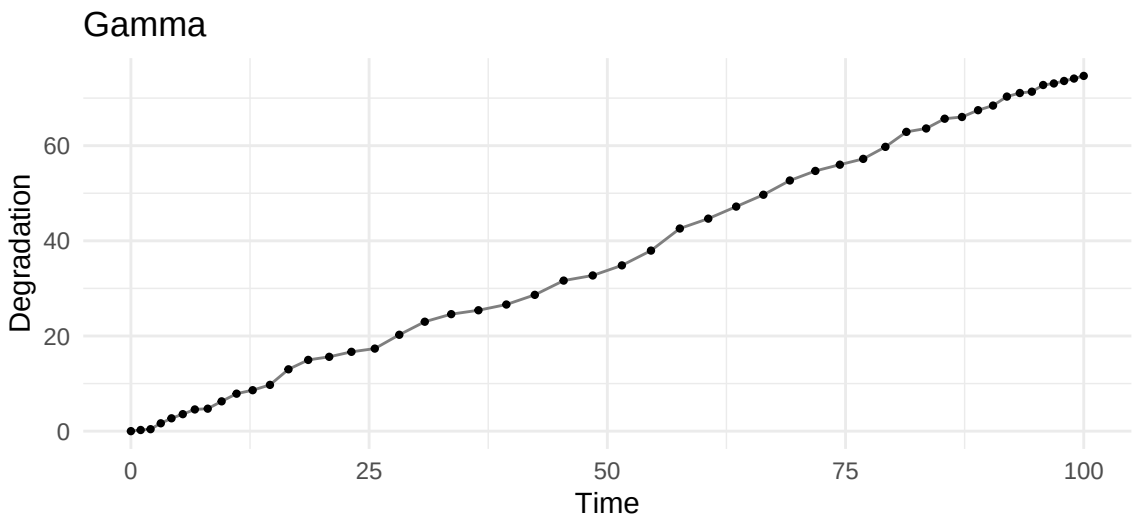
```
set.seed(41)
x <- sim_gamma(t4, a = 2, b = 3)
head(x)
```

```
##             t        X1
## [1,] 0.000000 0.0000000
## [2,] 1.023202 0.2387756
## [3,] 2.063114 0.4122480
## [4,] 3.136169 1.6505655
## [5,] 4.258259 2.6990597
## [6,] 5.444469 3.5686811
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 0.239 0.173 1.238 1.048 0.870 0.991 0.162 1.542 1.612 0.731 1.119 3.271
## [13] 1.976 0.650 1.048 0.680 2.907 2.739 1.594 0.817 1.206 2.034 2.995 1.077
## [25] 2.122 3.104 4.642 2.076 2.531 2.490 2.996 2.010 1.311 1.223 2.526 3.151
## [37] 0.712 2.053 0.363 1.427 0.973 1.887 0.754 0.275 1.407 0.321 0.524 0.505
## [49] 0.562
```

```
plot_traj(x, "Gamma")
```



Modèle 2 : Wiener

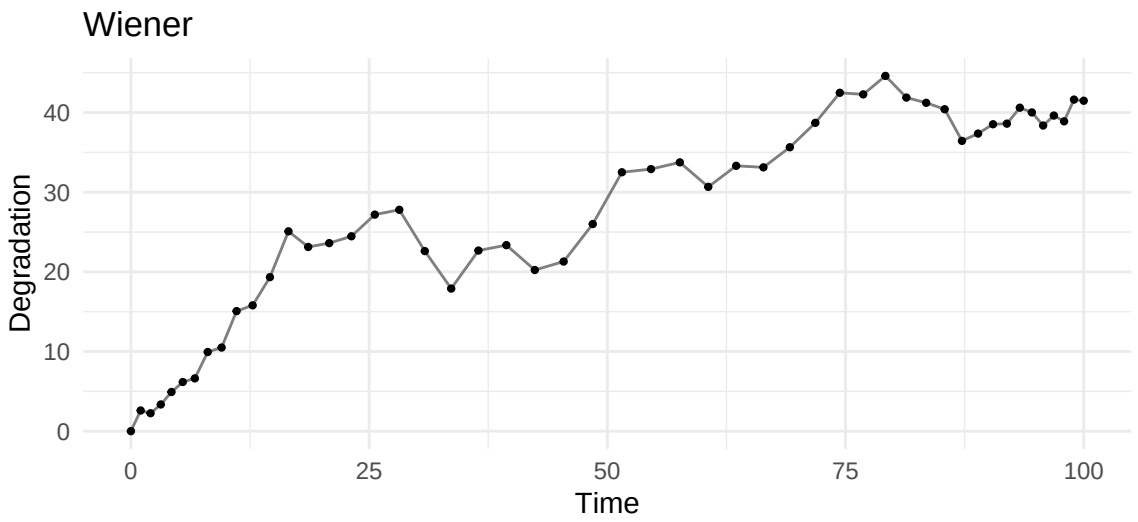
```
set.seed(42)
x <- sim_wiener(t4, mu = 0.5, sigma = 1.5)
head(x)
```

```
##             t       X1
## [1,] 0.000000 0.000000
## [2,] 1.023202 2.591759
## [3,] 2.063114 2.247929
## [4,] 3.136169 3.348695
## [5,] 4.258259 4.915316
## [6,] 5.444469 6.168873
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  2.592 -0.344  1.101  1.567  1.254  0.453  3.317  0.557  4.577  0.722
## [11]  3.543  5.752 -1.963  0.481  0.860  2.721  0.601 -5.181 -4.710  4.777
## [21]  0.677 -3.124  1.065  4.709  6.504  0.398  0.842 -3.077  2.646 -0.194
## [31]  2.524  3.068  3.774 -0.204  2.322 -2.721 -0.658 -0.808 -3.970  0.915
## [41]  1.172  0.075  2.002 -0.594 -1.642  1.249 -0.724  2.729 -0.143
```

```
plot_traj(x, "Wiener")
```



Modèle 3 : inverse gaussien

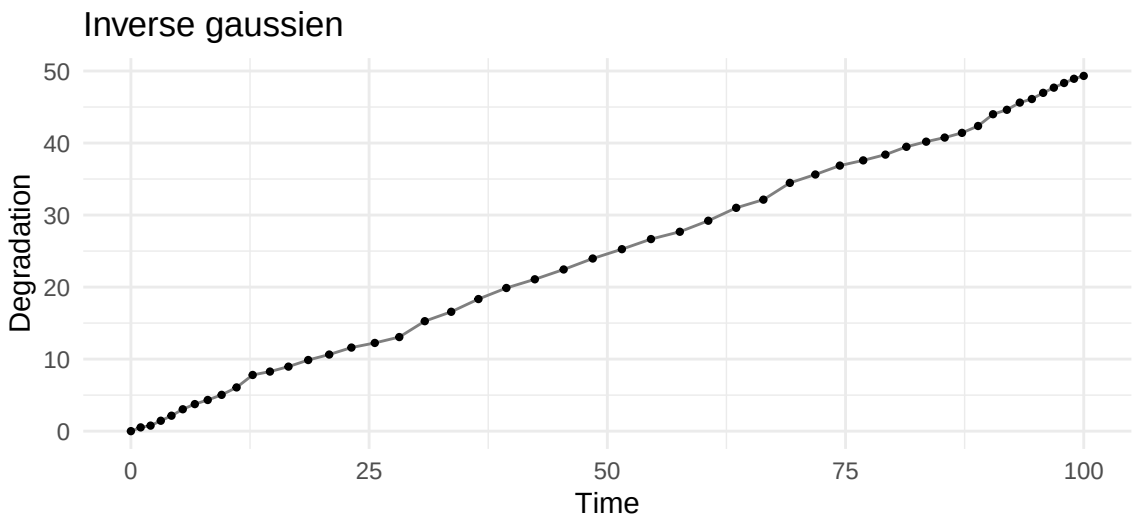
```
set.seed(43)
x <- sim_ig(t4, mu = 0.5, eta = 2)
head(x)
```

```
##             t        X1
## [1,] 0.000000 0.0000000
## [2,] 1.023202 0.5211761
## [3,] 2.063114 0.7657921
## [4,] 3.136169 1.4437941
## [5,] 4.258259 2.1422958
## [6,] 5.444469 3.0379037
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 0.521 0.245 0.678 0.699 0.896 0.715 0.574 0.711 1.031 1.731 0.470 0.688
## [13] 0.917 0.760 0.968 0.642 0.816 2.205 1.305 1.768 1.534 1.208 1.364 1.522
## [25] 1.296 1.403 1.017 1.530 1.790 1.136 2.327 1.164 1.238 0.723 0.799 1.087
## [37] 0.712 0.568 0.654 0.953 1.636 0.614 1.001 0.496 0.858 0.720 0.647 0.586
## [49] 0.400
```

```
plot_traj(x, "Inverse gaussien")
```



Modèle 4 : variance-gamma

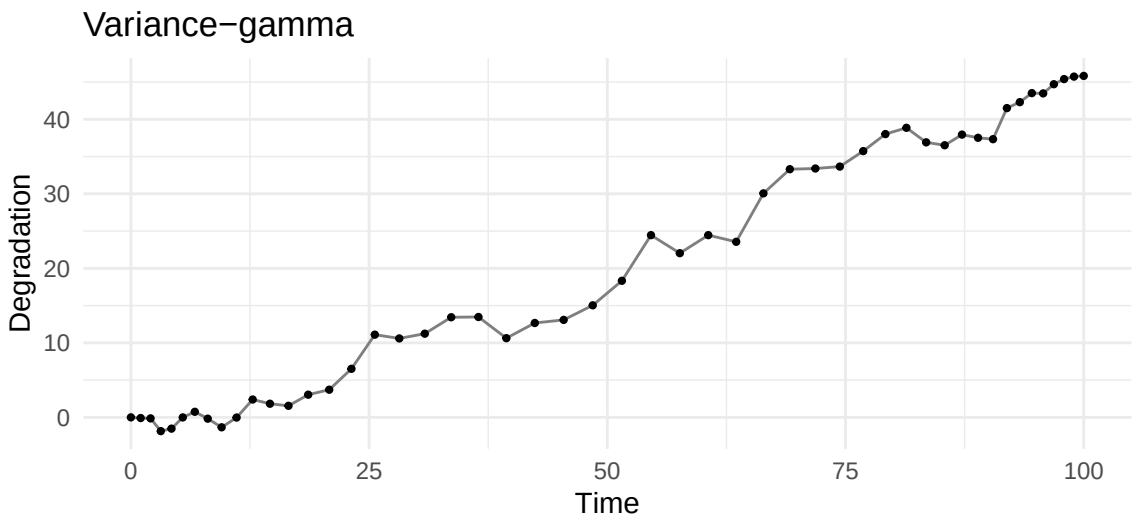
```
set.seed(44)
x <- sim_vg(t4, theta = 0.5, sigma = 1.5, nu = 2)
head(x)
```

```
##             t           X1
## [1,] 0.000000  0.000000000
## [2,] 1.023202 -0.091184314
## [3,] 2.063114 -0.136224319
## [4,] 3.136169 -1.849759658
## [5,] 4.258259 -1.513628228
## [6,] 5.444469 -0.007804398
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] -0.091 -0.045 -1.714  0.336  1.506  0.755 -0.924 -1.157  1.309  2.420
## [11] -0.566 -0.278  1.491  0.668  2.805  4.583 -0.505  0.638  2.203  0.036
## [21] -2.845  2.040  0.415  1.956  3.293  6.125 -2.421  2.422 -0.894  6.502
## [31]  3.243  0.091  0.259  2.078  2.285  0.840 -1.950 -0.391  1.428 -0.428
## [41] -0.184  4.170  0.797  1.216 -0.041  1.232  0.688  0.333  0.096
```

```
plot_traj(x, "Variance-gamma")
```



Modèle 5 : Meixner

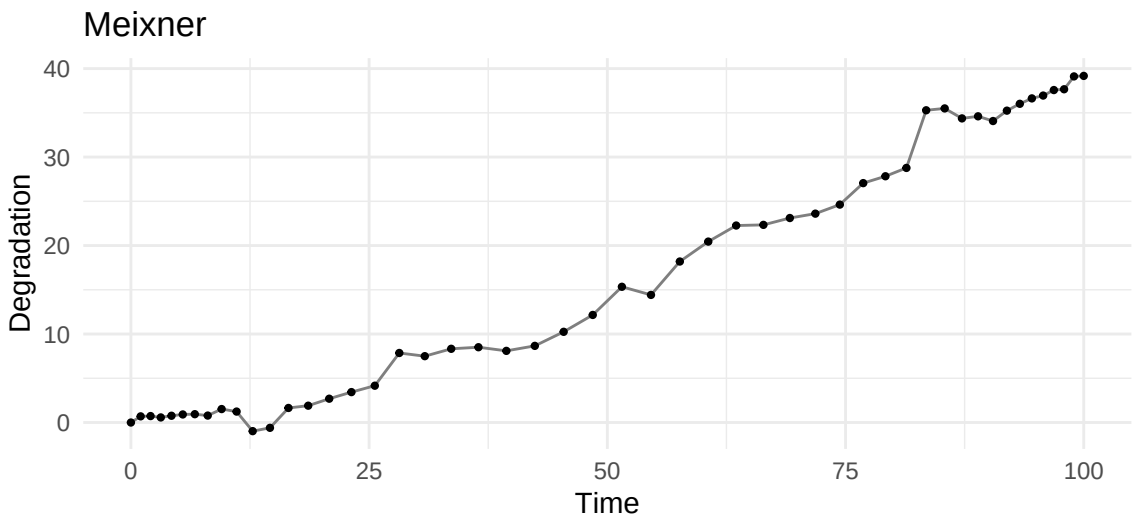
```
set.seed(45)
x <- sim_meixner(t4, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x)
```

```
##              t        X1
## [1,] 0.000000 0.0000000
## [2,] 1.023202 0.6957971
## [3,] 2.063114 0.7273165
## [4,] 3.136169 0.5743888
## [5,] 4.258259 0.7643220
## [6,] 5.444469 0.9113741
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  0.696  0.032 -0.153  0.190  0.147  0.030 -0.156  0.736 -0.284 -2.212
## [11]  0.375  2.245  0.257  0.789  0.746  0.724  3.699 -0.361  0.834  0.177
## [21] -0.411  0.565  1.589  1.906  3.183 -0.919  3.775  2.251  1.816  0.075
## [31]  0.774  0.493  1.023  2.427  0.777  0.941  6.516  0.209 -1.126  0.235
## [41] -0.533  1.172  0.774  0.611  0.320  0.622  0.089  1.452  0.058
```

```
plot_traj(x, "Meixner")
```



Modèle 6 : normal inverse gaussien (NIG)

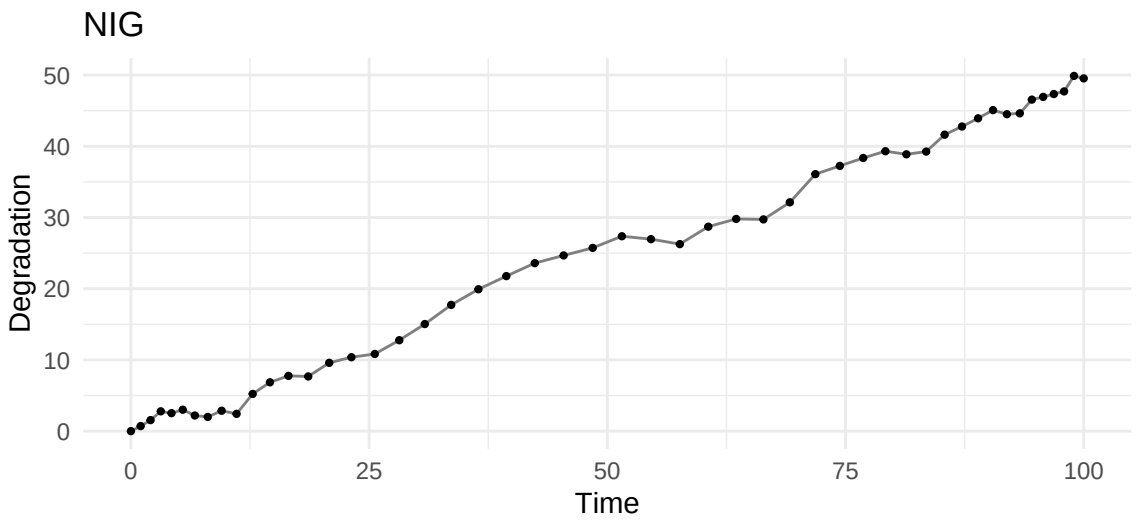
```
set.seed(46)
x <- sim_nig(t4, alpha = 2, beta = 0.5, delta = 1, mu = 0.2)
head(x)
```

```
##             t        X1
## [1,] 0.000000 0.0000000
## [2,] 1.023202 0.7077617
## [3,] 2.063114 1.5470937
## [4,] 3.136169 2.7812536
## [5,] 4.258259 2.5236187
## [6,] 5.444469 3.0063542
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  0.708  0.839  1.234 -0.258  0.483 -0.813 -0.197  0.868 -0.436  2.802
## [11]  1.627  0.904 -0.081  1.913  0.787  0.457  1.933  2.268  2.692  2.195
## [21]  1.846  1.822  1.080  1.061  1.633 -0.411 -0.693  2.449  1.086 -0.065
## [31]  2.397  3.958  1.156  1.116  0.949 -0.436  0.384  2.375  1.146  1.150
## [41]  1.153 -0.579  0.125  1.927  0.381  0.399  0.370  2.179 -0.348
```

```
plot_traj(x, "NIG")
```



Modèle 7 : Poisson composé

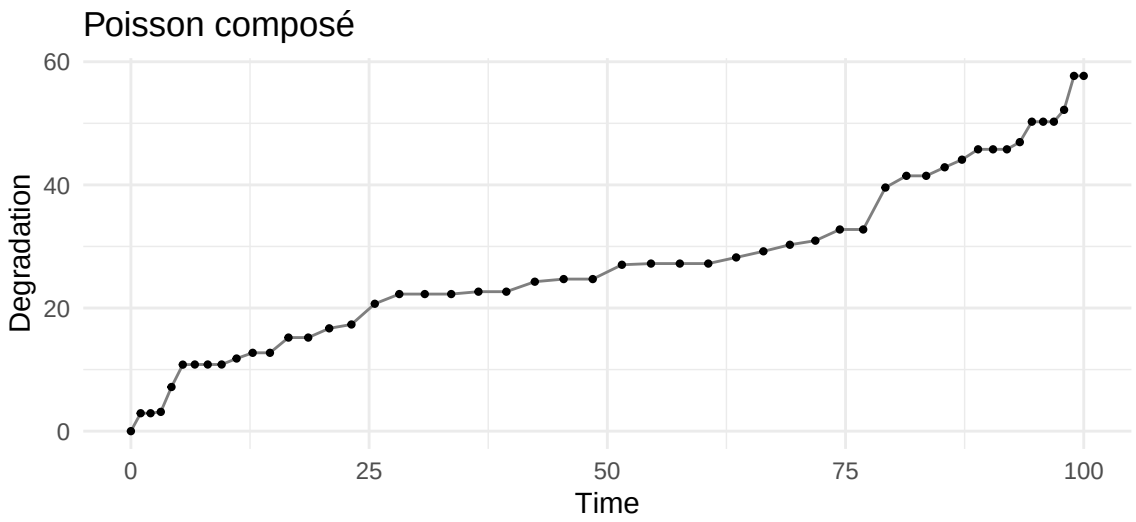
```
set.seed(47)
x <- sim_cpoisson(t4, lambda = 0.5)
head(x)
```

```
##             t        X1
## [1,] 0.000000  0.000000
## [2,] 1.023202  2.908973
## [3,] 2.063114  2.908973
## [4,] 3.136169  3.130890
## [5,] 4.258259  7.169680
## [6,] 5.444469 10.802231
```

```
round(diff(x[, "X1"]), 3)
```

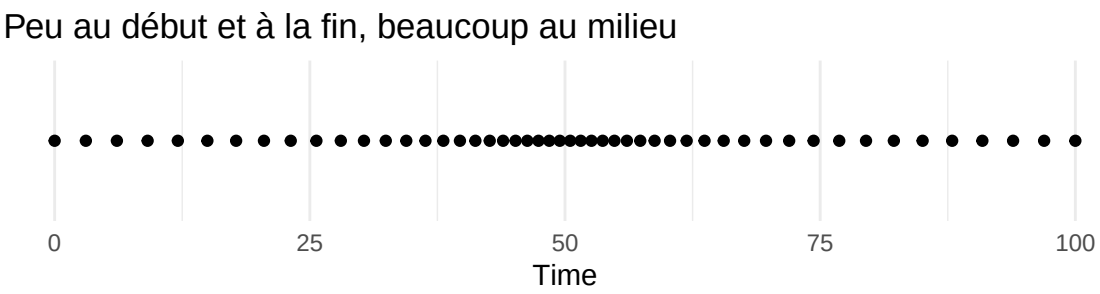
```
##  [1] 2.909 0.000 0.222 4.039 3.633 0.022 0.000 0.000 0.972 0.928 0.000 2.475
## [13] 0.000 1.499 0.625 3.365 1.576 0.000 0.000 0.388 0.000 1.619 0.427 0.000
## [25] 2.324 0.195 0.000 0.000 0.997 0.994 1.065 0.664 1.816 0.000 6.813 1.904
## [37] 0.000 1.386 1.238 1.681 0.000 0.000 1.168 3.319 0.000 0.000 1.925 5.514
## [49] 0.000
```

```
plot_traj(x, "Poisson composé")
```


Poisson composé
Degradation
Time
0
20
40
60
0
25
50
75
100

Cas 5 : mixtes, beaucoup au milieu

```
t5 <- inspection_times("dec-inc", n = 49, tmax = 100)$time
plot_plan(t5, "Peu au début et à la fin, beaucoup au milieu")
```



Modèle 1 : gamma

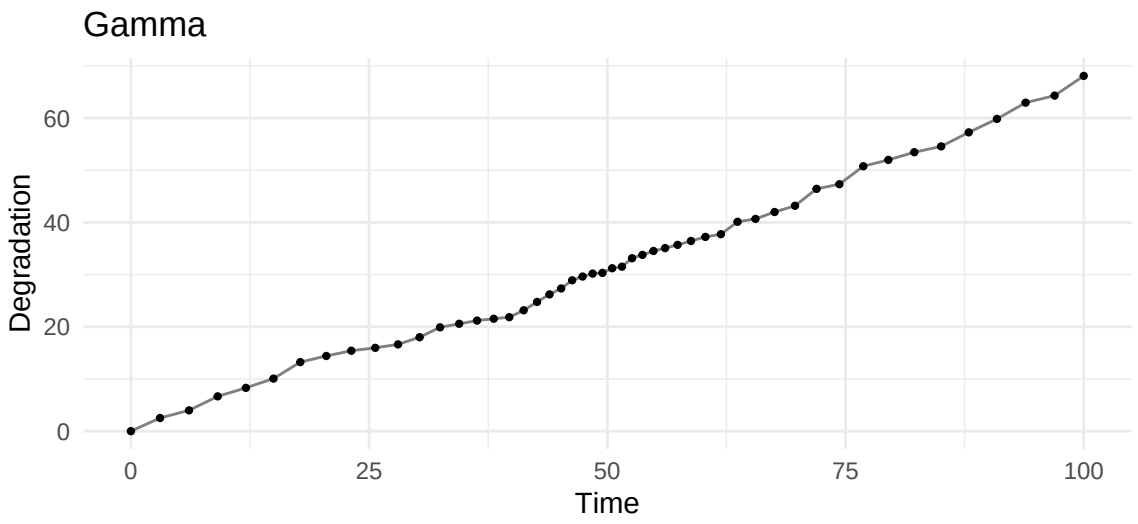
```
set.seed(51)
x <- sim_gamma(t5, a = 2, b = 3)
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  3.058430  2.519601
## [3,]  6.100152  3.987164
## [4,]  9.108729  6.665784
## [5,] 12.068272  8.292320
## [6,] 14.963694 10.070391
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 2.520 1.468 2.679 1.627 1.778 3.158 1.175 1.006 0.554 0.665 1.371 1.895
## [13] 0.685 0.608 0.352 0.299 1.324 1.598 1.450 1.134 1.562 0.721 0.566 0.130
## [25] 0.889 0.302 1.622 0.635 0.756 0.550 0.620 0.752 0.771 0.525 2.374 0.557
## [37] 1.318 1.194 3.247 0.886 3.445 1.214 1.475 1.112 2.691 2.577 3.117 1.351
## [49] 3.777
```

```
plot_traj(x, "Gamma")
```



Modèle 2 : Wiener

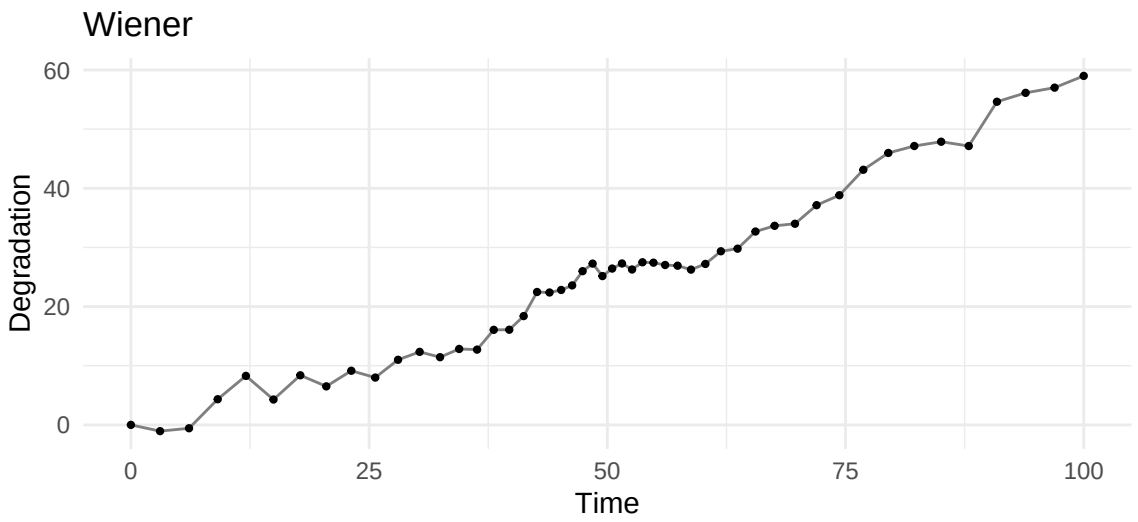
```
set.seed(52)
x <- sim_wiener(t5, mu = 0.5, sigma = 1.5)
head(x)
```

```
##              t         X1
## [1,]  0.000000  0.0000000
## [2,]  3.058430 -1.0578153
## [3,]  6.100152 -0.5699274
## [4,]  9.108729  4.3388401
## [5,] 12.068272  8.2823679
## [6,] 14.963694  4.2856276
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] -1.058  0.488  4.909  3.944 -3.997  4.100 -1.875  2.642 -1.152  3.007
## [11]  1.338 -0.901  1.392 -0.112  3.346  0.019  2.305  4.074 -0.085  0.427
## [21]  0.772  2.408  1.275 -2.110  1.275  0.858 -1.016  1.220 -0.069 -0.393
## [31] -0.119 -0.661  0.959  2.143  0.439  2.897  0.967  0.348  3.141  1.684
## [41]  4.302  2.844  1.176  0.720 -0.726  7.486  1.505  0.880  1.986
```

```
plot_traj(x, "Wiener")
```



Modèle 3 : inverse gaussien

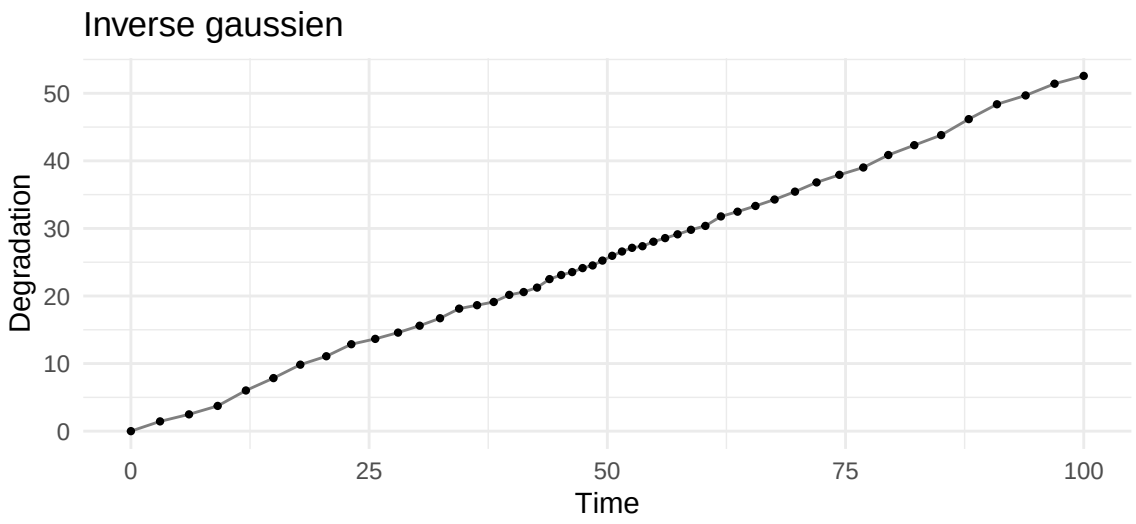
```
set.seed(53)
x <- sim_ig(t5, mu = 0.5, eta = 2)
head(x)
```

```
##              t       X1
## [1,]  0.000000 0.000000
## [2,]  3.058430 1.444313
## [3,]  6.100152 2.483767
## [4,]  9.108729 3.733955
## [5,] 12.068272 6.018209
## [6,] 14.963694 7.847824
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 1.444 1.039 1.250 2.284 1.830 1.989 1.245 1.780 0.792 0.928 1.020 1.107
## [13] 1.421 0.510 0.477 1.050 0.417 0.657 1.258 0.607 0.431 0.591 0.393 0.719
## [25] 0.708 0.637 0.543 0.239 0.664 0.536 0.561 0.669 0.575 1.404 0.700 0.851
## [37] 0.951 1.164 1.370 1.121 1.085 1.846 1.457 1.493 2.361 2.198 1.309 1.731
## [49] 1.159
```

```
plot_traj(x, "Inverse gaussien")
```



Modèle 4 : variance-gamma

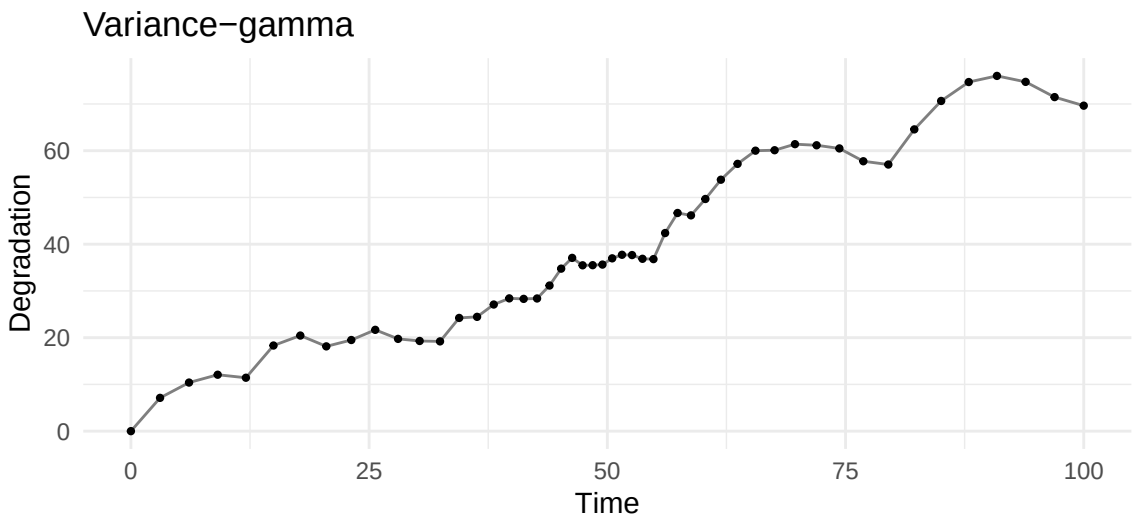
```
set.seed(54)
x <- sim_vg(t5, theta = 0.5, sigma = 1.5, nu = 2)
head(x)
```

```
##              t       X1
## [1,]  0.000000  0.00000
## [2,]  3.058430  7.11978
## [3,]  6.100152 10.39530
## [4,]  9.108729 12.06922
## [5,] 12.068272 11.42444
## [6,] 14.963694 18.32119
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  7.120  3.276  1.674 -0.645  6.897  2.131 -2.314  1.358  2.177 -1.934
## [11] -0.449 -0.086  5.017  0.232  2.639  1.314 -0.109  0.086  2.769  3.617
## [21]  2.300 -1.587  0.027  0.129  1.335  0.760 -0.060 -0.799 -0.069  5.582
## [31]  4.289 -0.510  3.505  4.123  3.393  2.820  0.097  1.296 -0.235 -0.673
## [41] -2.745 -0.702  7.520  6.081  4.049  1.312 -1.277 -3.260 -1.833
```

```
plot_traj(x, "Variance-gamma")
```



Modèle 5 : Meixner

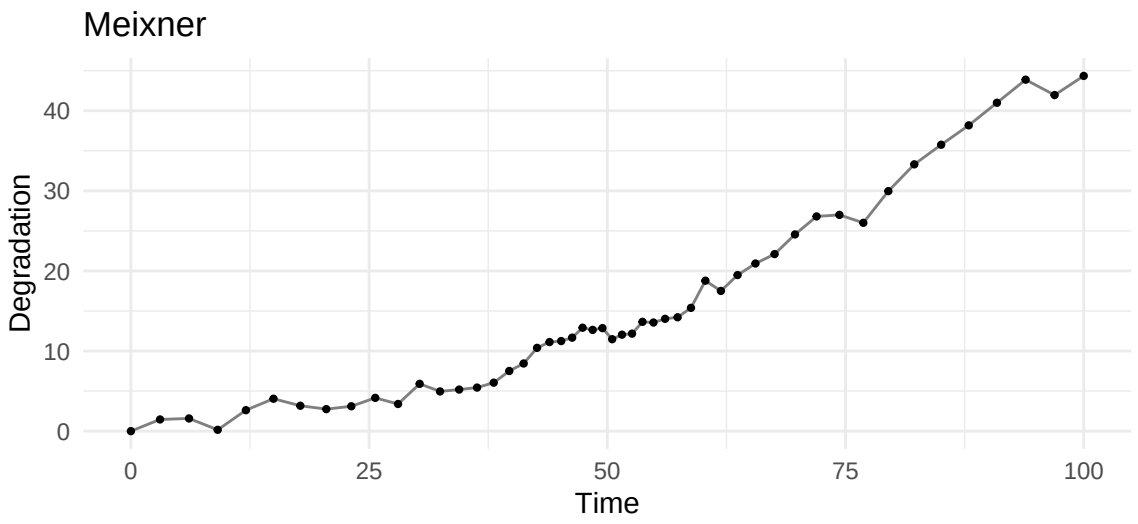
```
set.seed(55)
x <- sim_meixner(t5, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x)
```

```
##              t        X1
## [1,]  0.000000 0.0000000
## [2,]  3.058430 1.4599132
## [3,]  6.100152 1.5847059
## [4,]  9.108729 0.1640323
## [5,] 12.068272 2.6159170
## [6,] 14.963694 4.0469879
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  1.460  0.125 -1.421  2.452  1.431 -0.877 -0.421  0.356  1.050 -0.765
## [11]  2.517 -0.950  0.235  0.246  0.613  1.459  0.936  1.952  0.722  0.126
## [21]  0.425  1.241 -0.272  0.219 -1.384  0.567  0.128  1.475 -0.086  0.465
## [31]  0.192  1.176  3.391 -1.281  1.985  1.440  1.176  2.454  2.247  0.194
## [41] -0.990  3.956  3.343  2.447  2.423  2.820  2.880 -1.921  2.393
```

```
plot_traj(x, "Meixner")
```



Modèle 6 : normal inverse gaussien (NIG)

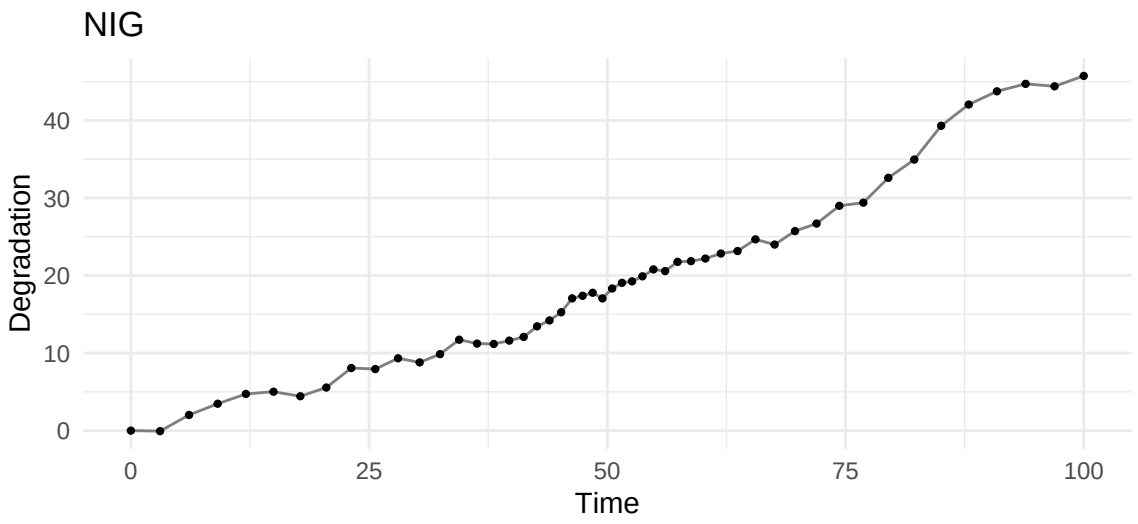
```
set.seed(56)
x <- sim_nig(t5, alpha = 2, beta = 0.5, delta = 1, mu = 0.2)
head(x)
```

```
##              t          X1
## [1,]  0.000000  0.00000000
## [2,]  3.058430 -0.06698059
## [3,]  6.100152  2.01676731
## [4,]  9.108729  3.46802520
## [5,] 12.068272  4.72674140
## [6,] 14.963694  5.01892995
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] -0.067  2.084  1.451  1.259  0.292 -0.587  1.124  2.518 -0.134  1.389
## [11] -0.530  1.066  1.860 -0.500 -0.049  0.430  0.477  1.366  0.763  1.055
## [21]  1.788  0.330  0.394 -0.727  1.275  0.731  0.190  0.651  0.890 -0.217
## [31]  1.193  0.091  0.336  0.646  0.324  1.503 -0.670  1.736  0.970  2.292
## [41]  0.405  3.201  2.356  4.357  2.739  1.706  0.959 -0.314  1.343
```

```
plot_traj(x, "NIG")
```



Modèle 7 : Poisson composé

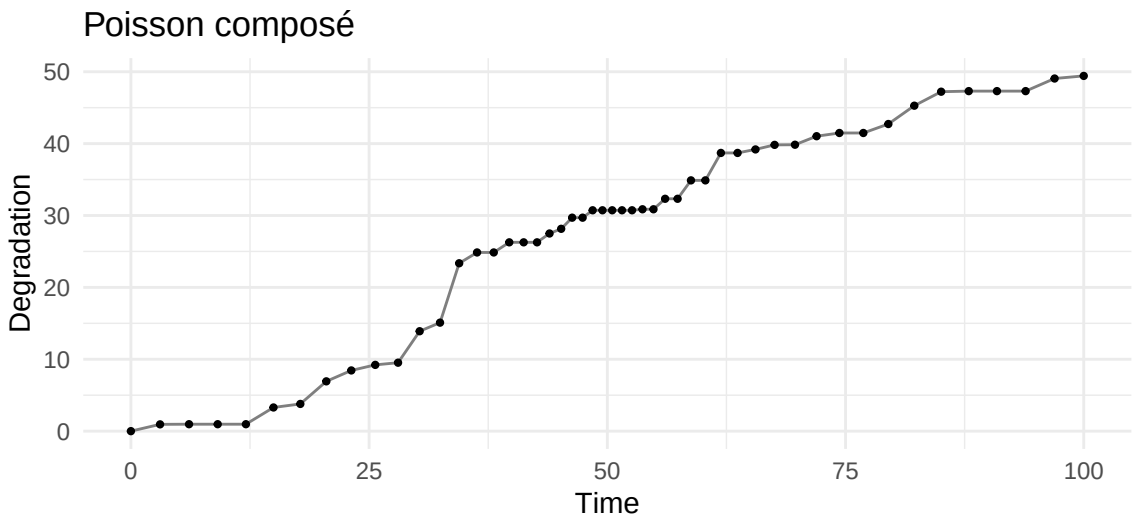
```
set.seed(57)
x <- sim_cpoisson(t5, lambda = 0.5)
head(x)
```

```
##              t        X1
## [1,]  0.000000 0.0000000
## [2,]  3.058430 0.9436195
## [3,]  6.100152 0.9640602
## [4,]  9.108729 0.9640602
## [5,] 12.068272 0.9640602
## [6,] 14.963694 3.2923329
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 0.944 0.020 0.000 0.000 2.328 0.493 3.142 1.517 0.774 0.305 4.377 1.195
## [13] 8.258 1.505 0.000 1.409 0.000 0.000 1.230 0.651 1.548 0.000 1.029 0.000
## [25] 0.000 0.000 0.000 0.141 0.000 1.459 0.000 2.561 0.000 3.817 0.000 0.489
## [37] 0.634 0.019 1.184 0.449 0.000 1.247 2.551 1.945 0.081 0.000 0.000 1.754
## [49] 0.364
```

```
plot_traj(x, "Poisson composé")
```

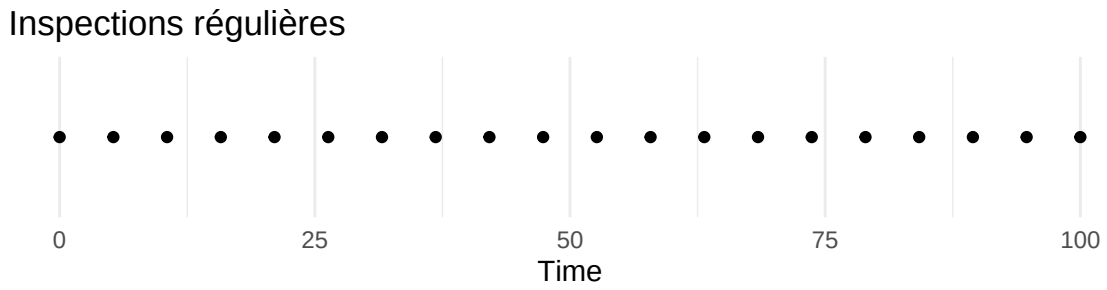

Poisson composé
Degradation
Time
0
10
20
30
40
50
0
25
50
75
100

II - Plusieurs trajectoires

10 trajectoires, chacune inspectée 20 fois entre 0 et 100, aux mêmes instants. `x` est une matrice : une colonne `t`, puis `X1`, ..., `X10`. On affiche les premières lignes des trois premières trajectoires.

Cas 1 : inspections régulières

```
p1 <- inspection_times("regular", n = 19, n_traj = 10, tmax = 100)
plot_plan(p1$time[p1$id == 1], "Inspections régulières")
```



Modèle 1 : gamma

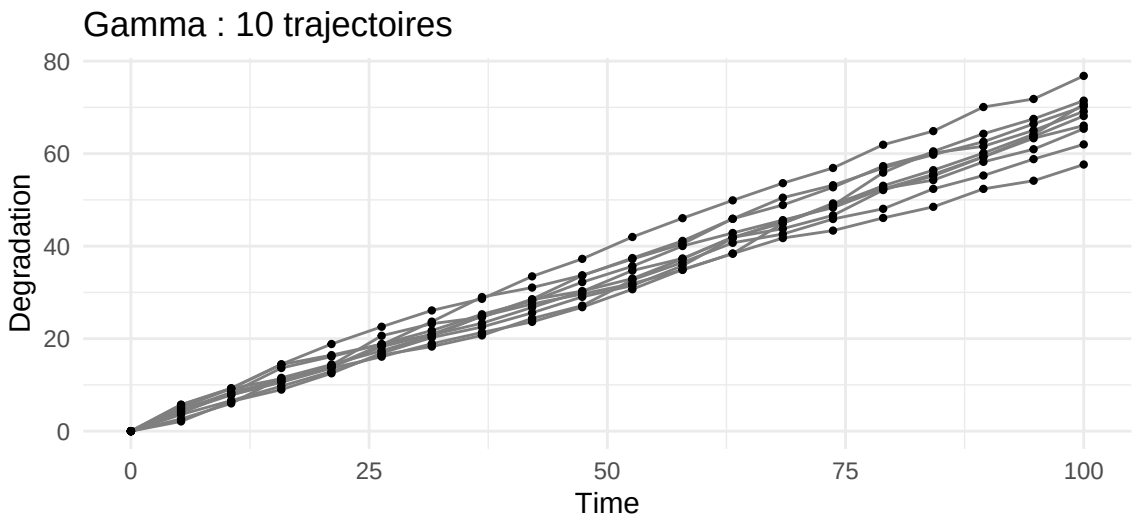
```
set.seed(111)
x <- sim_gamma(p1, a = 2, b = 3)
head(x[, 1:4])
```

```
##              t        X1        X2        X3
## [1,]  0.000000  0.000000  0.000000  0.000000
## [2,]  5.263158  3.594986  4.263145  5.730951
## [3,] 10.526316  6.597123  7.835829  9.278414
## [4,] 15.789474  9.725184 10.737855 14.460063
## [5,] 21.052632 12.889366 14.284863 18.836785
## [6,] 26.315789 16.476598 20.618768 22.568129
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 3.595 3.002 3.128 3.164 3.587 1.813 2.416 3.656 2.771 5.574 3.771 4.237
## [13] 1.893 3.261 2.201 4.311 2.890 3.545 3.176
```

```
plot_traj(x, "Gamma : 10 trajectoires")
```



Modèle 2 : Wiener

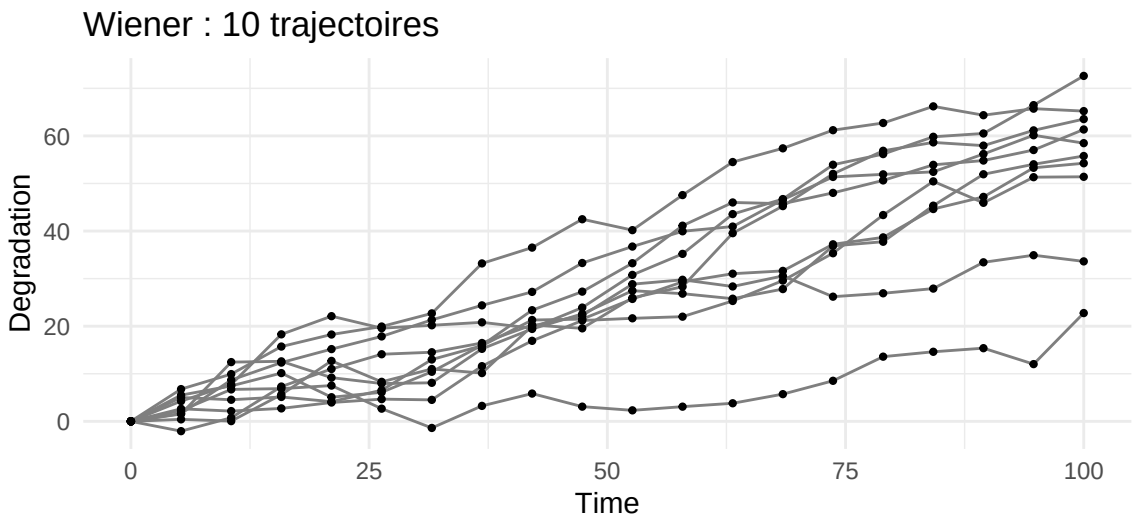
```
set.seed(112)
x <- sim_wiener(p1, mu = 0.5, sigma = 1.5)
head(x[, 1:4])
```

```
##              t        X1        X2        X3
## [1,]  0.000000  0.000000  0.000000  0.000000
## [2,]  5.263158  1.550234  4.253663  6.772753
## [3,] 10.526316 12.452393  7.741697  9.929144
## [4,] 15.789474 12.612256 18.289009 15.721810
## [5,] 21.052632  9.185158 22.104626 18.251965
## [6,] 26.315789  7.944378 19.582038 19.929558
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  1.550 10.902  0.160 -3.427 -1.241  0.155  7.140  4.187  3.155  4.880
## [11] -0.628 -1.054  2.026  9.074  0.875  7.583  6.600  2.099  1.737
```

```
plot_traj(x, "Wiener : 10 trajectoires")
```



Modèle 3 : inverse gaussien

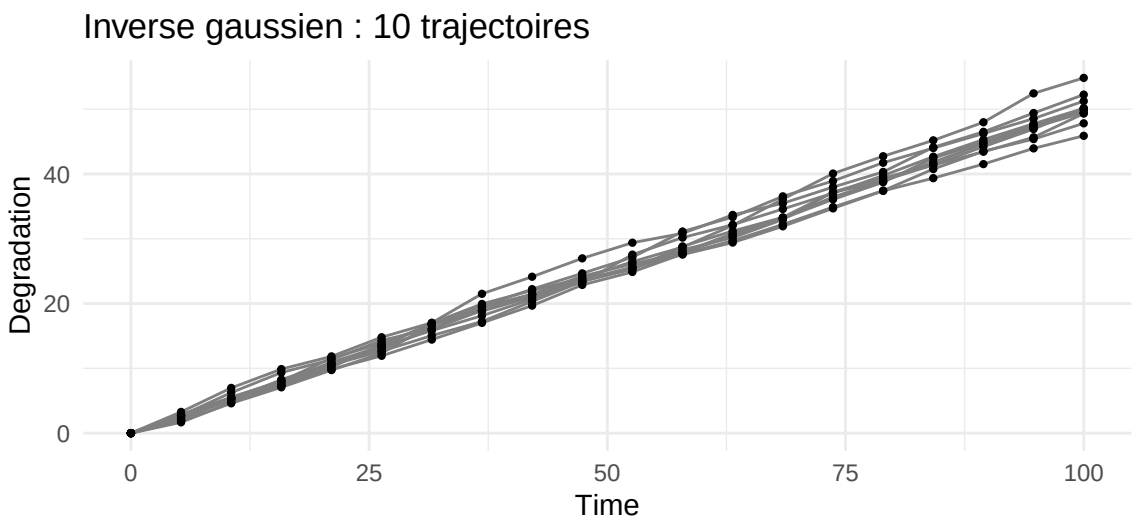
```
set.seed(113)
x <- sim_ig(p1, mu = 0.5, eta = 2)
head(x[, 1:4])
```

```
##              t        X1        X2        X3
## [1,]  0.000000  0.000000  0.000000  0.000000
## [2,]  5.263158  2.709183  2.353263  3.271314
## [3,] 10.526316  6.256515  5.193081  6.976841
## [4,] 15.789474  9.353974  7.519534  9.883285
## [5,] 21.052632 11.340768 10.560528 11.844324
## [6,] 26.315789 14.312931 12.650435 14.798366
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 2.709 3.547 3.097 1.987 2.972 1.808 3.437 2.668 2.435 2.477 3.981 2.226
## [13] 3.206 2.367 2.820 2.262 2.239 2.309 2.704
```

```
plot_traj(x, "Inverse gaussien : 10 trajectoires")
```



Modèle 4 : variance-gamma

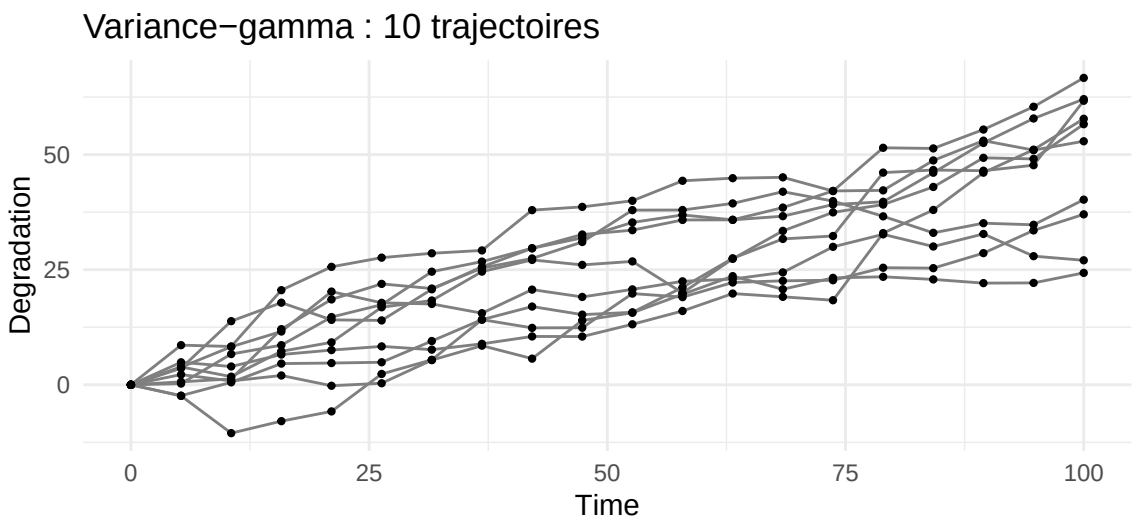
```
set.seed(114)
x <- sim_vg(p1, theta = 0.5, sigma = 1.5, nu = 2)
head(x[, 1:4])
```

```
##              t        X1         X2         X3
## [1,]  0.000000  0.000000   0.000000  0.0000000
## [2,]  5.263158  3.823209  -2.413243  2.2232642
## [3,] 10.526316  8.259109 -10.504474  0.8303106
## [4,] 15.789474 11.577788  -7.898515  2.0181671
## [5,] 21.052632 20.242764  -5.784190 -0.2135451
## [6,] 26.315789 17.803908   2.350151  0.3350213
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  3.823  4.436  3.319  8.665 -2.439 -0.243 -2.009  5.120 -1.587  1.640
## [11]  1.750  0.492  1.450  5.549  2.712 -2.639  2.726 -4.825 -0.898
```

```
plot_traj(x, "Variance-gamma : 10 trajectoires")
```



Modèle 5 : Meixner

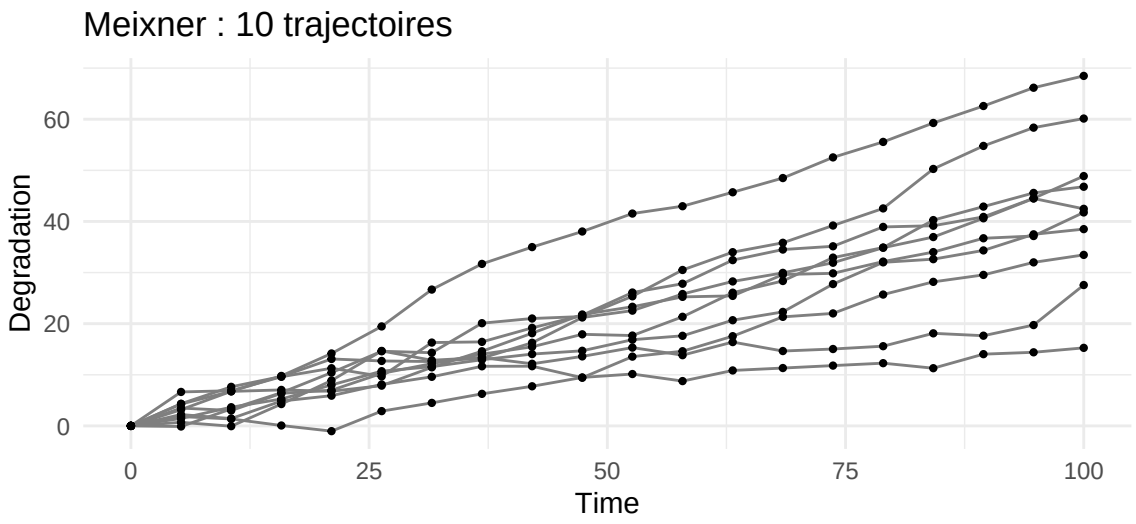
```
set.seed(115)
x <- sim_meixner(p1, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x[, 1:4])
```

```
##              t        X1        X2       X3
## [1,]  0.000000  0.000000  0.000000 0.000000
## [2,]  5.263158  3.484525  1.464709 4.285754
## [3,] 10.526316  3.010297  3.654886 6.750845
## [4,] 15.789474  6.373638  5.254512 7.040562
## [5,] 21.052632  6.969703  8.015926 6.776480
## [6,] 26.315789 10.268732 10.705041 7.873784
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  3.485 -0.474  3.363  0.596  3.299  1.904  1.857  1.446  2.441 -0.242
## [11]  3.670  4.717  2.288  4.571  1.964  5.378  2.641  2.675  1.212
```

```
plot_traj(x, "Meixner : 10 trajectoires")
```



Modèle 6 : normal inverse gaussien (NIG)

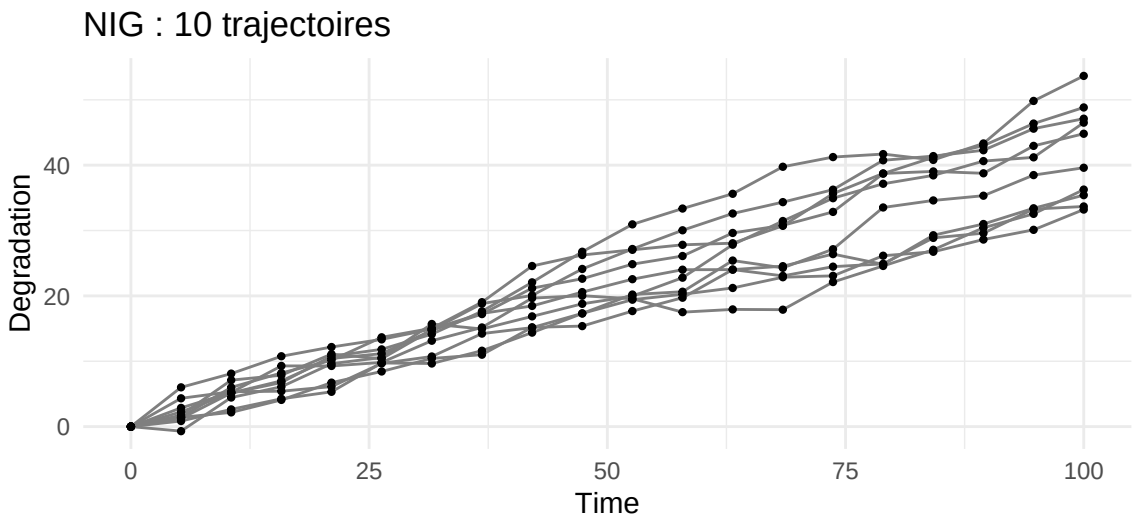
```
set.seed(116)
x <- sim_nig(p1, alpha = 2, beta = 0.5, delta = 1, mu = 0.2)
head(x[, 1:4])
```

```
##              t        X1        X2        X3
## [1,]  0.000000  0.000000 0.0000000  0.000000
## [2,]  5.263158  1.756169 0.8573535  1.240381
## [3,] 10.526316  7.109808 2.6351308  5.143635
## [4,] 15.789474  7.867051 4.2568225  7.003330
## [5,] 21.052632 11.092615 5.3272078 10.433357
## [6,] 26.315789 10.384142 9.7366396 11.159962
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  1.756  5.354  0.757  3.226 -0.708  4.406  4.031  1.248  4.052  3.055
## [11]  2.865  2.565  1.742  1.906  4.496  0.642  0.896  3.300  1.525
```

```
plot_traj(x, "NIG : 10 trajectoires")
```



Modèle 7 : Poisson composé

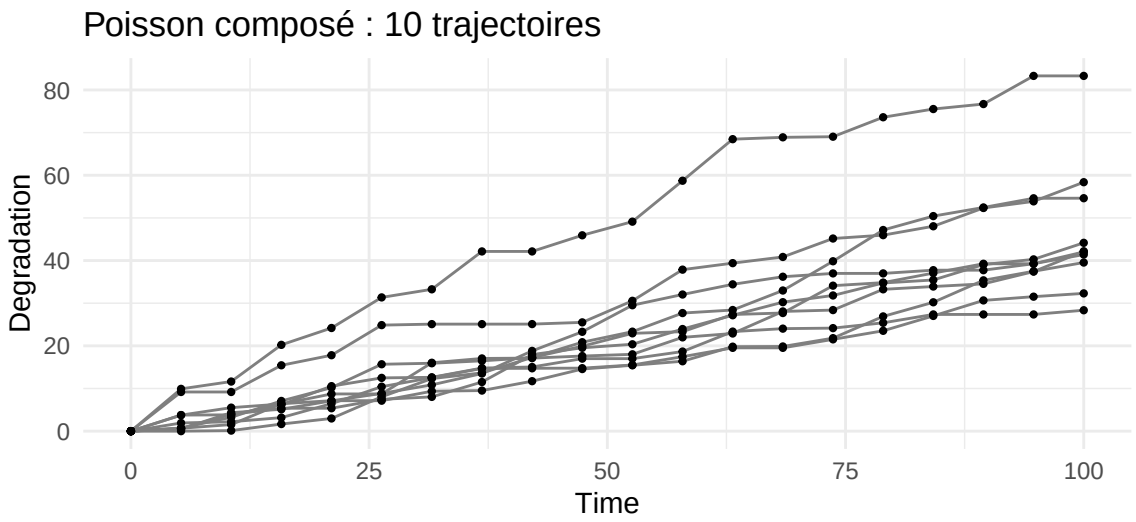
```
set.seed(117)
x <- sim_cpoisson(p1, lambda = 0.5)
head(x[, 1:4])
```

```
##              t        X1        X2       X3
## [1,]  0.000000  0.000000  0.000000 0.000000
## [2,]  5.263158  9.186037  9.908346 3.758788
## [3,] 10.526316  9.186037 11.629852 5.510407
## [4,] 15.789474 15.439410 20.233175 6.360477
## [5,] 21.052632 17.802966 24.185785 8.724386
## [6,] 26.315789 24.862176 31.356840 8.724386
```

```
round(diff(x[, "X1"]), 3)
```

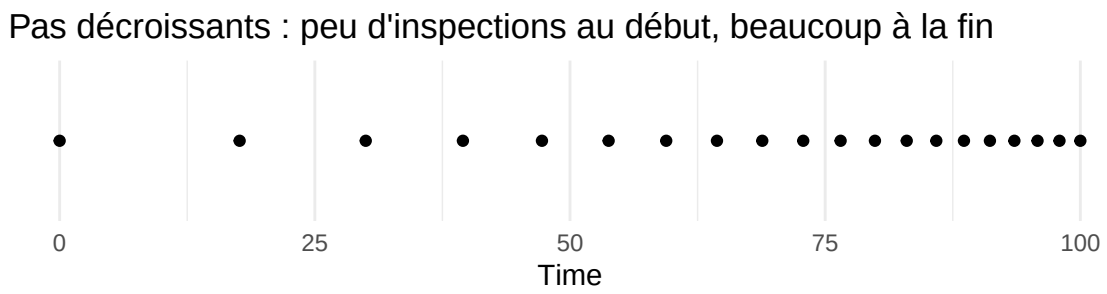
```
##  [1] 9.186 0.000 6.253 2.364 7.059 0.228 0.000 0.010 0.407 5.049 7.311 1.526
## [13] 1.453 4.322 0.790 2.084 4.289 1.546 4.496
```

```
plot_traj(x, "Poisson composé : 10 trajectoires")
```


Poisson composé : 10 trajectoires
Degradation
Time
0
20
40
60
80
0
25
50
75
100

Cas 2 : pas décroissants

```
p2 <- inspection_times("decreasing", n = 19, n_traj = 10, tmax = 100)
plot_plan(p2$time[p2$id == 1], "Pas décroissants : peu d'inspections au début, beaucoup à la fin")
```



Modèle 1 : gamma

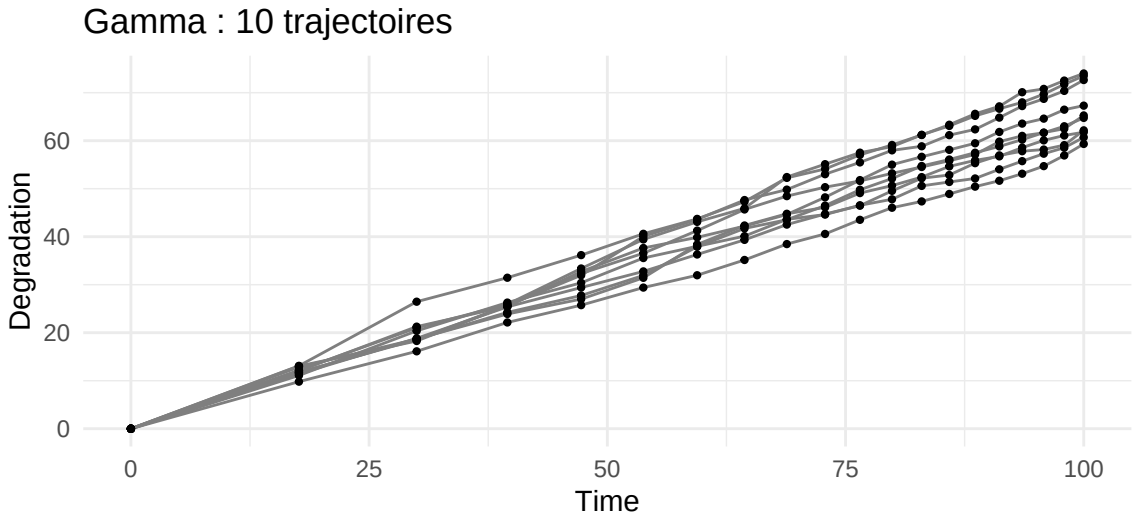
```
set.seed(121)
x <- sim_gamma(p2, a = 2, b = 3)
head(x[, 1:4])
```

```
##             t       X1       X2       X3
## [1,]  0.00000  0.00000  0.00000  0.00000
## [2,] 17.63450 11.09319 13.09138 12.17468
## [3,] 29.98974 20.37922 26.45837 21.00282
## [4,] 39.50887 26.27595 31.45178 25.91770
## [5,] 47.25388 30.39401 36.16545 33.37539
## [6,] 53.78331 35.55453 40.60457 39.43310
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 11.093  9.286  5.897  4.118  5.161  2.439  2.122  3.501  0.988  1.938
## [11]  1.249  2.791  0.818  0.728  1.909  1.699  1.554  1.240  2.170
```

```
plot_traj(x, "Gamma : 10 trajectoires")
```



Modèle 2 : Wiener

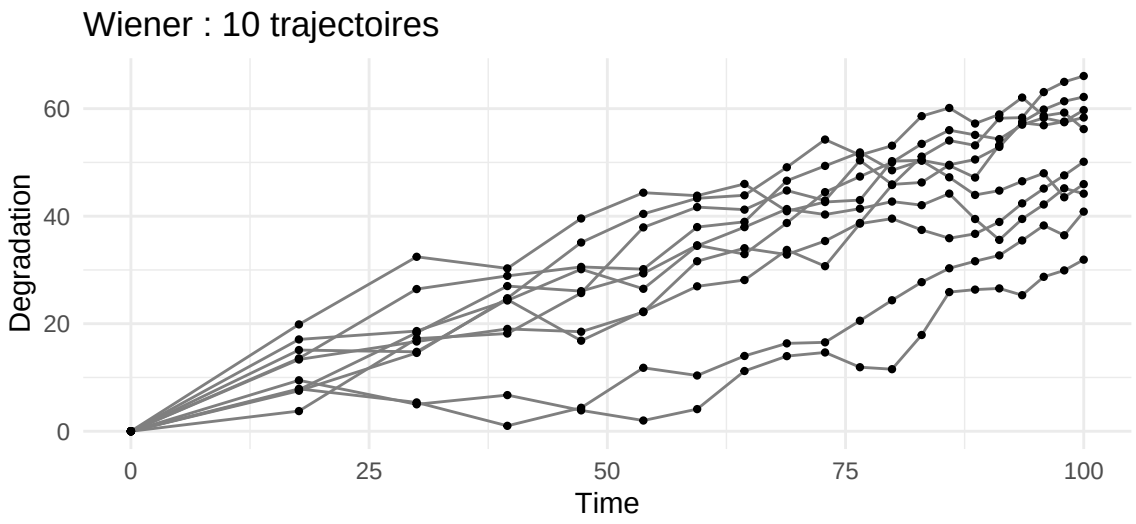
```
set.seed(122)
x <- sim_wiener(p2, mu = 0.5, sigma = 1.5)
head(x[, 1:4])
```

```
##             t       X1       X2       X3
## [1,]  0.00000  0.00000  0.00000 0.000000
## [2,] 17.63450 17.07338 13.34670 9.472999
## [3,] 29.98974 18.63306 16.68864 5.008735
## [4,] 39.50887 24.31601 19.02940 6.723054
## [5,] 47.25388 30.13363 18.50651 3.880899
## [6,] 53.78331 26.49122 22.16033 1.993831
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 17.073  1.560  5.683  5.818 -3.642  7.985  3.485  3.364 -1.019  1.093
## [11]  1.319 -0.671  2.149 -4.725 -3.880  3.891  2.691  3.011 -1.001
```

```
plot_traj(x, "Wiener : 10 trajectoires")
```



Modèle 3 : inverse gaussien

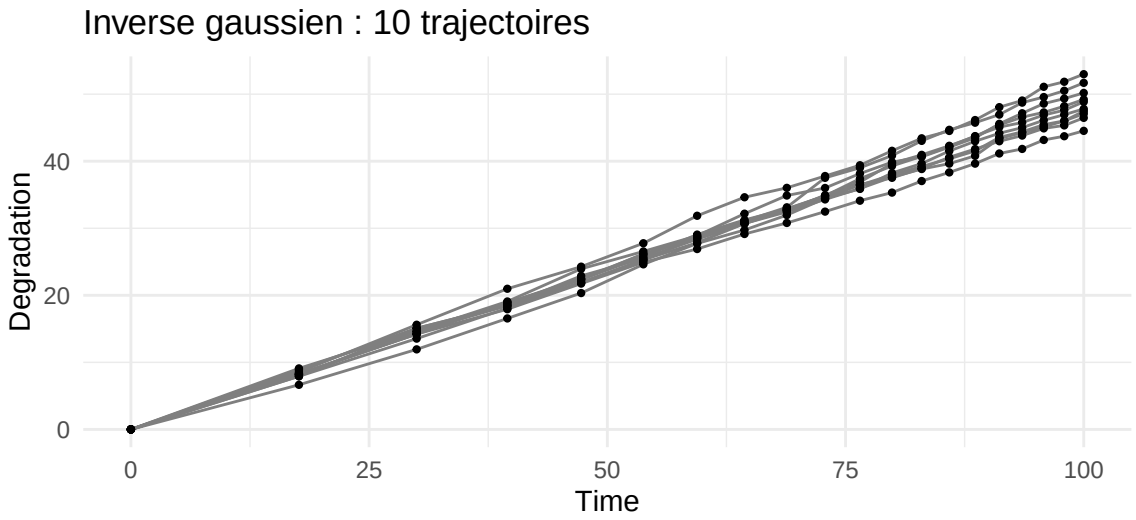
```
set.seed(123)
x <- sim_ig(p2, mu = 0.5, eta = 2)
head(x[, 1:4])
```

```
##             t        X1        X2        X3
## [1,]  0.00000  0.000000  0.000000  0.000000
## [2,] 17.63450  8.248148  8.126903  8.224491
## [3,] 29.98974 14.226786 14.592250 14.294230
## [4,] 39.50887 17.926372 18.442009 18.890120
## [5,] 47.25388 21.750129 21.919203 22.507345
## [6,] 53.78331 24.933294 25.267645 26.109115
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 8.248 5.979 3.700 3.824 3.183 1.971 2.241 1.646 1.691 1.629 1.207 1.724
## [13] 1.288 1.310 1.514 0.678 1.337 0.556 0.800
```

```
plot_traj(x, "Inverse gaussien : 10 trajectoires")
```



Modèle 4 : variance-gamma

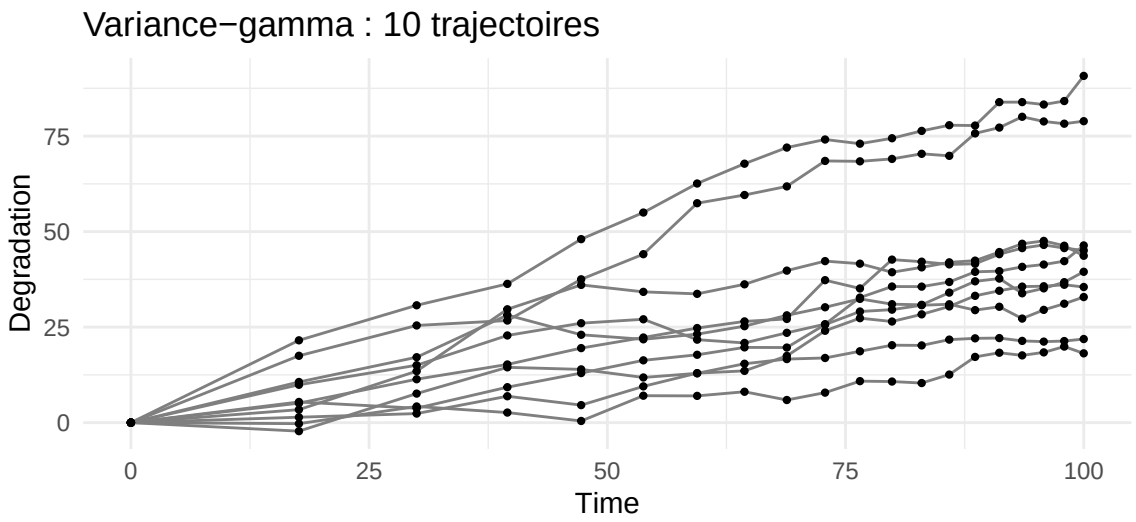
```
set.seed(124)
x <- sim_vg(p2, theta = 0.5, sigma = 1.5, nu = 2)
head(x[, 1:4])
```

```
##             t       X1        X2         X3
## [1,]  0.00000 0.000000  0.000000  0.0000000
## [2,] 17.63450 1.419466  9.911696 -0.2668853
## [3,] 29.98974 2.373640 15.029232  4.2291881
## [4,] 39.50887 6.909912 22.816187  2.6387083
## [5,] 47.25388 4.586146 26.025171  0.4401348
## [6,] 53.78331 9.492108 27.054773  7.0495887
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  1.419  0.954  4.536 -2.324  4.906  3.462  0.593  3.917  6.583  3.300
## [11] -0.881  1.881  2.065  2.783  1.321  1.093  0.091  0.383 -0.582
```

```
plot_traj(x, "Variance-gamma : 10 trajectoires")
```



Modèle 5 : Meixner

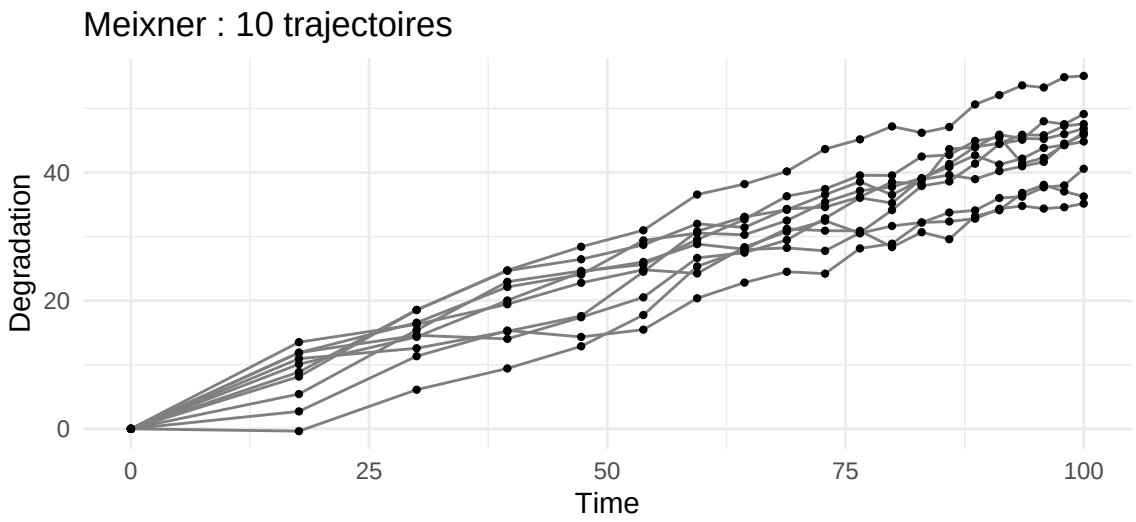
```
set.seed(125)
x <- sim_meixner(p2, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x[, 1:4])
```

```
##             t       X1       X2       X3
## [1,]  0.00000  0.00000  0.00000  0.00000
## [2,] 17.63450 10.94291 10.08426 11.87138
## [3,] 29.98974 12.57221 14.35015 16.56729
## [4,] 39.50887 15.20837 20.02635 22.14132
## [5,] 47.25388 17.60455 24.46681 24.07085
## [6,] 53.78331 24.51319 26.04517 29.39927
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 10.943  1.629  2.636  2.396  6.909  6.317  2.256  1.088  2.372  2.017
## [11] -2.046  2.401  0.692 -0.640  1.255  0.749  0.685  2.860  1.391
```

```
plot_traj(x, "Meixner : 10 trajectoires")
```



Modèle 6 : normal inverse gaussien (NIG)

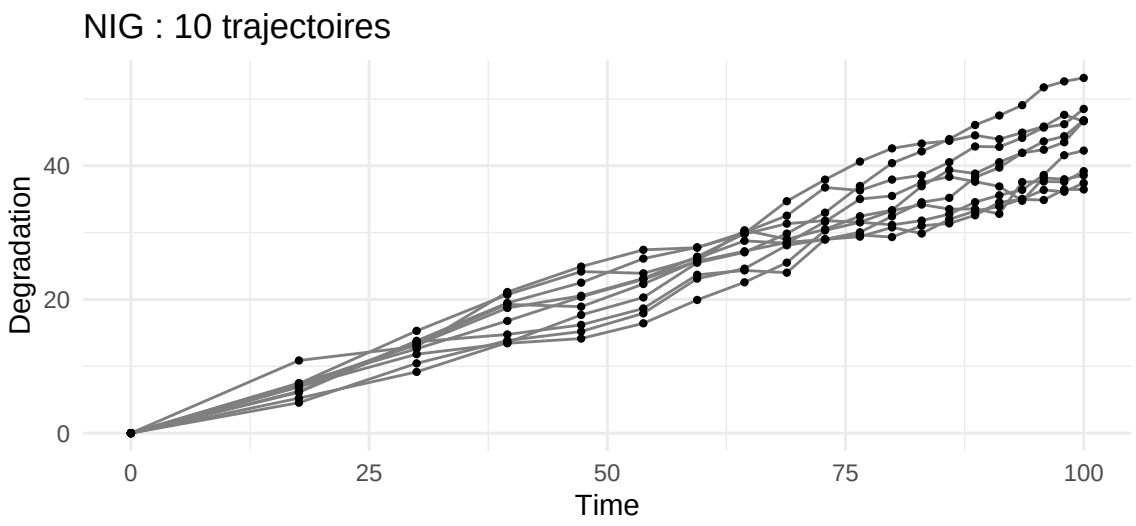
```
set.seed(126)
x <- sim_nig(p2, alpha = 2, beta = 0.5, delta = 1, mu = 0.2)
head(x[, 1:4])
```

```
##             t        X1        X2        X3
## [1,]  0.00000  0.000000  0.000000  0.000000
## [2,] 17.63450  6.957771  6.907165  6.107049
## [3,] 29.98974 11.809841 13.533811 13.650982
## [4,] 39.50887 13.438438 19.252362 14.758983
## [5,] 47.25388 14.146197 18.918417 16.195415
## [6,] 53.78331 16.428054 22.308552 18.648659
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  6.958  4.852  1.629  0.708  2.282  3.493  2.642  2.937  4.817  1.231
## [11]  1.722  0.922 -0.681  0.029 -0.743  4.757  0.077 -0.045  1.602
```

```
plot_traj(x, "NIG : 10 trajectoires")
```



Modèle 7 : Poisson composé

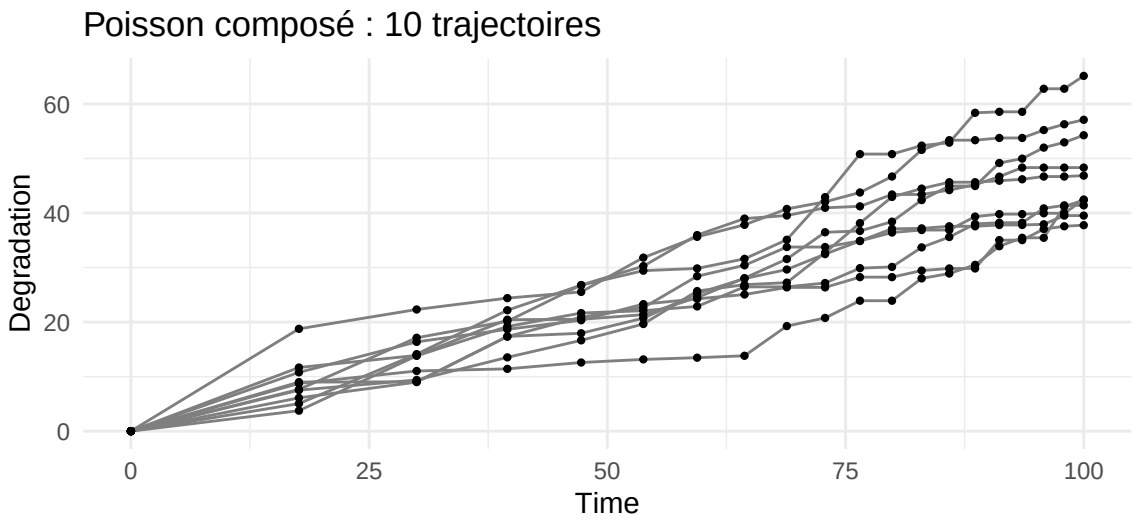
```
set.seed(127)
x <- sim_cpoisson(p2, lambda = 0.5)
head(x[, 1:4])
```

```
##             t        X1        X2        X3
## [1,]  0.00000  0.000000  0.000000  0.000000
## [2,] 17.63450  7.513850  8.746215  7.648505
## [3,] 29.98974  9.384705 11.031886 17.120379
## [4,] 39.50887 13.531190 11.420454 20.151459
## [5,] 47.25388 16.650181 12.585142 26.786477
## [6,] 53.78331 19.645973 13.154995 30.262848
```

```
round(diff(x[, "X1"]), 3)
```

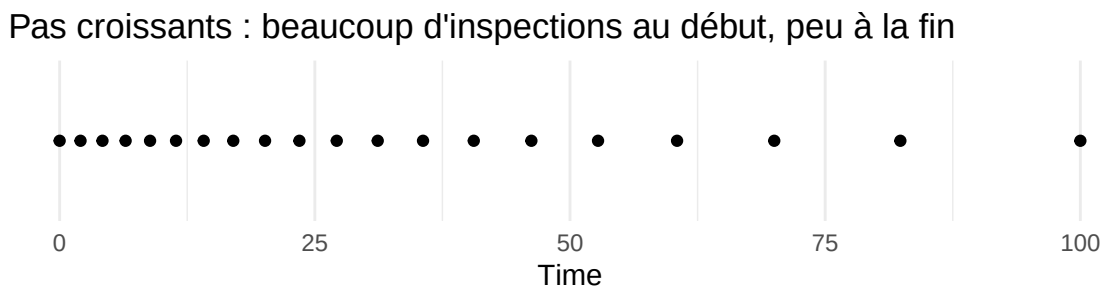
```
##  [1] 7.514 1.871 4.146 3.119 2.996 6.038 1.170 0.377 5.492 5.420 4.804 1.528
## [13] 1.169 0.000 0.274 0.266 0.499 0.000 0.170
```

```
plot_traj(x, "Poisson composé : 10 trajectoires")
```


Poisson composé : 10 trajectoires
Degradation
Time
0
20
40
60
0
25
50
75
100

Cas 3 : pas croissants

```
p3 <- inspection_times("increasing", n = 19, n_traj = 10, tmax = 100)
plot_plan(p3$time[p3$id == 1], "Pas croissants : beaucoup d'inspections au début, peu à la fin")
```



Modèle 1 : gamma

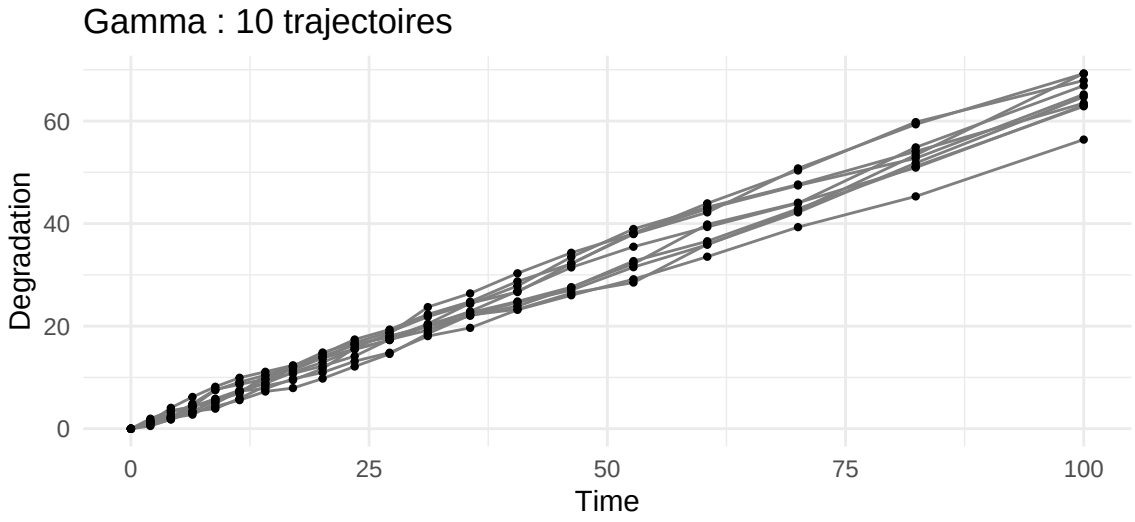
```
set.seed(131)
x <- sim_gamma(p3, a = 2, b = 3)
head(x[, 1:4])
```

```
##              t        X1       X2        X3
## [1,]  0.000000 0.0000000 0.000000 0.0000000
## [2,]  2.044686 0.7352141 1.345022 0.5348972
## [3,]  4.194814 1.8045992 2.541255 1.8324493
## [4,]  6.461852 3.1807416 3.395325 4.3396604
## [5,]  8.859250 4.3182935 3.906419 7.5080730
## [6,] 11.402919 5.5752760 5.900023 9.0922130
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  0.735  1.069  1.376  1.138  1.257  1.682  0.669  1.854  2.351  2.459
## [11]  3.957  3.619  2.461  2.394  4.478  4.378  6.684 10.811 15.900
```

```
plot_traj(x, "Gamma : 10 trajectoires")
```



Modèle 2 : Wiener

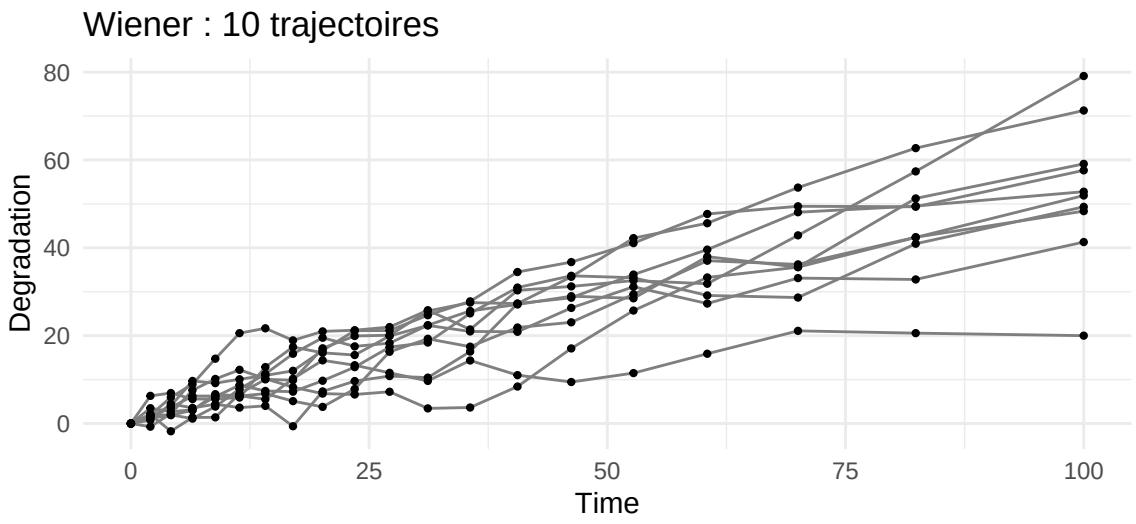
```
set.seed(132)
x <- sim_wiener(p3, mu = 0.5, sigma = 1.5)
head(x[, 1:4])
```

```
##              t       X1       X2        X3
## [1,]  0.000000 0.000000 0.000000  0.000000
## [2,]  2.044686 2.039259 3.487764  1.565567
## [3,]  4.194814 1.893723 2.714529  4.031155
## [4,]  6.461852 3.004754 3.097378  9.687832
## [5,]  8.859250 6.681576 6.134371  9.187424
## [6,] 11.402919 6.249524 8.737552 10.076563
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  2.039 -0.146  1.111  3.677 -0.432  0.563 -1.712 -1.315  4.074  8.477
## [11]  2.953 -1.798  4.368  1.184  6.371  7.649 -0.835  6.184  5.940
```

```
plot_traj(x, "Wiener : 10 trajectoires")
```



Modèle 3 : inverse gaussien

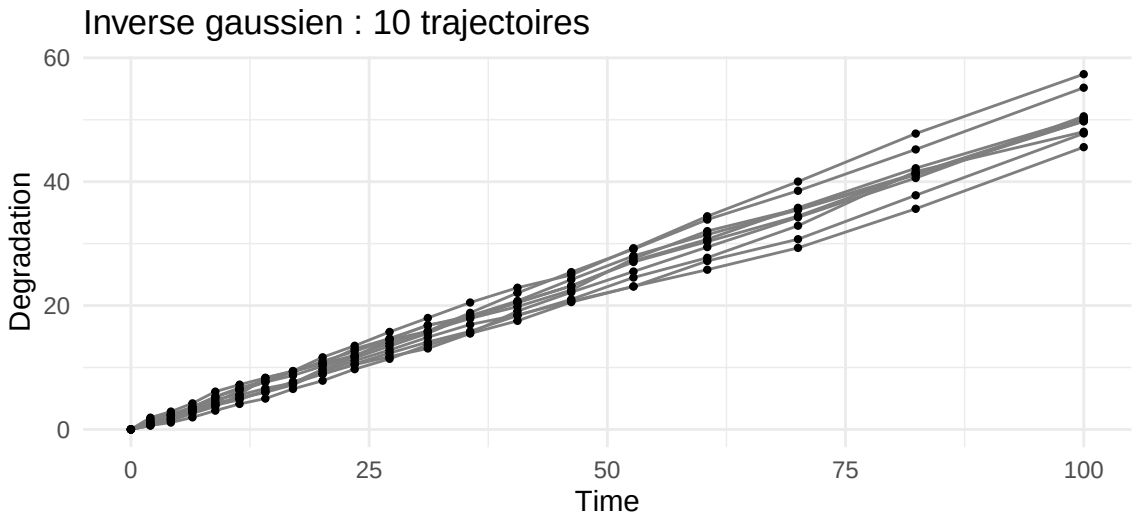
```
set.seed(133)
x <- sim_ig(p3, mu = 0.5, eta = 2)
head(x[, 1:4])
```

```
##              t        X1       X2        X3
## [1,]  0.000000 0.0000000 0.000000 0.0000000
## [2,]  2.044686 0.9905369 1.247812 0.7661222
## [3,]  4.194814 2.2006585 2.255756 1.5841530
## [4,]  6.461852 3.0720847 3.668087 3.5023778
## [5,]  8.859250 4.3903777 5.198115 4.8321986
## [6,] 11.402919 6.2607374 6.557040 5.6310647
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 0.991 1.210 0.871 1.318 1.870 1.374 1.075 1.513 1.607 2.607 1.424 2.964
## [13] 3.217 3.362 3.691 4.790 4.651 6.676 9.967
```

```
plot_traj(x, "Inverse gaussien : 10 trajectoires")
```



Modèle 4 : variance-gamma

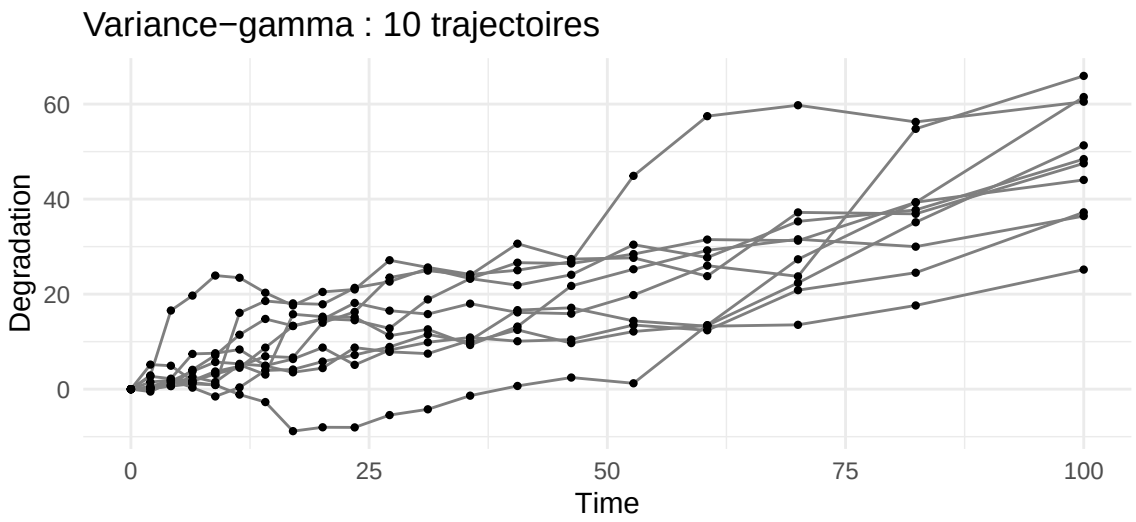
```
set.seed(134)
x <- sim_vg(p3, theta = 0.5, sigma = 1.5, nu = 2)
head(x[, 1:4])
```

```
##              t        X1        X2         X3
## [1,]  0.000000 0.0000000  0.000000  0.0000000
## [2,]  2.044686 0.1856678  2.895859  0.1090180
## [3,]  4.194814 1.6761704 16.557590  0.6399658
## [4,]  6.461852 3.7614427 19.695813  1.1370570
## [5,]  8.859250 5.7122171 23.913441  0.8924149
## [6,] 11.402919 5.3310732 23.453153 16.0690973
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  0.186  1.491  2.085  1.951 -0.381 -0.445 -1.340  0.890  4.315 -0.895
## [11] -0.380  2.808  6.344  0.500 -2.759 -1.119  0.293  4.076  7.547
```

```
plot_traj(x, "Variance-gamma : 10 trajectoires")
```



Modèle 5 : Meixner

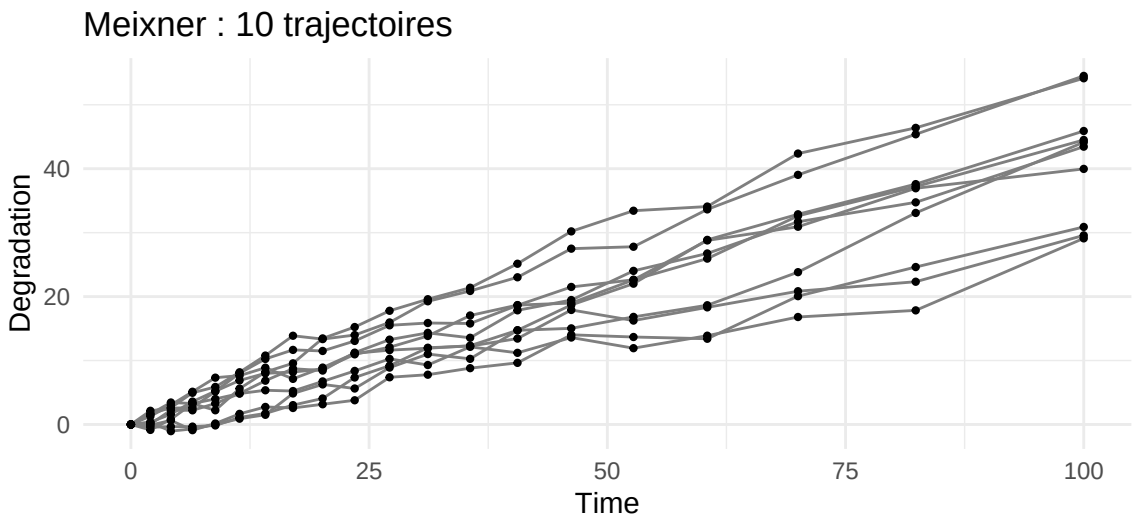
```
set.seed(135)
x <- sim_meixner(p3, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x[, 1:4])
```

```
##              t          X1        X2       X3
## [1,]  0.000000  0.00000000 0.0000000 0.000000
## [2,]  2.044686  0.31631140 0.0555588 1.291301
## [3,]  4.194814 -1.04229570 2.2365168 3.047314
## [4,]  6.461852 -0.68799770 2.2140710 5.117104
## [5,]  8.859250 -0.01440197 3.2786644 7.320675
## [6,] 11.402919  0.90549898 4.8513695 7.637460
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  0.316 -1.359  0.354  0.674  0.920  0.606  3.341  1.412 -0.639  3.271
## [11]  2.111 -0.742  4.465  0.301  1.791  1.821  5.175  9.265 11.060
```

```
plot_traj(x, "Meixner : 10 trajectoires")
```



Modèle 6 : normal inverse gaussien (NIG)

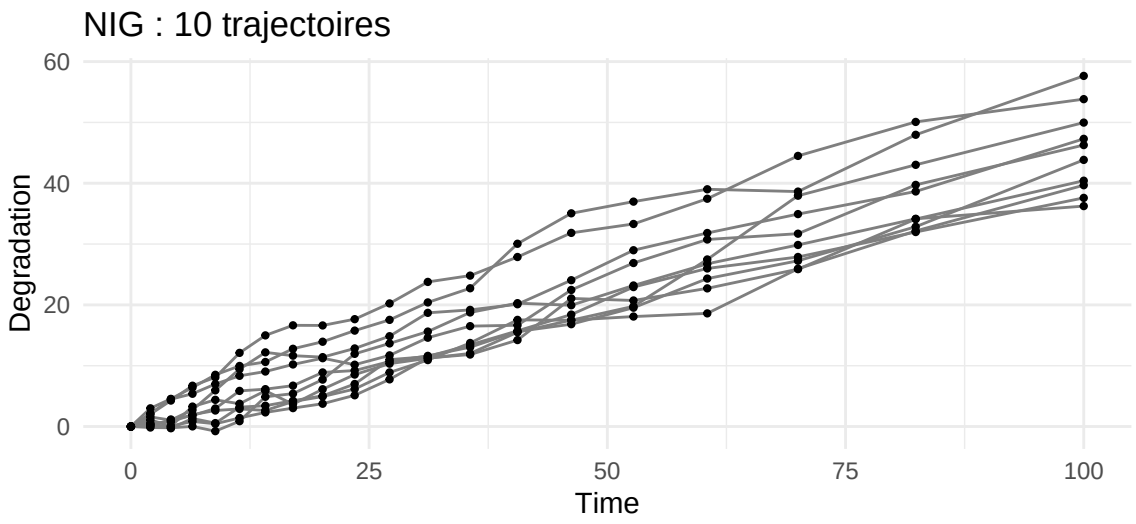
```
set.seed(136)
x <- sim_nig(p3, alpha = 2, beta = 0.5, delta = 1, mu = 0.2)
head(x[, 1:4])
```

```
##              t       X1       X2         X3
## [1,]  0.000000 0.000000 0.000000 0.00000000
## [2,]  2.044686 2.136377 1.573151 1.08127517
## [3,]  4.194814 4.492436 1.051712 0.07437418
## [4,]  6.461852 5.385379 2.729506 0.84028619
## [5,]  8.859250 6.980514 5.946594 0.44599024
## [6,] 11.402919 8.338229 9.441637 1.40854862
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  2.136  2.356  0.893  1.595  1.358  0.702  1.165  1.028 -1.074  1.542
## [11]  2.901  1.901  0.146  5.803  4.428  3.867  0.956  8.023  6.550
```

```
plot_traj(x, "NIG : 10 trajectoires")
```



Modèle 7 : Poisson composé

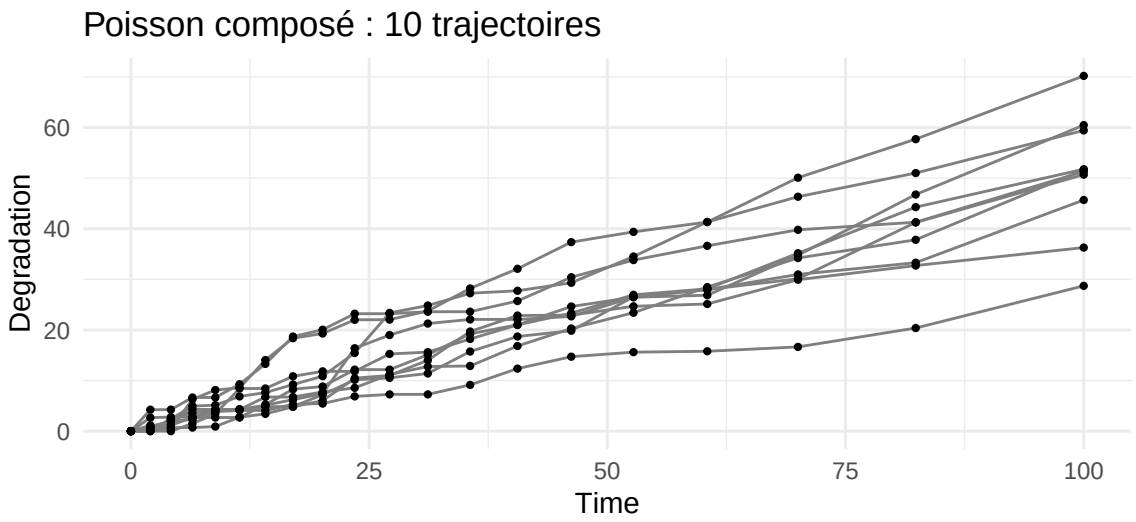
```
set.seed(137)
x <- sim_cpoisson(p3, lambda = 0.5)
head(x[, 1:4])
```

```
##              t        X1        X2        X3
## [1,]  0.000000 0.0000000 0.0000000 0.0000000
## [2,]  2.044686 0.3064054 0.8433624 0.7062454
## [3,]  4.194814 0.5146710 1.9689993 0.7062454
## [4,]  6.461852 6.2893287 4.2862981 0.7062454
## [5,]  8.859250 8.1244537 4.2862981 0.9071791
## [6,] 11.402919 8.5623822 4.2862981 2.7595776
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  0.306  0.208  5.775  1.835  0.438  5.482  4.321  0.932  2.706  0.037
## [11]  1.708  4.458  3.880  5.270  2.031  1.935  8.771  7.621 12.501
```

```
plot_traj(x, "Poisson composé : 10 trajectoires")
```


Poisson composé : 10 trajectoires
Degradation
0
20
40
60
0
25
50
75
100
Time

Cas 4 : mixtes, beaucoup au début et à la fin

```
p4 <- inspection_times("inc-dec", n = 19, n_traj = 10, tmax = 100)
plot_plan(p4$time[p4$id == 1], "Beaucoup au début et à la fin, peu au milieu")
```



Modèle 1 : gamma

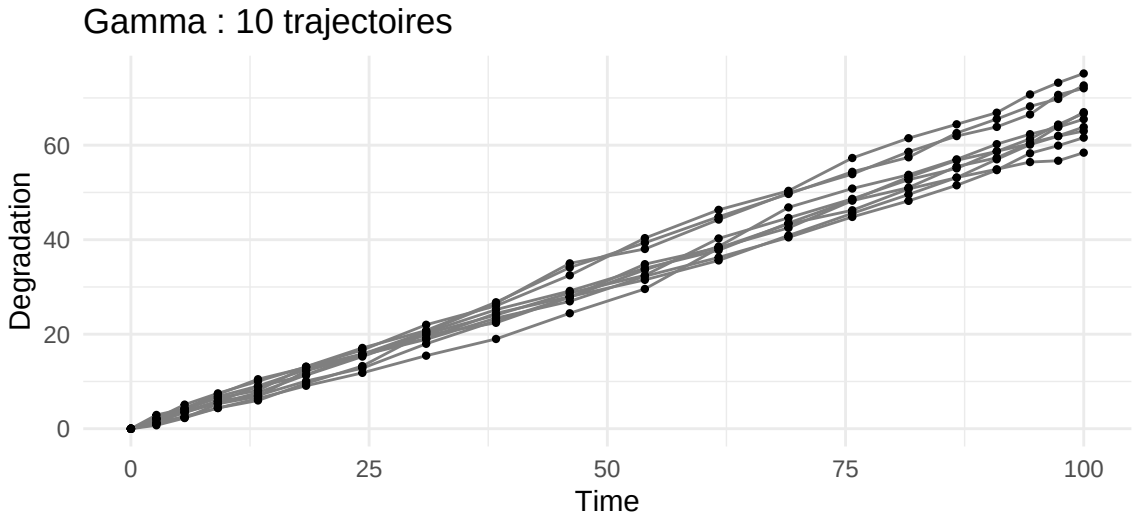
```
set.seed(141)
x <- sim_gamma(p4, a = 2, b = 3)
head(x[, 1:4])
```

```
##              t        X1        X2         X3
## [1,]  0.000000  0.000000  0.000000  0.0000000
## [2,]  2.679282  2.019488  1.957203  0.7280362
## [3,]  5.638566  3.739231  3.377459  2.2594259
## [4,]  9.127515  5.788542  5.847097  5.2725973
## [5,] 13.338389  8.920814  7.039203  6.8201854
## [6,] 18.385222 11.314482 10.082758 12.2277362
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 2.019 1.720 2.049 3.132 2.394 3.955 4.798 3.988 4.667 3.740 7.802 4.352
## [13] 4.021 4.054 2.393 3.642 1.555 3.672 3.027
```

```
plot_traj(x, "Gamma : 10 trajectoires")
```



Modèle 2 : Wiener

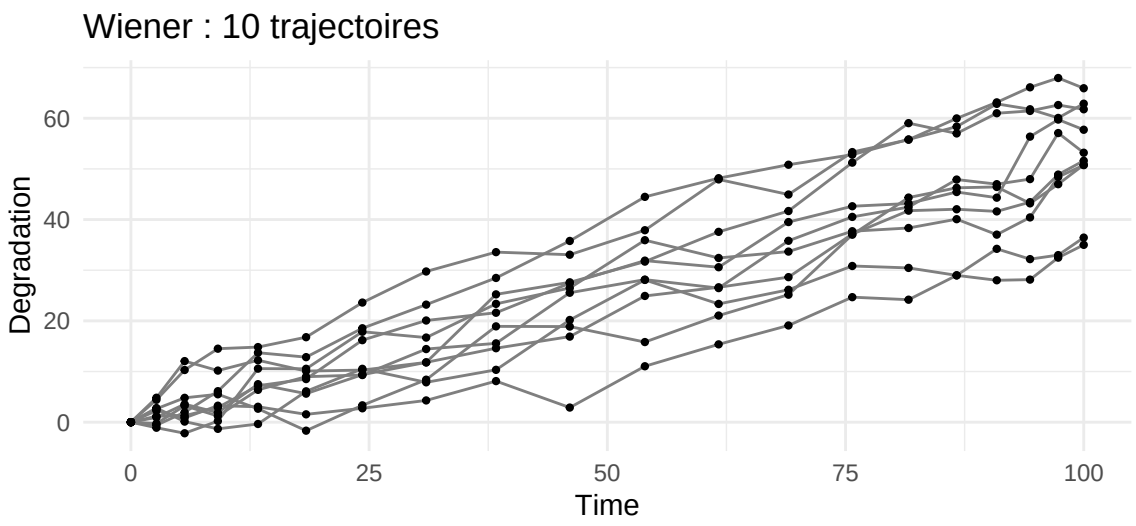
```
set.seed(142)
x <- sim_wiener(p4, mu = 0.5, sigma = 1.5)
head(x[, 1:4])
```

```
##              t        X1         X2         X3
## [1,]  0.000000  0.000000  0.0000000  0.0000000
## [2,]  2.679282  4.420328  2.7880040 -1.0967920
## [3,]  5.638566 10.297168  0.1124737 -2.1863975
## [4,]  9.127515 14.508139 -1.3165312  0.2204537
## [5,] 13.338389 14.823545 -0.3581420 10.5689549
## [6,] 18.385222 16.776834  6.0893315 10.5384770
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  4.420  5.877  4.211  0.315  1.953  6.835  6.136  3.821 -0.510  4.836
## [11] 10.029 -2.980  8.377  2.451  4.197  3.168  2.972  1.838 -2.020
```

```
plot_traj(x, "Wiener : 10 trajectoires")
```



Modèle 3 : inverse gaussien

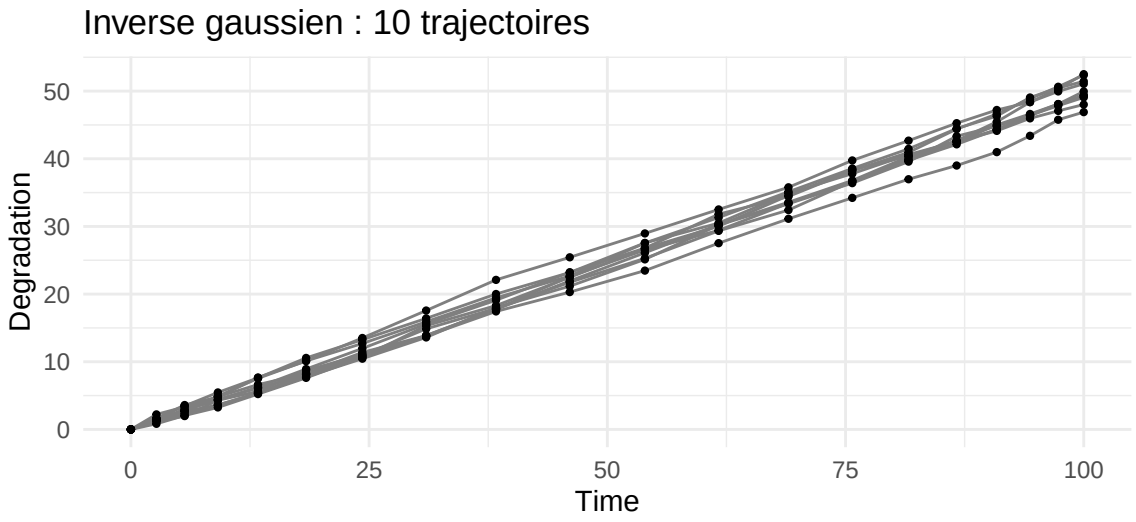
```
set.seed(143)
x <- sim_ig(p4, mu = 0.5, eta = 2)
head(x[, 1:4])
```

```
##              t        X1        X2        X3
## [1,]  0.000000 0.0000000  0.000000  0.000000
## [2,]  2.679282 0.8299491  1.622756  1.326539
## [3,]  5.638566 2.0564501  3.603001  2.677501
## [4,]  9.127515 3.2484546  4.775965  4.910051
## [5,] 13.338389 5.2230838  7.672906  7.548331
## [6,] 18.385222 7.6192727 10.108601 10.121467
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 0.830 1.227 1.192 1.975 2.396 3.059 4.649 2.974 4.183 4.270 3.289 4.527
## [13] 3.804 2.025 2.271 2.789 2.904 1.958 2.179
```

```
plot_traj(x, "Inverse gaussien : 10 trajectoires")
```



Modèle 4 : variance-gamma

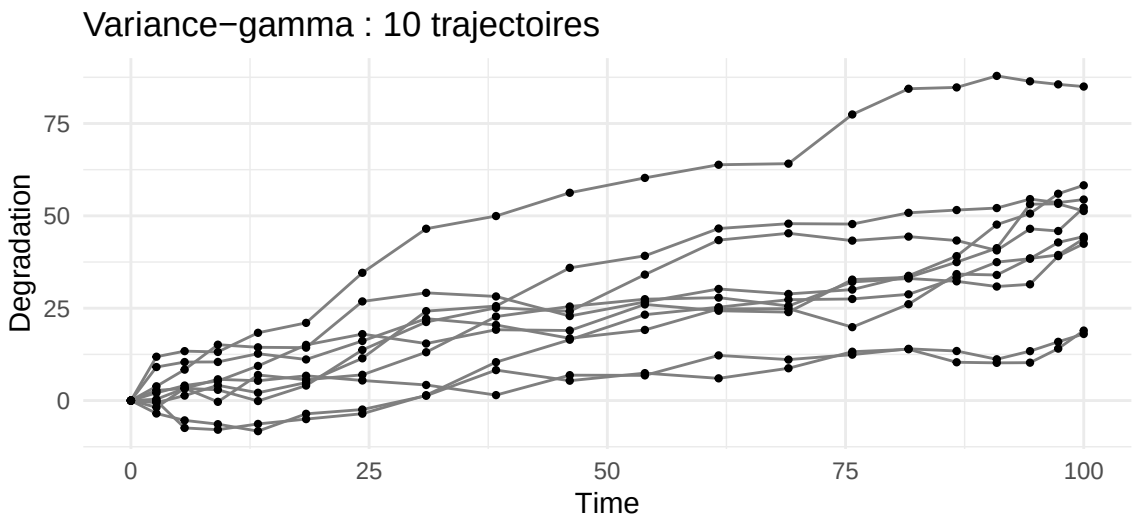
```
set.seed(144)
x <- sim_vg(p4, theta = 0.5, sigma = 1.5, nu = 2)
head(x[, 1:4])
```

```
##              t        X1       X2         X3
## [1,]  0.000000  0.000000  0.00000  0.0000000
## [2,]  2.679282  9.077875 11.86655  0.3547826
## [3,]  5.638566 10.424350 13.37995  3.3131547
## [4,]  9.127515 10.455647 13.16475  2.9083206
## [5,] 13.338389 12.685364 18.34240 -0.1231431
## [6,] 18.385222 11.128890 21.02556  4.0621666
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  9.078  1.346  0.031  2.230 -1.556  5.007  6.030 -1.710 -3.607  2.271
## [11]  5.697  0.059 -5.000  6.207  8.150 -0.239  4.484  0.931  4.392
```

```
plot_traj(x, "Variance-gamma : 10 trajectoires")
```



Modèle 5 : Meixner

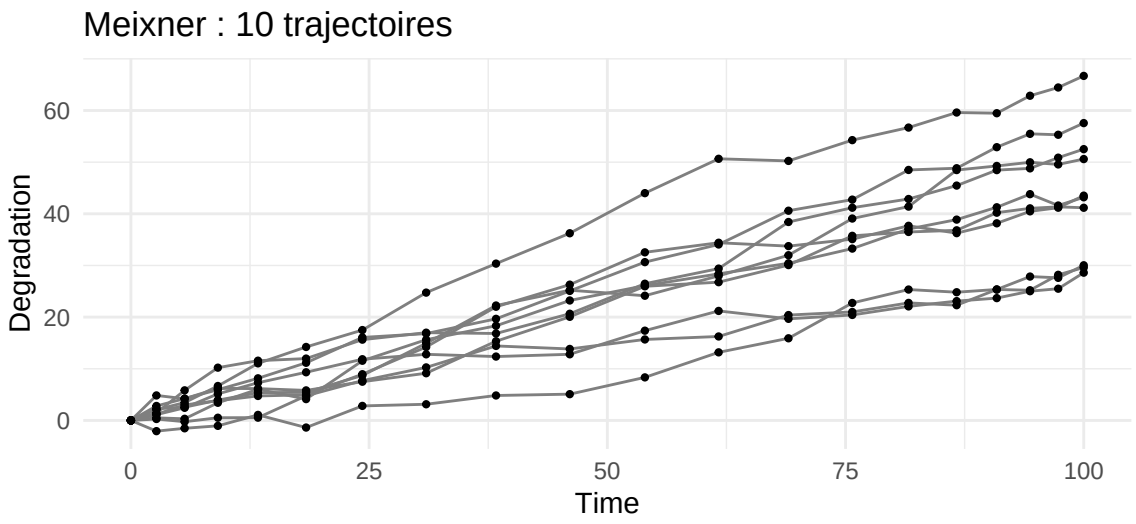
```
set.seed(145)
x <- sim_meixner(p4, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x[, 1:4])
```

```
##              t        X1         X2       X3
## [1,]  0.000000  0.000000  0.0000000 0.000000
## [2,]  2.679282  2.090297  0.2763308 4.837432
## [3,]  5.638566  3.453719 -0.2519603 4.196843
## [4,]  9.127515  6.665677  0.5233437 6.211717
## [5,] 13.338389 11.044790  0.5365556 6.157052
## [6,] 18.385222 14.221212  4.7681305 5.814885
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  2.090  1.363  3.212  4.379  3.176  3.270  7.253  5.610  5.880  7.764
## [11]  6.651 -0.408  4.022  2.430  2.914 -0.137  3.384  1.598  2.244
```

```
plot_traj(x, "Meixner : 10 trajectoires")
```



Modèle 6 : normal inverse gaussien (NIG)

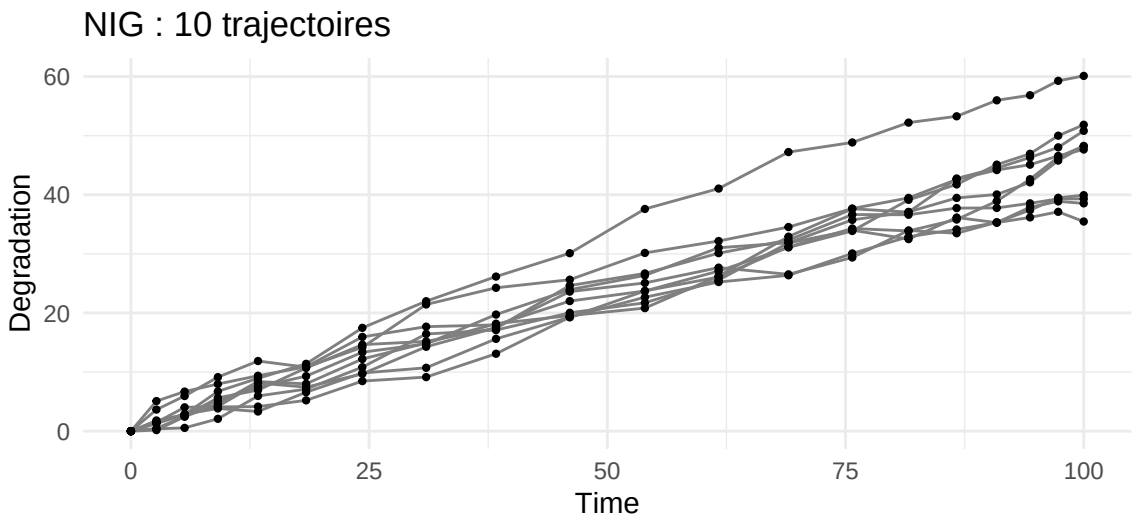
```
set.seed(146)
x <- sim_nig(p4, alpha = 2, beta = 0.5, delta = 1, mu = 0.2)
head(x[, 1:4])
```

```
##              t         X1        X2       X3
## [1,]  0.000000  0.0000000 0.0000000 0.000000
## [2,]  2.679282  0.4502579 0.1747548 1.471945
## [3,]  5.638566  2.5315490 2.4365730 4.028428
## [4,]  9.127515  5.6370511 4.6249024 4.185283
## [5,] 13.338389  6.9426752 8.3819989 8.142720
## [6,] 18.385222 10.6933017 8.0105974 7.446320
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  0.450  2.081  3.106  1.306  3.751  3.938  0.557  2.245  7.202  2.049
## [11]  3.432  2.222  4.311 -0.033  1.124  0.028  0.779  0.747  0.014
```

```
plot_traj(x, "NIG : 10 trajectoires")
```



Modèle 7 : Poisson composé

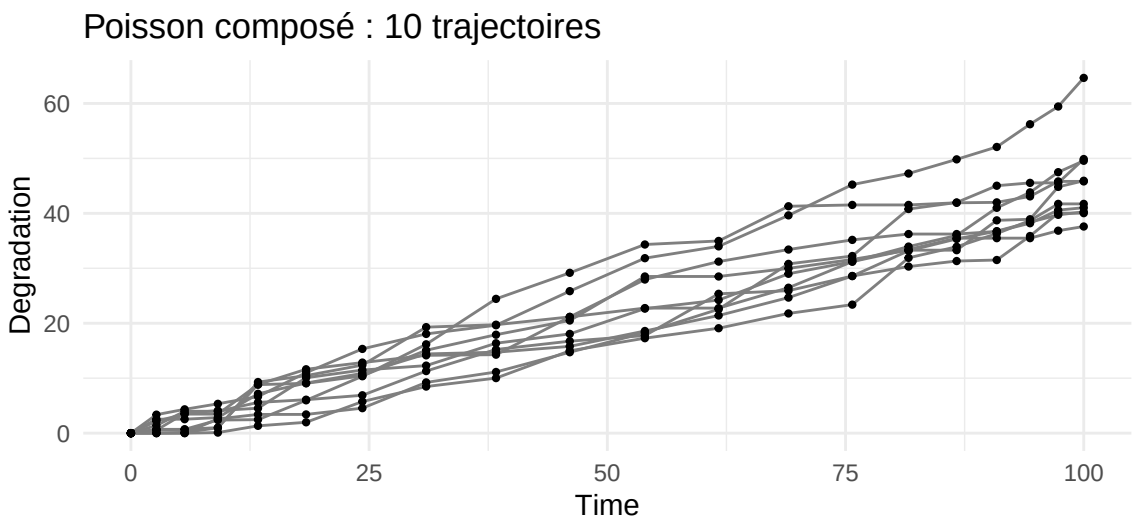
```
set.seed(147)
x <- sim_cpoisson(p4, lambda = 0.5)
head(x[, 1:4])
```

```
##              t          X1         X2       X3
## [1,]  0.000000  0.00000000  0.0000000 0.000000
## [2,]  2.679282  0.01603795  0.5471488 2.538561
## [3,]  5.638566  0.13710933  3.6011246 2.538561
## [4,]  9.127515  0.96794307  4.1083816 2.832254
## [5,] 13.338389  9.29701299  4.5344724 7.159487
## [6,] 18.385222 10.46385631 10.0833594 9.112500
```

```
round(diff(x[, "X1"]), 3)
```

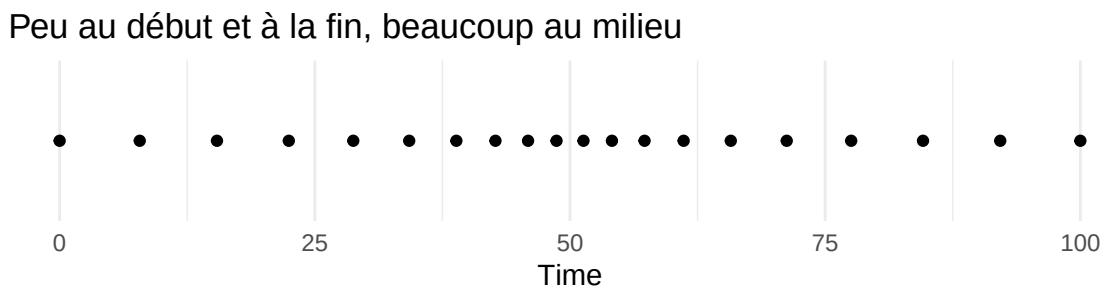
```
##  [1] 0.016 0.121 0.831 8.329 1.167 1.921 6.908 0.440 1.421 6.803 3.261 2.193
## [13] 1.758 1.065 0.000 0.470 1.499 2.381 0.442
```

```
plot_traj(x, "Poisson composé : 10 trajectoires")
```


Poisson composé : 10 trajectoires
Degradation
Time
0
20
40
60
0
25
50
75
100

Cas 5 : mixtes, beaucoup au milieu

```
p5 <- inspection_times("dec-inc", n = 19, n_traj = 10, tmax = 100)
plot_plan(p5$time[p5$id == 1], "Peu au début et à la fin, beaucoup au milieu")
```



Modèle 1 : gamma

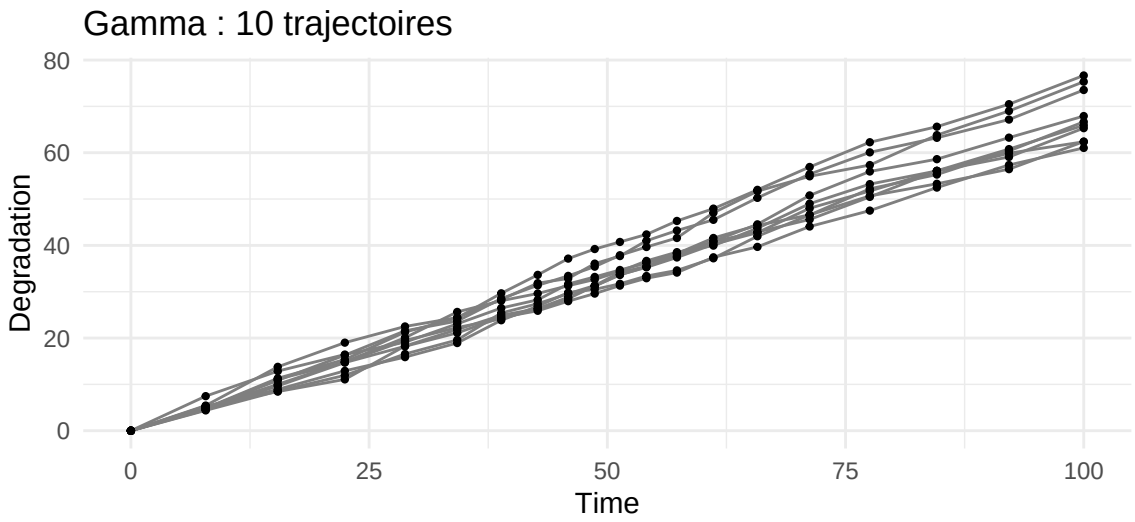
```
set.seed(151)
x <- sim_gamma(p5, a = 2, b = 3)
head(x[, 1:4])
```

```
##              t        X1        X2        X3
## [1,]  0.000000  0.000000  0.000000  0.000000
## [2,]  7.847034  4.997945  5.427388  4.508088
## [3,] 15.414065 10.000425 13.784730  9.666280
## [4,] 22.451433 14.887507 19.004520 14.705898
## [5,] 28.766874 20.068298 22.497879 18.221144
## [6,] 34.246357 25.636675 24.422123 21.106650
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 4.998 5.002 4.887 5.181 5.568 2.413 1.565 1.683 1.346 1.733 2.275 1.884
## [13] 1.411 3.759 5.296 4.215 2.849 3.014 6.223
```

```
plot_traj(x, "Gamma : 10 trajectoires")
```



Modèle 2 : Wiener

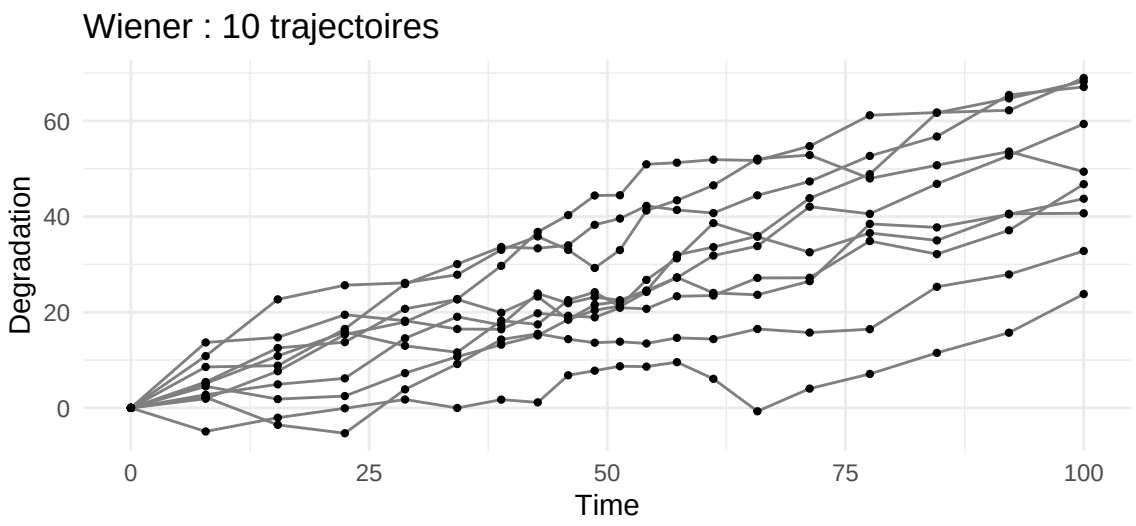
```
set.seed(152)
x <- sim_wiener(p5, mu = 0.5, sigma = 1.5)
head(x[, 1:4])
```

```
##              t       X1        X2        X3
## [1,]  0.000000  0.00000  0.000000  0.000000
## [2,]  7.847034 13.67857  5.409302  2.218293
## [3,] 15.414065 14.73747 12.520859 -3.556313
## [4,] 22.451433 19.47012 13.747583 -5.304637
## [5,] 28.766874 18.22984 20.709516  3.867991
## [6,] 34.246357 16.48403 22.677161  9.174728
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 13.679  1.059  4.733 -1.240 -1.746 -0.047  3.343 -0.521 -0.325  2.005
## [11] -0.223  2.626  0.149  3.679  0.044  7.674 -2.736  4.982  9.647
```

```
plot_traj(x, "Wiener : 10 trajectoires")
```



Modèle 3 : inverse gaussien

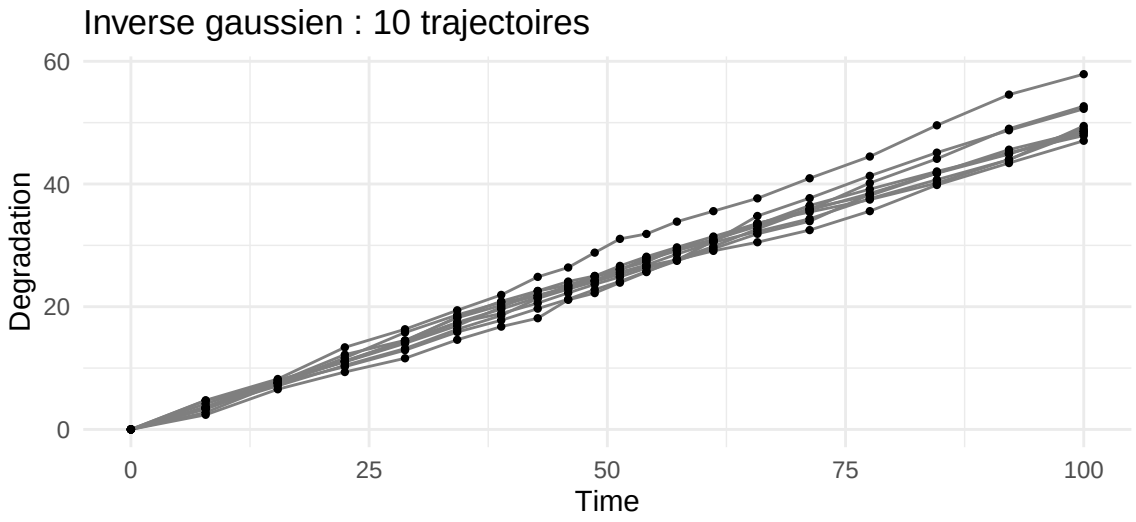
```
set.seed(153)
x <- sim_ig(p5, mu = 0.5, eta = 2)
head(x[, 1:4])
```

```
##              t        X1        X2        X3
## [1,]  0.000000  0.000000  0.000000  0.000000
## [2,]  7.847034  2.384445  4.692209  3.441015
## [3,] 15.414065  6.546014  7.482063  7.628125
## [4,] 22.451433  9.356540 11.657376 11.096211
## [5,] 28.766874 11.593604 15.796735 14.419819
## [6,] 34.246357 14.609611 18.657331 18.303713
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 2.384 4.162 2.811 2.237 3.016 2.145 1.369 2.998 1.100 1.696 1.731 2.111
## [13] 2.055 2.871 3.150 4.326 3.947 4.900 3.666
```

```
plot_traj(x, "Inverse gaussien : 10 trajectoires")
```



Modèle 4 : variance-gamma

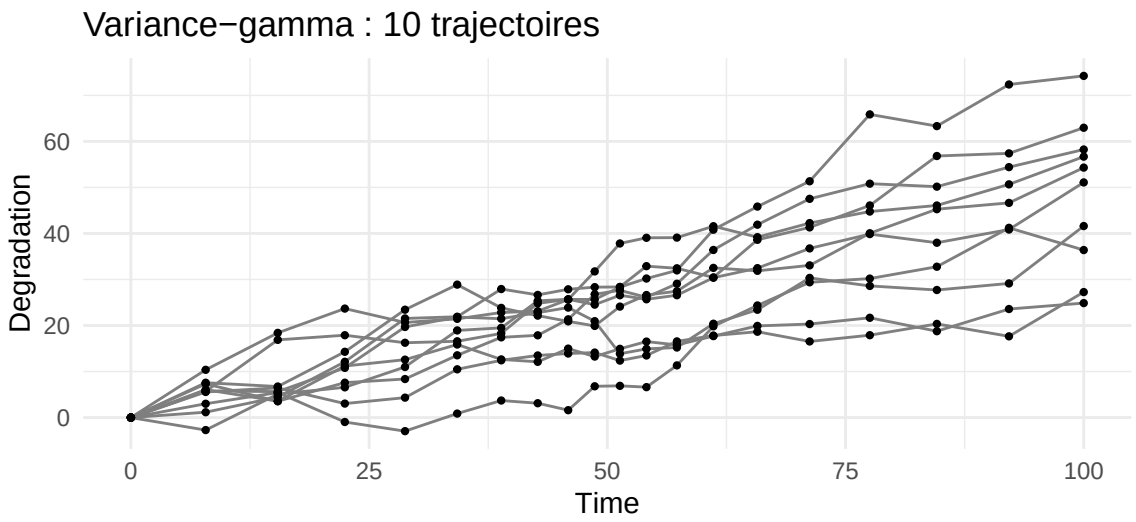
```
set.seed(154)
x <- sim_vg(p5, theta = 0.5, sigma = 1.5, nu = 2)
head(x[, 1:4])
```

```
##              t        X1       X2        X3
## [1,]  0.000000  0.000000  0.00000  0.000000
## [2,]  7.847034  5.779344 10.37584  7.568184
## [3,] 15.414065  5.639402 18.40278  6.745224
## [4,] 22.451433 10.796474 23.67842 14.289071
## [5,] 28.766874 19.688026 20.65997 23.435589
## [6,] 34.246357 21.839781 21.44797 28.884128
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  5.779 -0.140  5.157  8.892  2.152 -0.363  1.330  1.087 -2.950 -7.080
## [11]  1.063  0.295  5.170  3.033  6.921 -1.743 -0.884  1.426 12.469
```

```
plot_traj(x, "Variance-gamma : 10 trajectoires")
```



Modèle 5 : Meixner

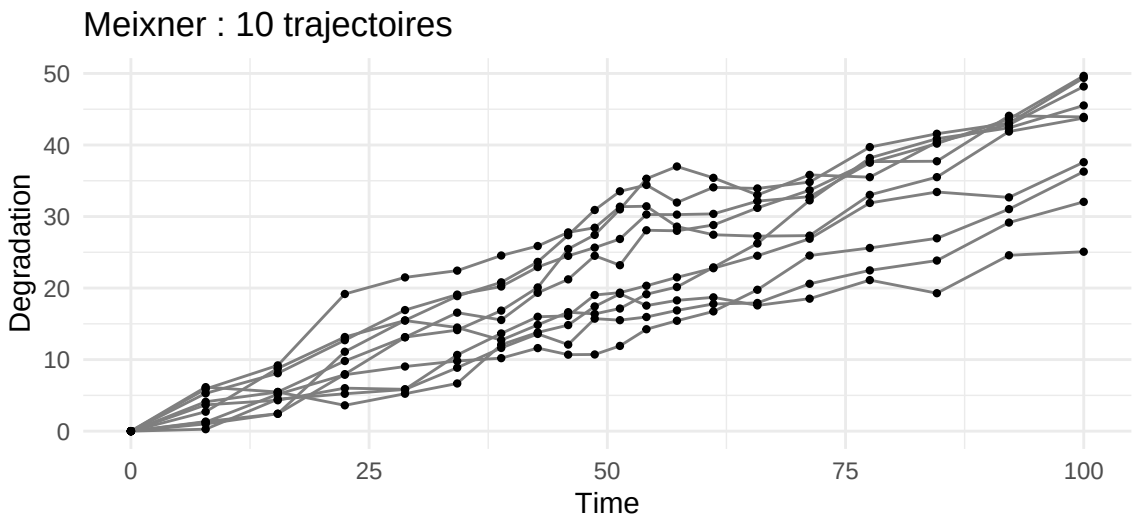
```
set.seed(155)
x <- sim_meixner(p5, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x[, 1:4])
```

```
##              t        X1       X2        X3
## [1,]  0.000000  0.000000 0.000000  0.000000
## [2,]  7.847034  5.286274 1.263592  5.887246
## [3,] 15.414065  8.093605 5.154572  9.193110
## [4,] 22.451433 12.712238 7.883720 19.169830
## [5,] 28.766874 16.922962 9.032652 21.488630
## [6,] 34.246357 19.086135 9.805271 22.436828
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  5.286  2.807  4.619  4.211  2.163  1.131  2.719  1.564  1.154  1.206
## [11]  3.420 -0.012  0.101  1.784  0.624  4.742  2.672  3.430  6.037
```

```
plot_traj(x, "Meixner : 10 trajectoires")
```



Modèle 6 : normal inverse gaussien (NIG)

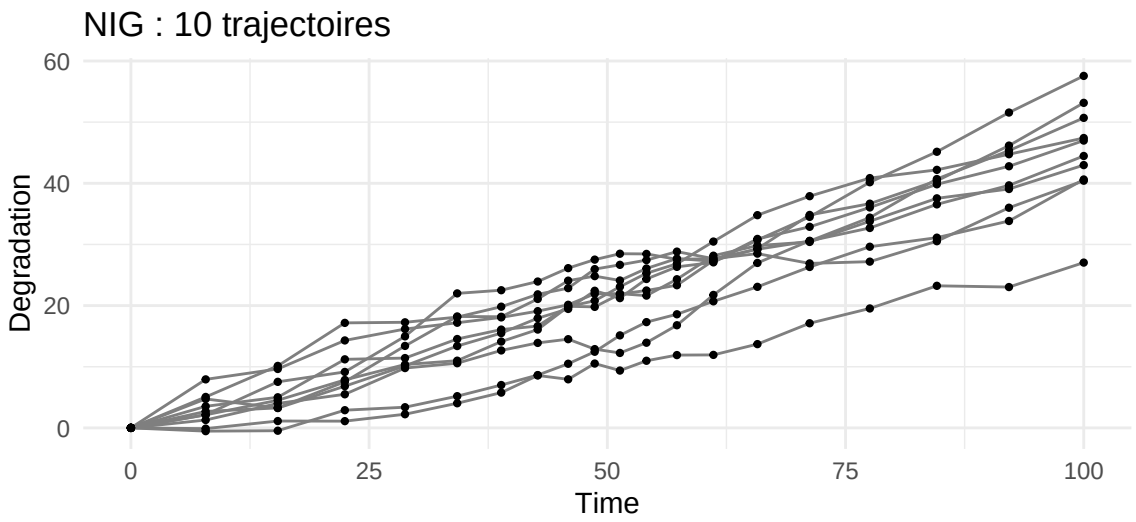
```
set.seed(156)
x <- sim_nig(p5, alpha = 2, beta = 0.5, delta = 1, mu = 0.2)
head(x[, 1:4])
```

```
##              t        X1        X2        X3
## [1,]  0.000000  0.000000  0.000000  0.000000
## [2,]  7.847034  2.692694  2.159865  7.934170
## [3,] 15.414065  3.269838  7.531751  9.637883
## [4,] 22.451433  7.546070  9.149866 14.301135
## [5,] 28.766874 13.402778 14.968463 16.167574
## [6,] 34.246357 18.223668 21.994793 17.193845
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  2.693  0.577  4.276  5.857  4.821 -0.054  2.916  2.990  0.741 -0.712
## [11]  1.962  1.624 -0.434  3.644  1.980  3.167  3.785  2.950  4.203
```

```
plot_traj(x, "NIG : 10 trajectoires")
```



Modèle 7 : Poisson composé

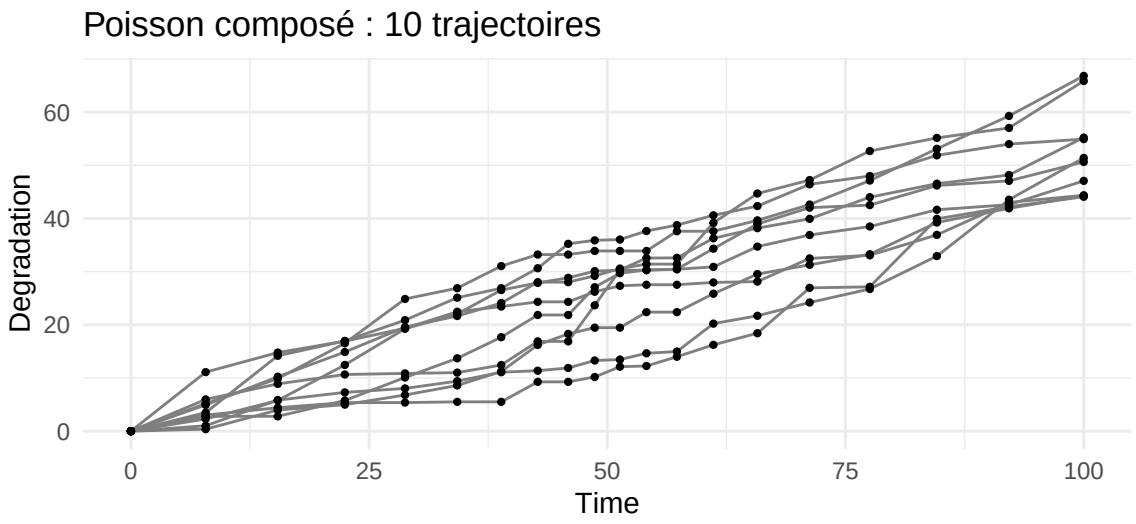
```
set.seed(157)
x <- sim_cpoisson(p5, lambda = 0.5)
head(x[, 1:4])
```

```
##              t        X1        X2       X3
## [1,]  0.000000  0.000000  0.000000  0.00000
## [2,]  7.847034  2.787838  5.188802 11.09909
## [3,] 15.414065  2.787838  9.871755 14.77085
## [4,] 22.451433  5.757662 16.520749 16.92100
## [5,] 28.766874 10.086708 24.855109 19.32408
## [6,] 34.246357 13.689676 26.880477 22.44915
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 2.788 0.000 2.970 4.329 3.603 3.977 4.177 0.000 5.223 2.633 0.605 0.144
## [13] 0.439 3.812 2.184 1.599 3.148 0.952 4.485
```

```
plot_traj(x, "Poisson composé : 10 trajectoires")
```



ewpage

III - Processus de Lévy à échelle de temps

Une fonction d'échelle de temps A (fonction croissante, continue à droite avec des limites à gauche, $A(0) = 0$ et $A(t) \to +\infty$) donne le processus de Lévy à échelle de temps $X_t = Y_{A(t)}$, où Y est un processus de Lévy. Le paramètre que A remplace (identifiabilité) n'est alors pas donné :

Processus	Fonction	$X_t - X_s$, $\Delta A = A(t) - A(s)$	Avec $A(t) = at$ ou μt
Gamma	`sim_gamma(t, b, A)`	$\Gamma(\Delta A, b)$	gamma de forme a
Wiener	`sim_wiener(t, sigma, A)`	$\mathcal{N}(\Delta A, \sigma^2 \Delta A)$	Wiener $(\mu, \sigma\sqrt{\mu})$
Inverse gaussien	`sim_ig(t, eta, A)`	$IG(\Delta A, \eta \Delta A^2)$	IG $(\mu, \eta\mu^2)$

Pour le Wiener, $X_t = \mu t + \sigma B_{\mu t}$ a la loi de $\mu t + \sigma\sqrt{\mu}\, B_t$, car $(B_{\mu t})_{t\geq 0}$ a la loi de $(\sqrt{\mu}\, B_t)_{t\geq 0}$.

On prend $A(t) = 0.02\, t^{1.5}$, de sorte que $A(100) = 20$, et les 10 trajectoires du cas 2 de la partie II (pas décroissants).

Modèle 1 : gamma à échelle de temps (Cha et Mercier, 2019, exemple 2)

```
A <- function(t) 0.02 * t^1.5
set.seed(301)
x <- sim_gamma(p2, b = 3, A = A)
x
```

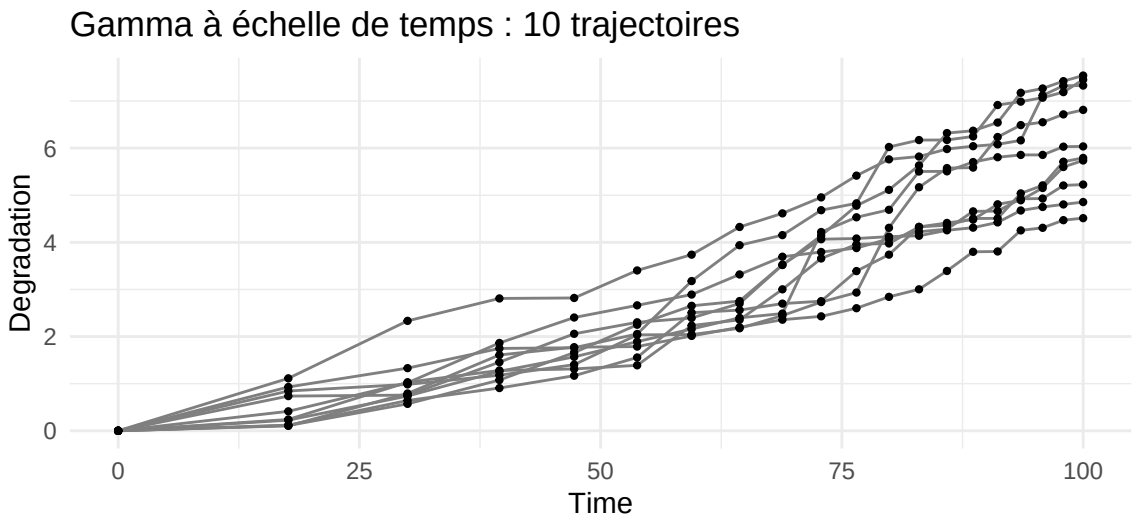
```
## 10 simulated trajectory(ies) of the time-scaled gamma process, same inspection times (matrix)
## parameters: b = 3
## time-scaling function: A = function (t)  0.02 * t^1.5
##
##             t       X1        X2        X3        X4       X5
## [1,]  0.00000 0.000000 0.0000000 0.0000000 0.0000000 0.000000
```

```
## [2,] 17.63450 0.413235 0.2260551 0.1117200 0.1195689 1.113368
## [3,] 29.98974 1.018565 0.7377684 0.5715126 0.6429133 2.333316
## [4,] 39.50887 1.863933 1.2619976 1.0769566 0.9074442 2.809108
## [5,] 47.25388 2.404166 1.5691255 1.6593942 1.1687358 2.819350
## [6,] 53.78331 2.662670 1.8916061 2.2509750 1.5558784 3.403399
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 0.413 0.605 0.845 0.540 0.259 0.229 0.427 0.376 0.100 0.084 0.196 0.150
## [13] 0.055 0.378 0.006 0.261 0.006 0.277 0.019
```

```
plot_traj(x, "Gamma à échelle de temps : 10 trajectoires")
```



Modèle 2 : Wiener à échelle de temps (Cha et Mercier, 2019, exemple 1)

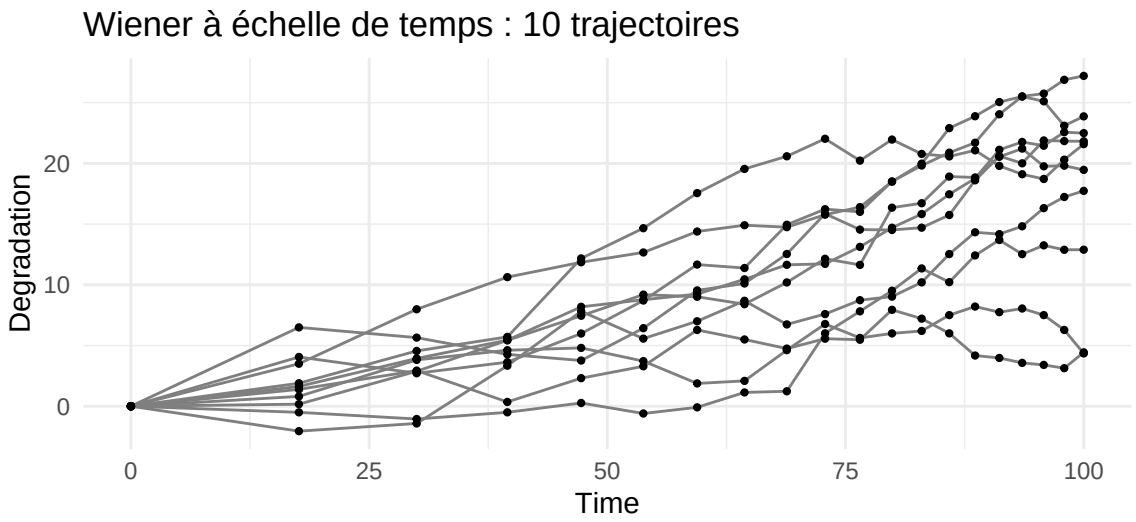
```
set.seed(302)
x <- sim_wiener(p2, sigma = 1.5, A = A)
head(x[, 1:4])
```

```
##             t        X1        X2        X3
## [1,]  0.00000  0.000000 0.0000000 0.0000000
## [2,] 17.63450  1.893327 0.1725442 0.8115244
## [3,] 29.98974  4.555056 2.8800225 3.8168637
## [4,] 39.50887  5.712354 5.4312405 4.6137487
## [5,] 47.25388 12.176191 7.4521673 4.8047256
## [6,] 53.78331 14.665366 9.1931509 3.7176109
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  1.893  2.662  1.157  6.464  2.489  2.890  1.989  1.042  1.442 -1.799
## [11]  1.733 -1.183 -0.204  0.501 -1.283 -0.688 -0.389  1.589  1.280
```

```
plot_traj(x, "Wiener à échelle de temps : 10 trajectoires")
```



Modèle 3 : inverse gaussien à échelle de temps (Cha et Mercier, 2019, exemple 3)

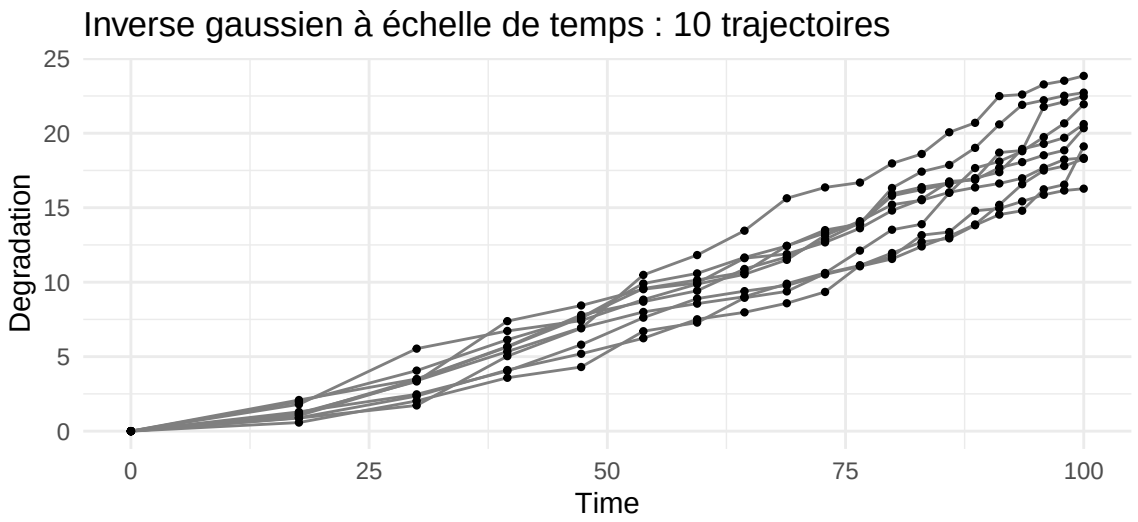
```
set.seed(303)
x <- sim_ig(p2, eta = 2, A = A)
head(x[, 1:4])
```

```
##             t       X1       X2         X3
## [1,]  0.00000 0.000000 0.000000  0.0000000
## [2,] 17.63450 1.974875 1.205460  0.8787256
## [3,] 29.98974 4.063401 3.334509  1.7250852
## [4,] 39.50887 6.136971 7.385009  5.0342163
## [5,] 47.25388 7.608008 8.435758  6.9013147
## [6,] 53.78331 9.573188 9.529091 10.4882354
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 1.975 2.089 2.074 1.471 1.965 0.571 0.526 1.771 0.861 0.613 1.884 0.414
## [13] 0.384 0.386 0.399 1.561 0.342 0.411 0.916
```

```
plot_traj(x, "Inverse gaussien à échelle de temps : 10 trajectoires")
```



Vérification : avec $A(t) = at$, on retrouve le processus gamma de forme a

Avec la même graine, le gamma à échelle de temps avec $A(t) = 2t$ et le gamma $(a, b) = (2, 3)$ donnent les mêmes trajectoires :

```
set.seed(304)
y1 <- sim_gamma(p2, b = 3, A = function(t) 2 * t)
set.seed(304)
y2 <- sim_gamma(p2, a = 2, b = 3)
all.equal(unclass(y1)[, -1], unclass(y2)[, -1], check.attributes = FALSE)
```

```
## [1] TRUE
```